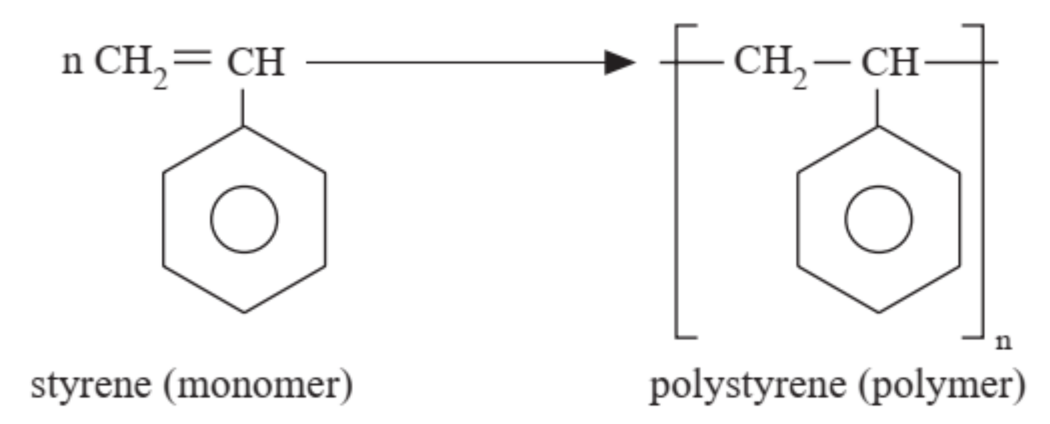
POLYMERS

Classification

What is polymer?

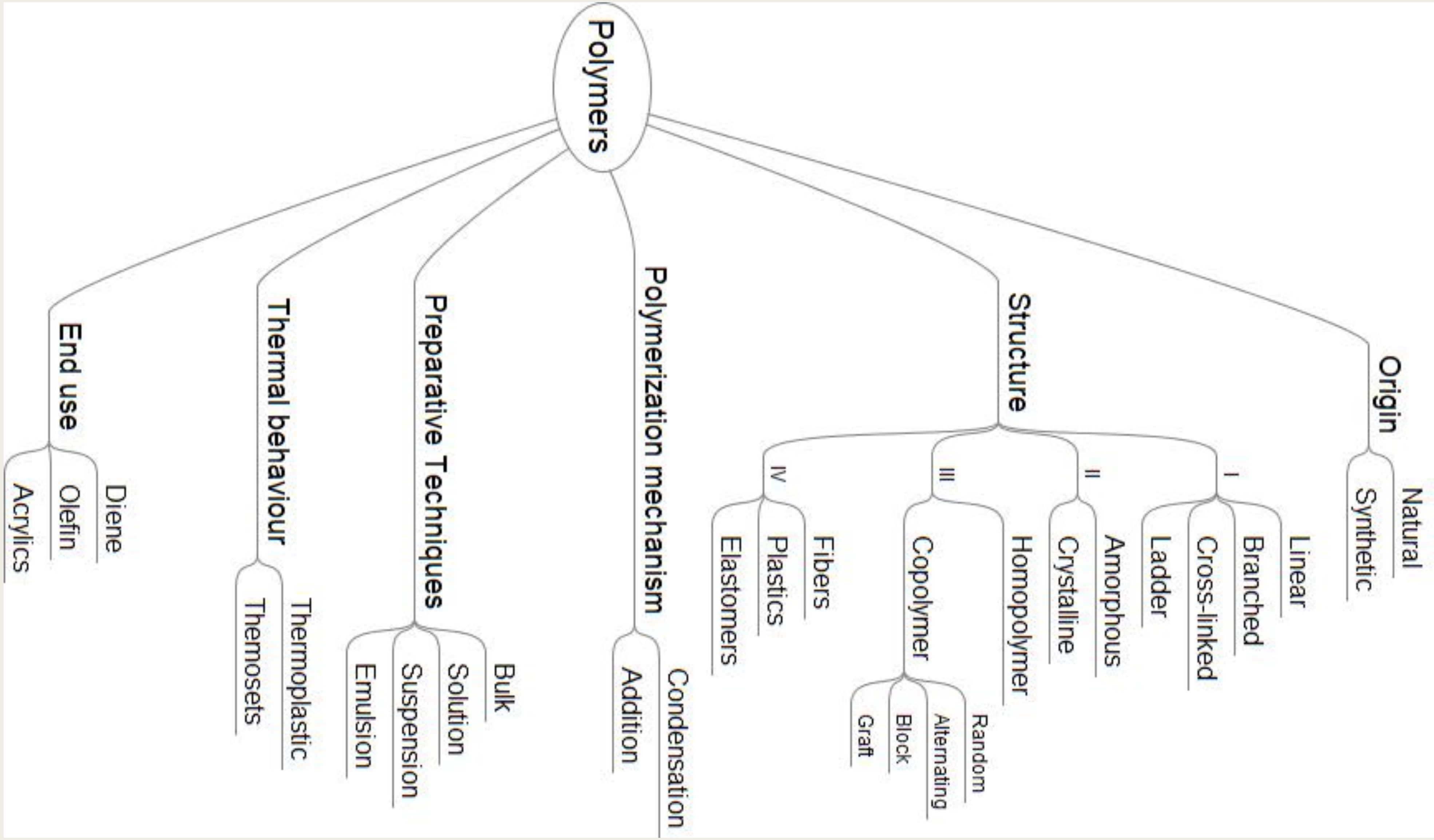
- Word polymer derived from classical Greek
  - *poly* meaning “many”
  - *meros* meaning “parts.”
- Polymer is a large molecule (macromolecule) built up by the repetition of small chemical units.
- The formation of the polymer polystyrene:



Historical Background

1845	SCHOENBEIM	- nitration reactions
1848	MENARD	- collodion films ;nitrated cellulose
1862	PARKES	- "Parkesine" ; added camphor (and other stuff) - *plasticizers*
1870's	HYATT	- billiard balls !! - extrusion
1884	CHARDONNET	- regenerated cellulose rayon,cellophane
1890's	CROSS, BEVAN, & BEADLE	
1905	BAEKELAND	- Bakelite, the first *thermoset*
1920	STAUDINGER	- the macromolecular hypothesis
> 1920	NEW *THERMOPLASTICS*	- (more on these later)

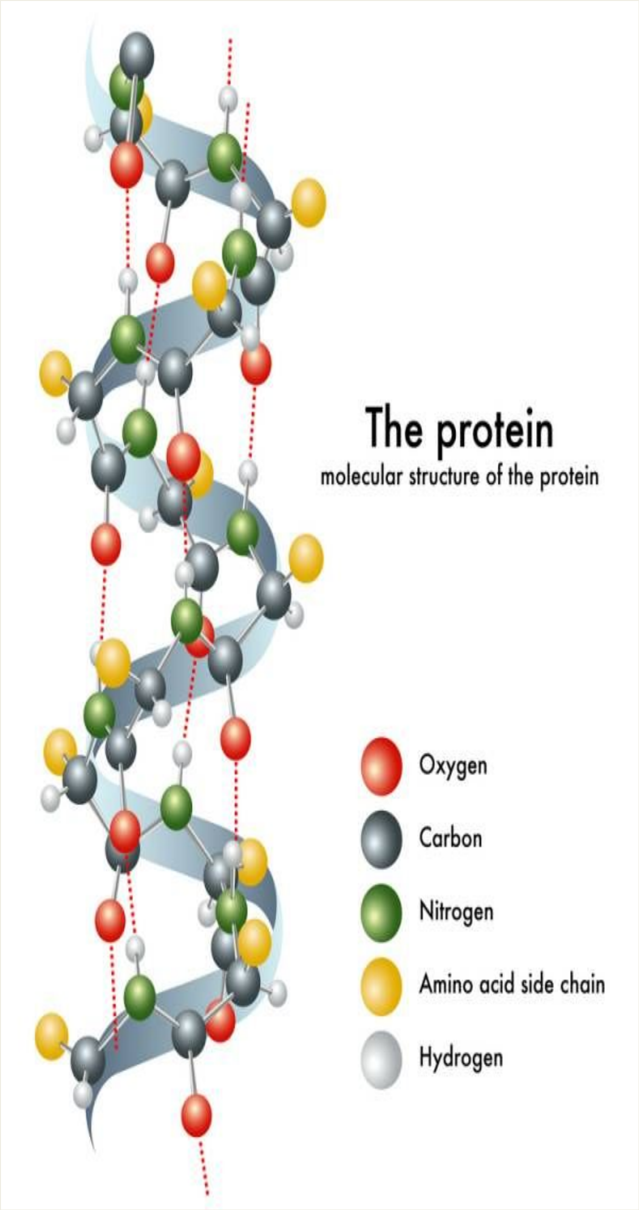
Classification



Origin

Natural

- Naturally occurring
- Examples
  - *Proteins*
  - *Enzymes*
  - *Nucleic acids*
  - *Cellulose*
  - *Natural rubber*
  - *Silk*
  - *Wool*
  - *etc*



Origin

Synthetic

- Man-made
- Examples

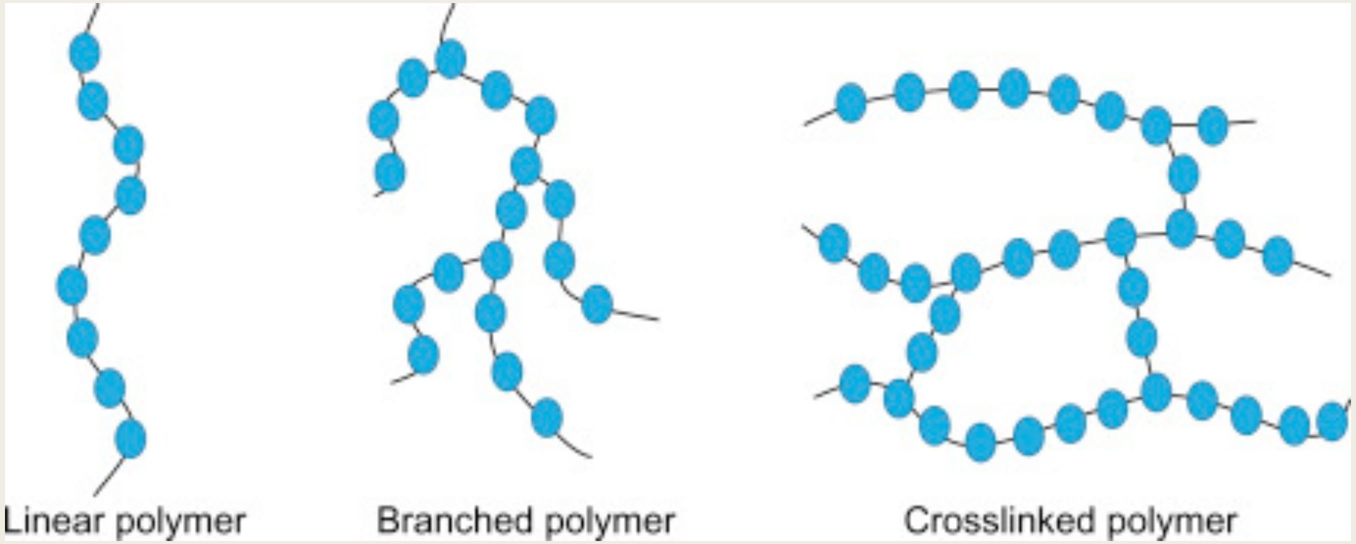
2

4

1

3

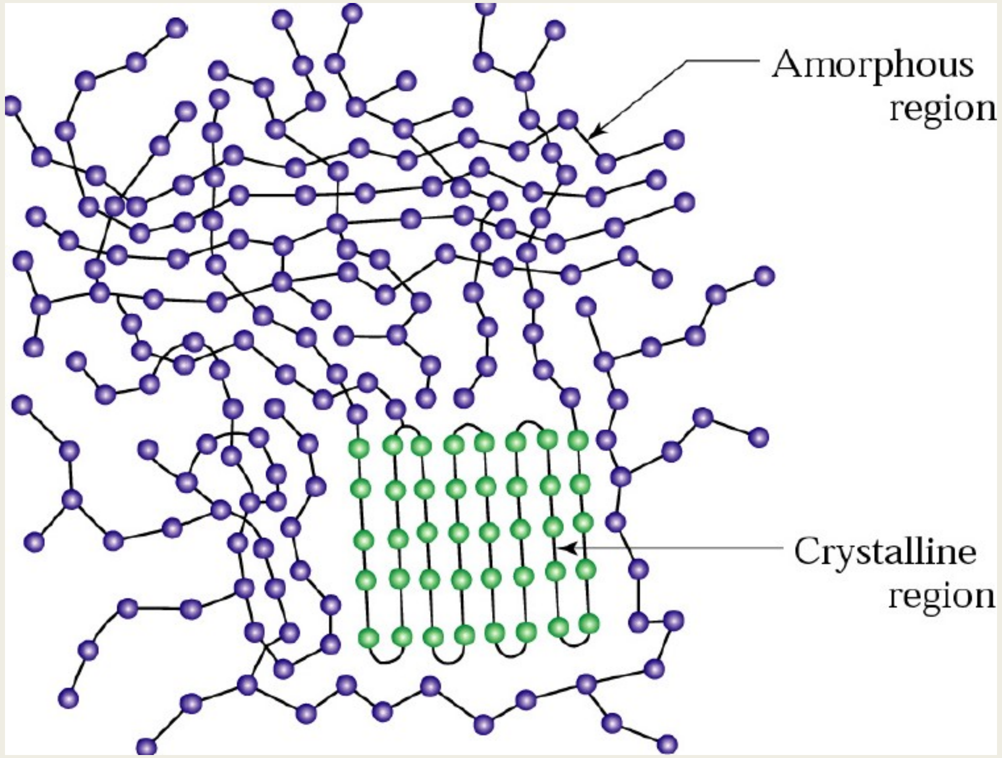
Structure I



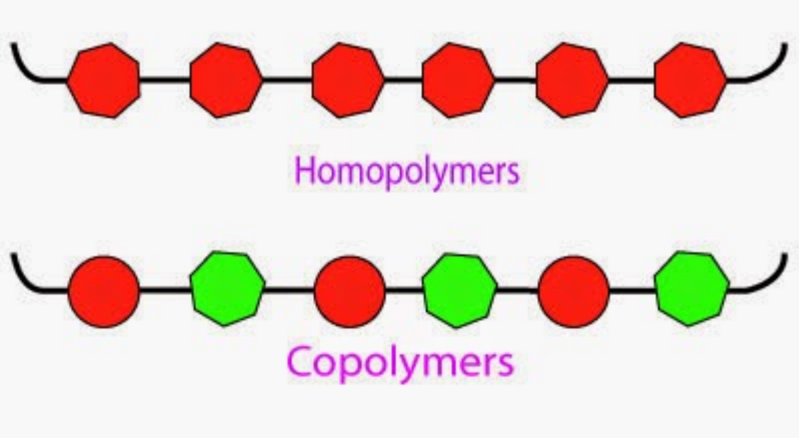
Linear polymer
Branched polymer
Crosslinked polymer

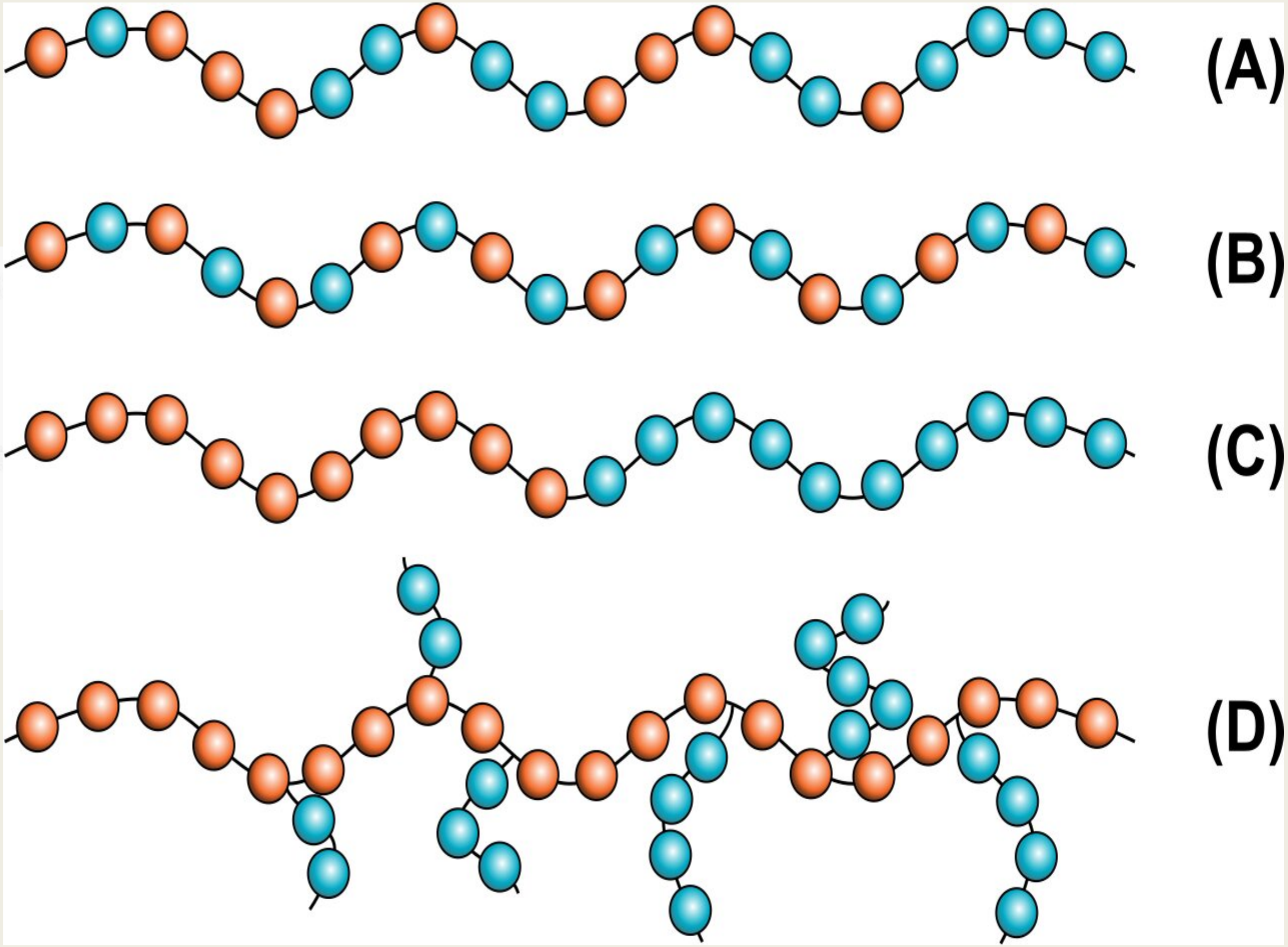
—A—A—A—A—A—A—A—A—
B B B B B B B B
—A—A—A—A—A—A—A—A—

Structure II



Structure III





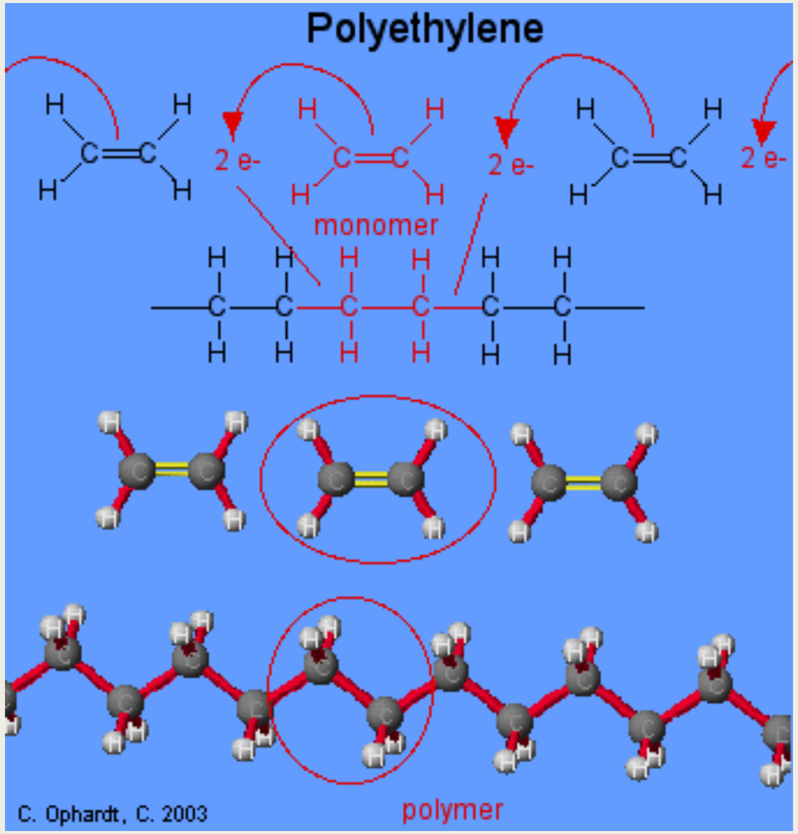
Structure IV

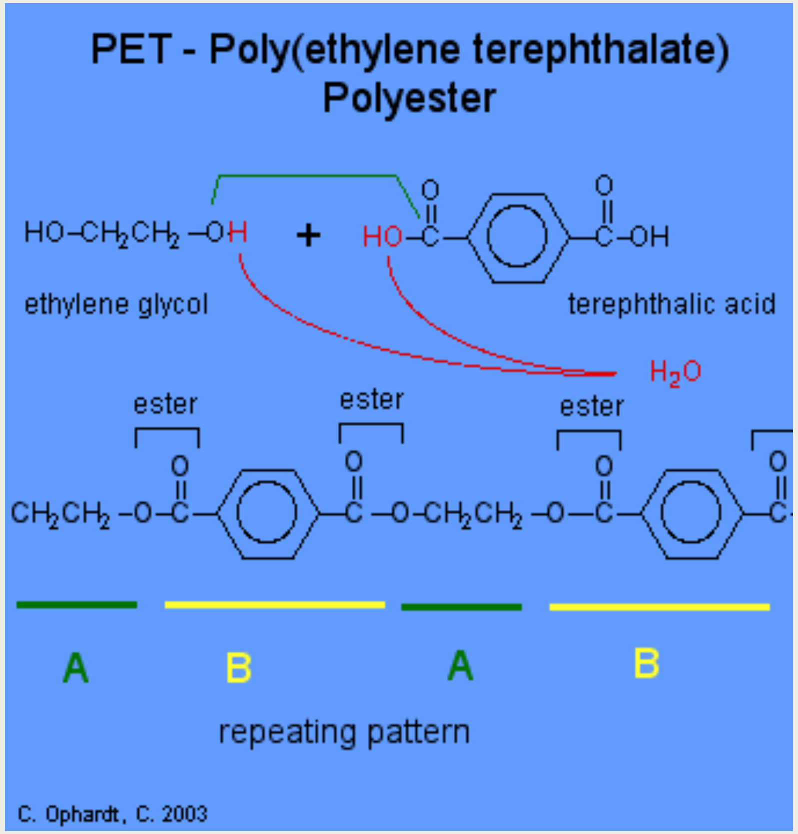
- Fibers are linear polymers with high symmetry and high intermolecular forces.
  - high modulus
  - high tensile strength
  - moderate extensibilities (usually less than 20%)

- Elastomers have molecules with irregular structure, weak intermolecular attractive forces, and very flexible polymer chains.
  - high extensibility (up to 1000%)
- *Plastics* fall between the structural extremes represented by fibers and elastomers

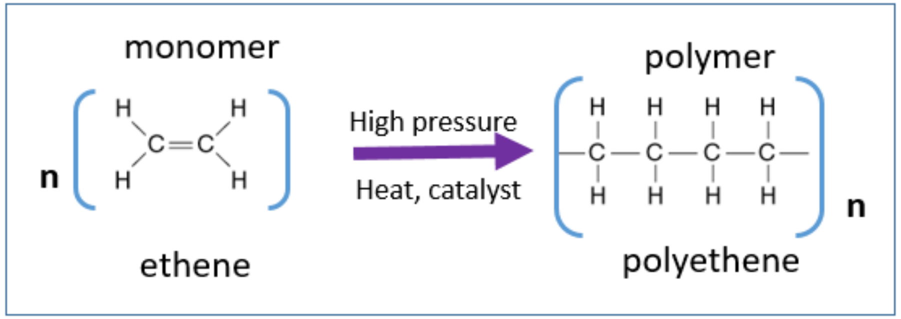
Polymerization Mechanism

- Addition
- Condensation

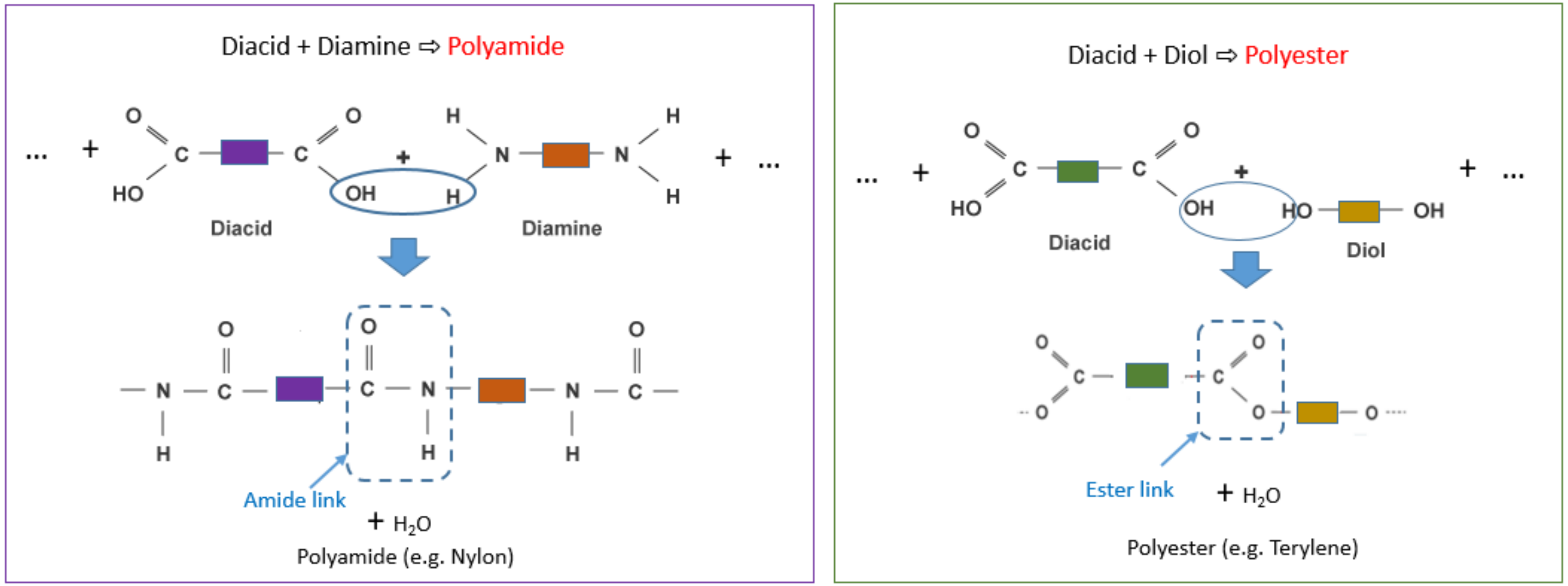




Addition Polymerisation



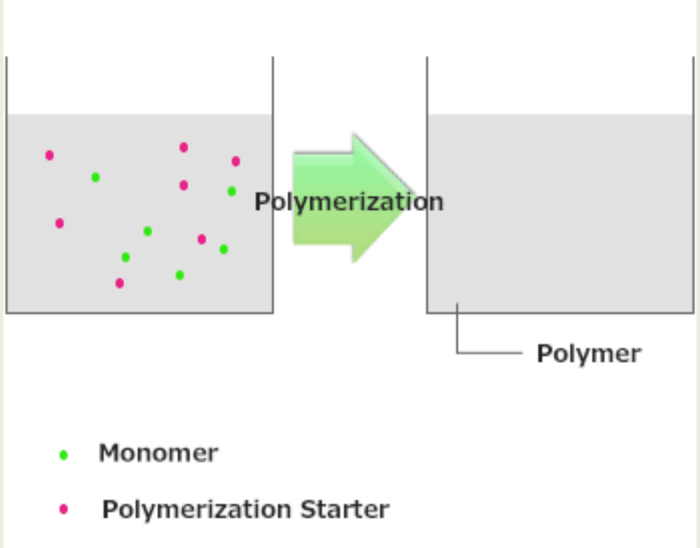
Condensation Polymerisation

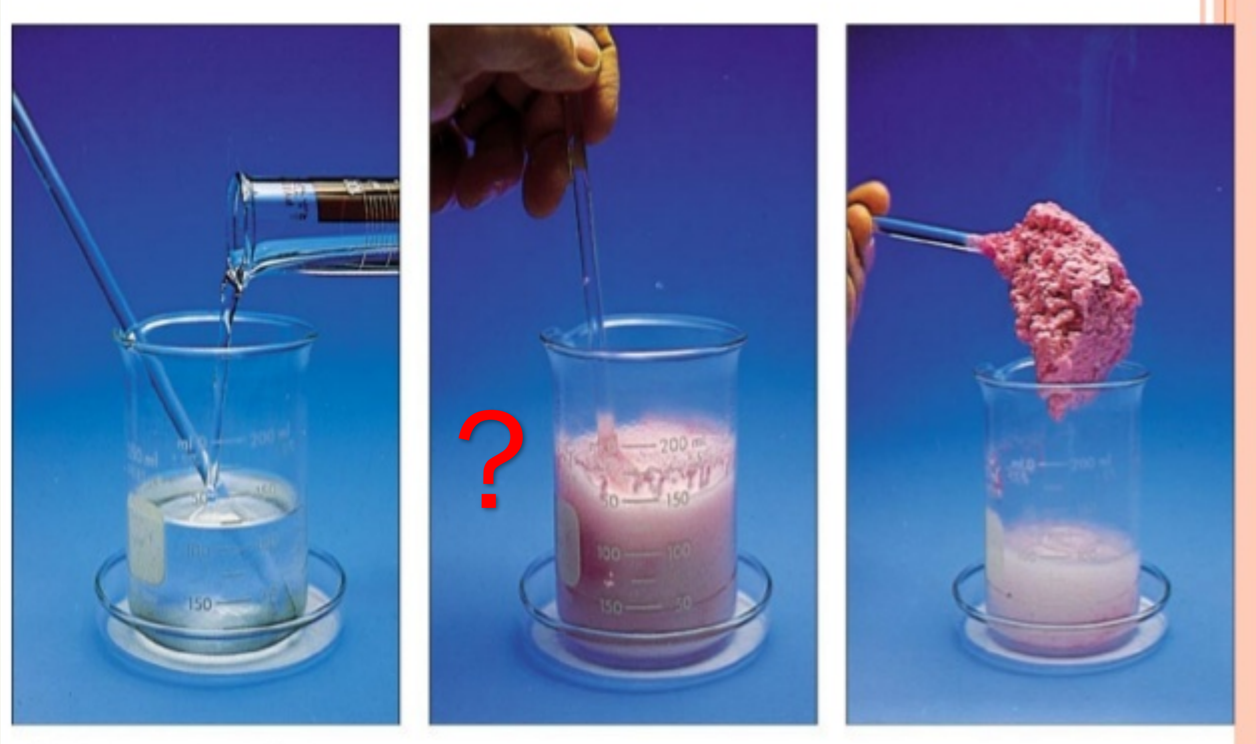


Preparative techniques

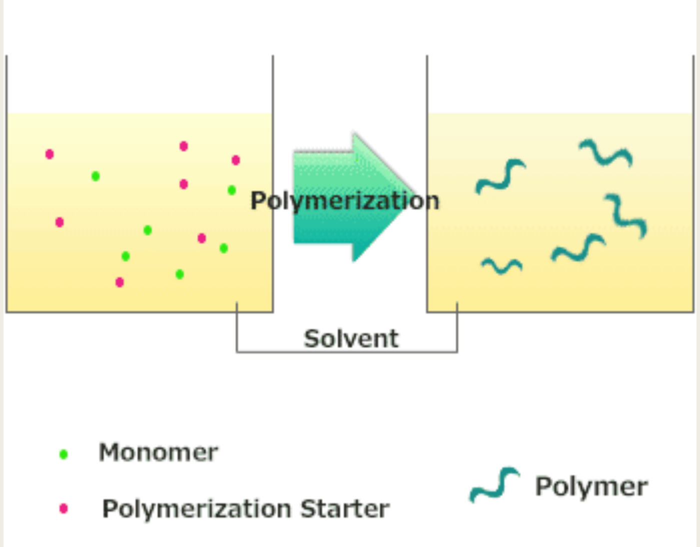
- ***Bulk polymerization,*** only the monomer (and possibly catalyst and initiator, but no solvent) is fed into the reactor. The monomer undergoes polymerization, at the end solid mass is removed as the polymer product.
- ***Solution polymerization*** involves polymerization of a monomer in a solvent in which both the monomer (reactant) and polymer (product) are soluble.
- ***Suspension polymerization*** refers to polymerization in an aqueous medium with the monomer as the dispersed phase.
- ***Emulsion polymerization*** is similar to suspension polymerization but the initiator is located in the aqueous phase (continuous phase)

Bulk Polymerization

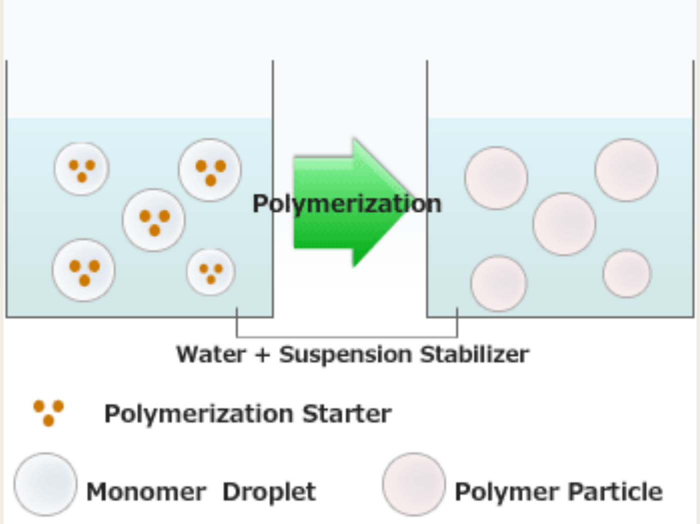




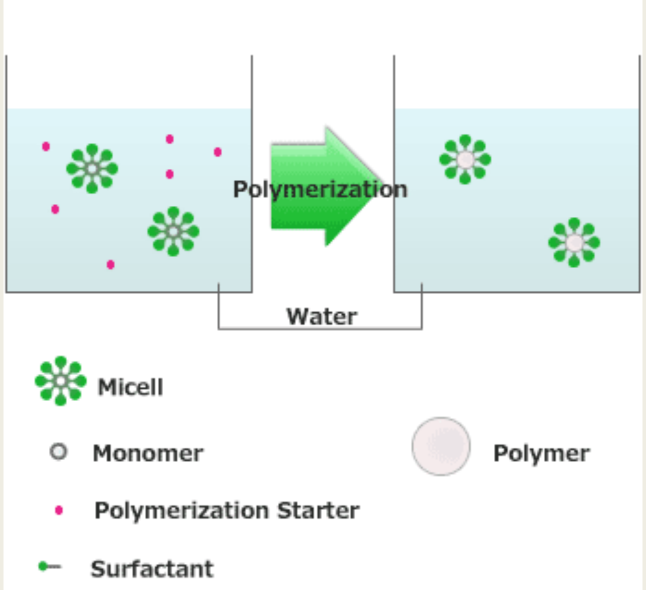
Solution Polymerization



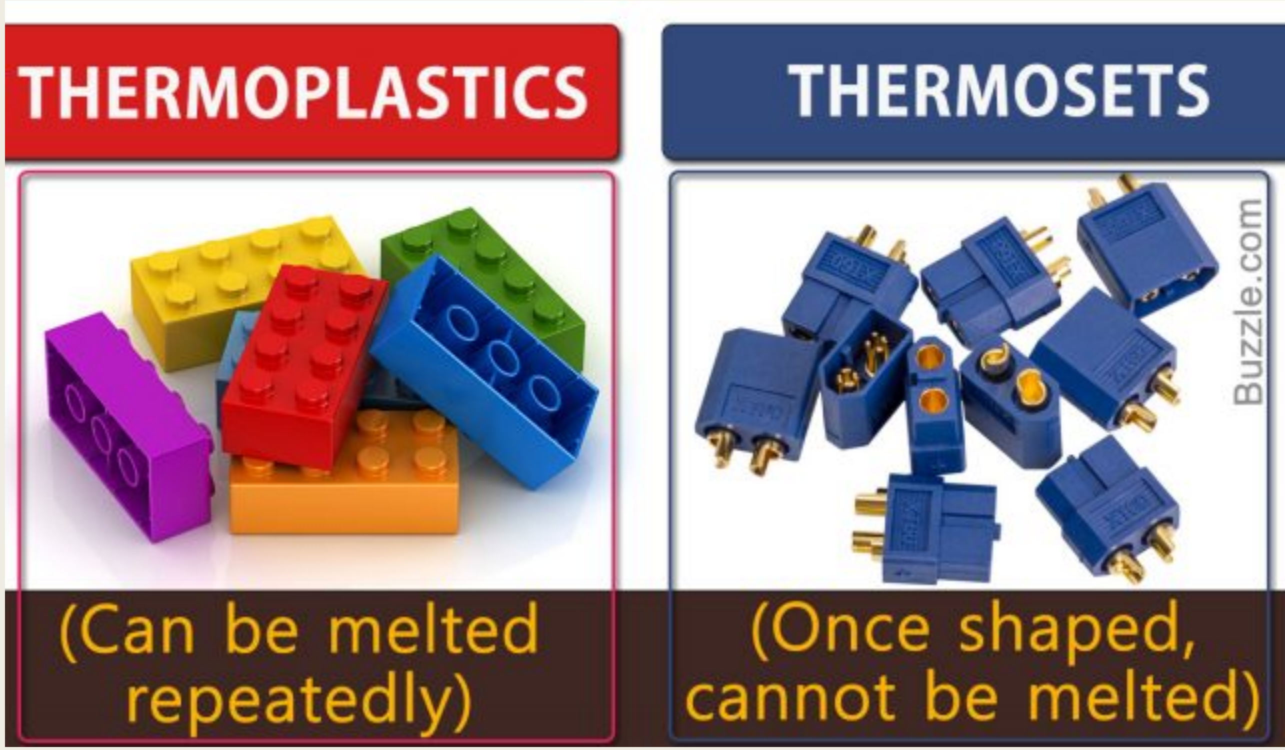
Suspension Polymerization

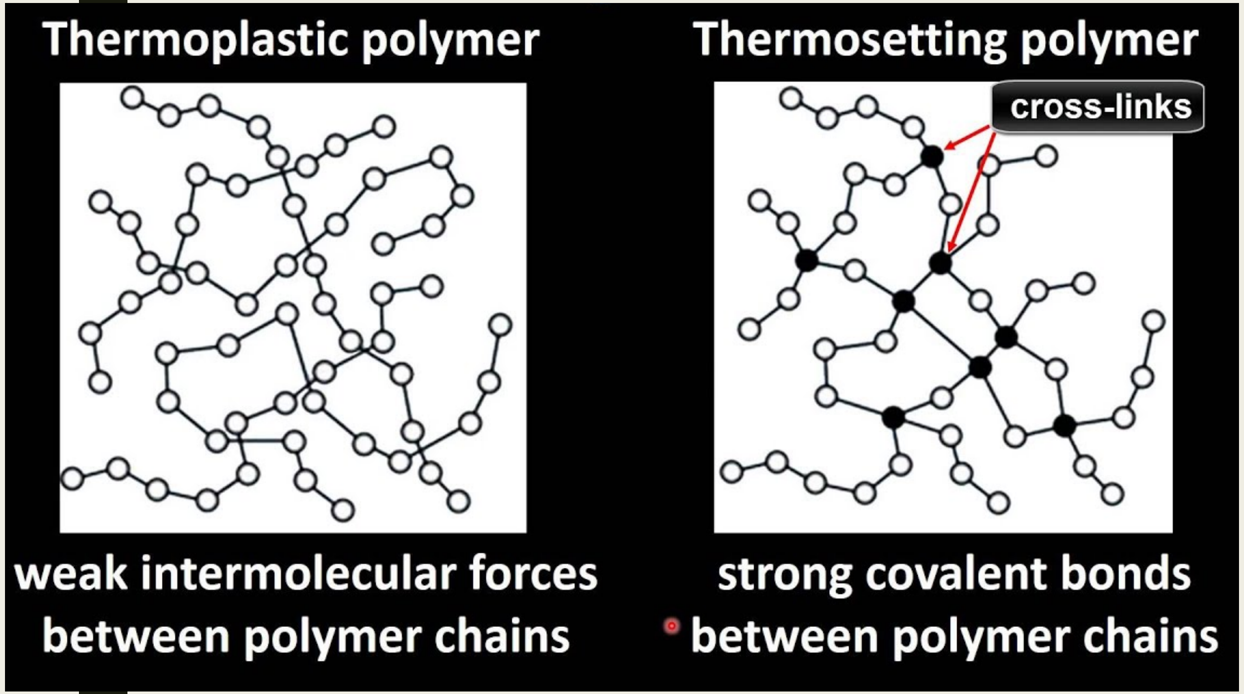


Emulsion Polymerization



Thermal Behavior





End Use

Association with a specific industry (end use)

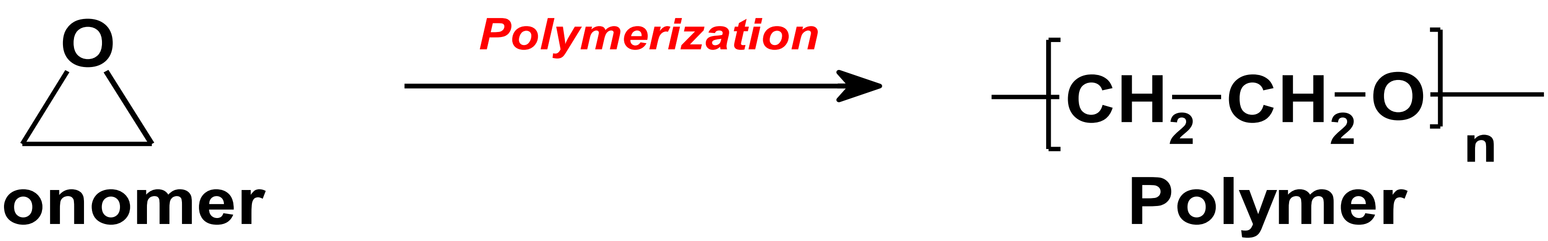
- Diene polymers (rubber industry)
- Olefin polymer (sheet, film, and fiber industries)
- Acrylics (coating and decorative materials).

Polymerization

Polymers

Polymers are large molecules made up of repeating units called Monomers. The synthetic process is Polymerization.

Polymers: Macromolecules formed by the covalent attachment of a set of small molecules termed monomers.



Polymers are classified as:

(1) Man-made or synthetic polymers that are synthesized in the laboratory; (nylon, poly-ethylene, poly-styrene)

(2) Biological polymer that are found in nature. Biological polymers: DNA, proteins, carbohydrates

Homopolymer: A polymer prepared from a single monomer

A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-AA--A-

B-B-B-B-B-B-B-B-BB-B-B-B-B-B-B-B-B-B-B-B-B

Copolymer: If two or more different monomers are employed.

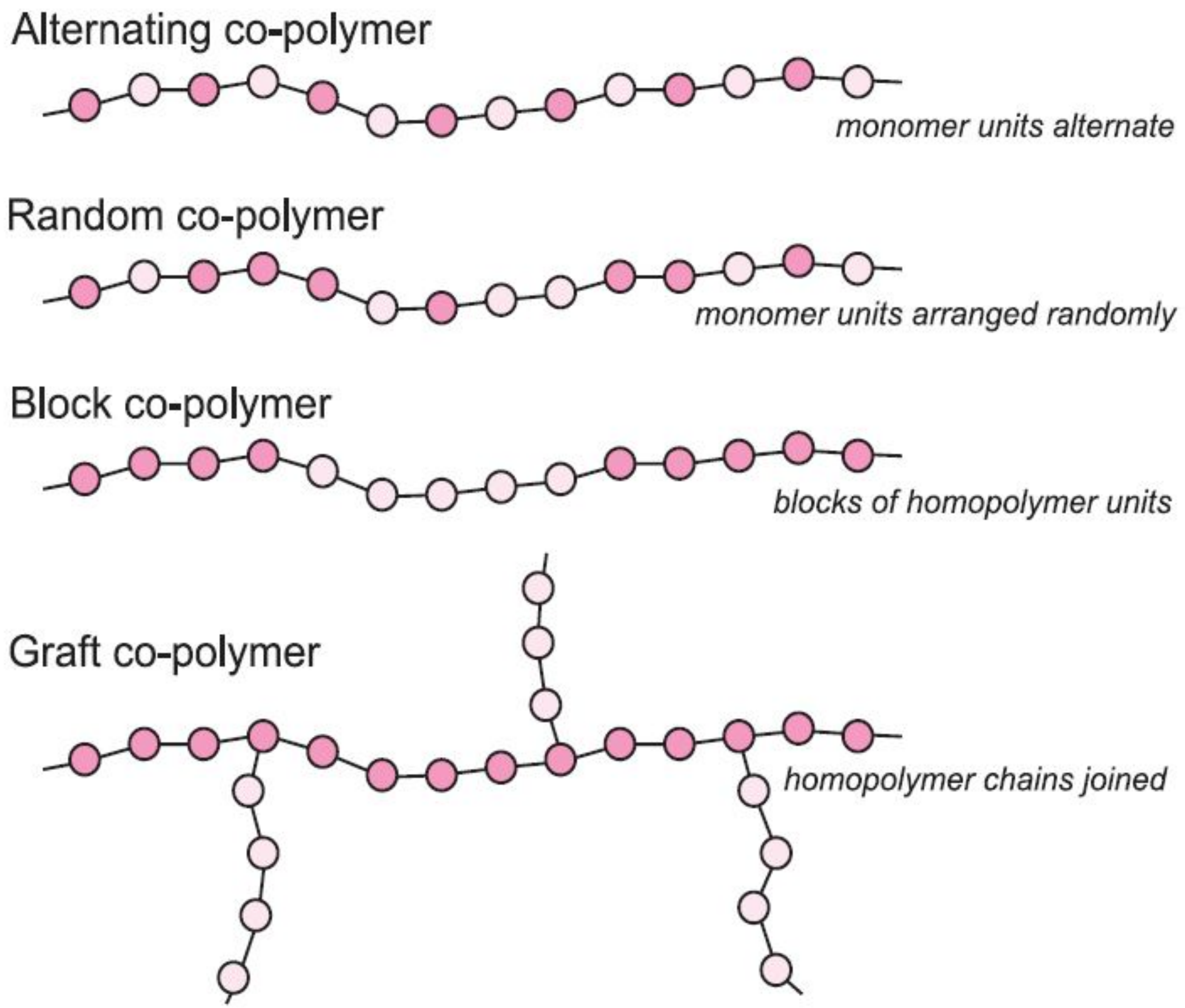
—A—B—A—B—A—B—A—B—A—B—A—B—A—B—

alternating copolymer

—A—A—B—A—B—B—A—B—A—A—B—B—B—A—

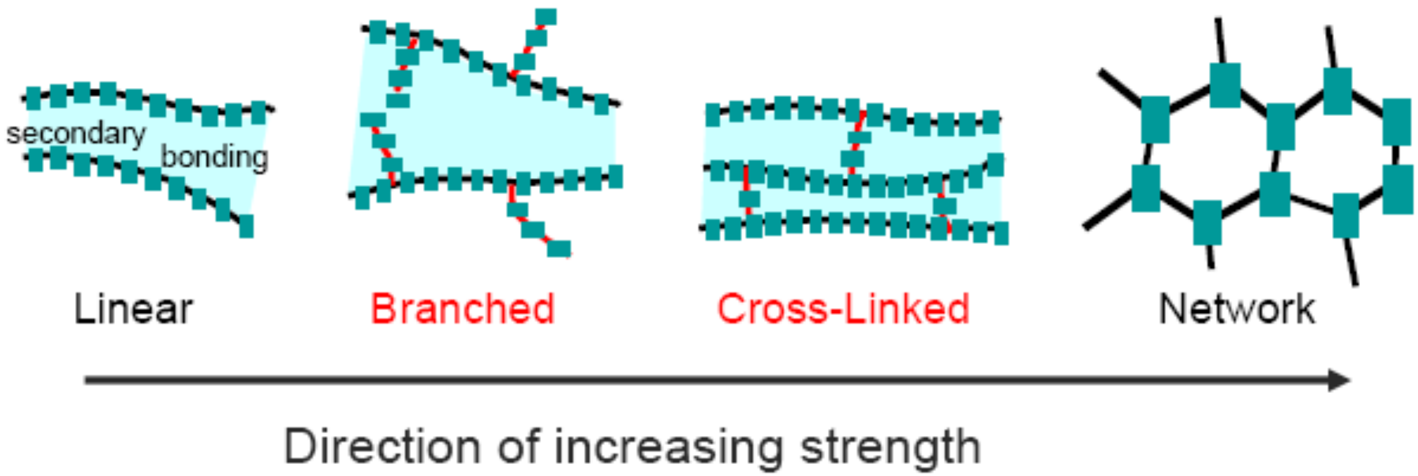
random copolymer

Homopolymer & Copolymer



Molecular Structures

- Covalent chain configurations and strength:



Adapted from Fig. 14.7, *Callister 7e.*

Polymerization

The chemical reaction in which high molecular mass molecules are formed from monomers.

There are two basic types of polymerization

1) **Chain-reaction (or addition) polymerization**

2) **Step-reaction (or condensation) polymerization**

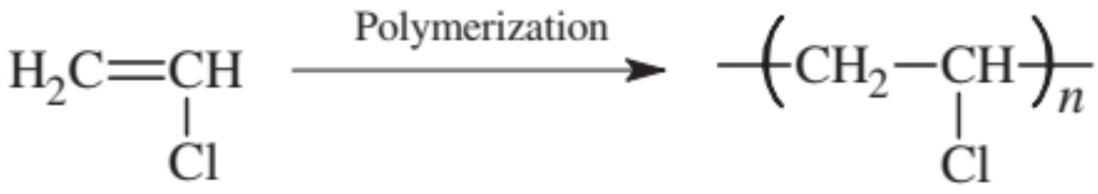
Methods for making polymers

Addition polymerization: monomers react to form a polymer <u>without net loss</u> of atoms.

This type of polymerization is a three step process involving two chemical components

1) Monomer

Can be regarded as one link in a polymer chain. It initially exists as simple units. In nearly all cases, the monomers have at least one carbon-carbon double bond. Ethylene is one example of a monomer used to make a common polymer.



Vinyl chloride

Poly(vinyl chloride)

2) **Catalyst:**

In chain-reaction polymerization, the catalyst can be a **free-radical peroxide** added in relatively low concentrations

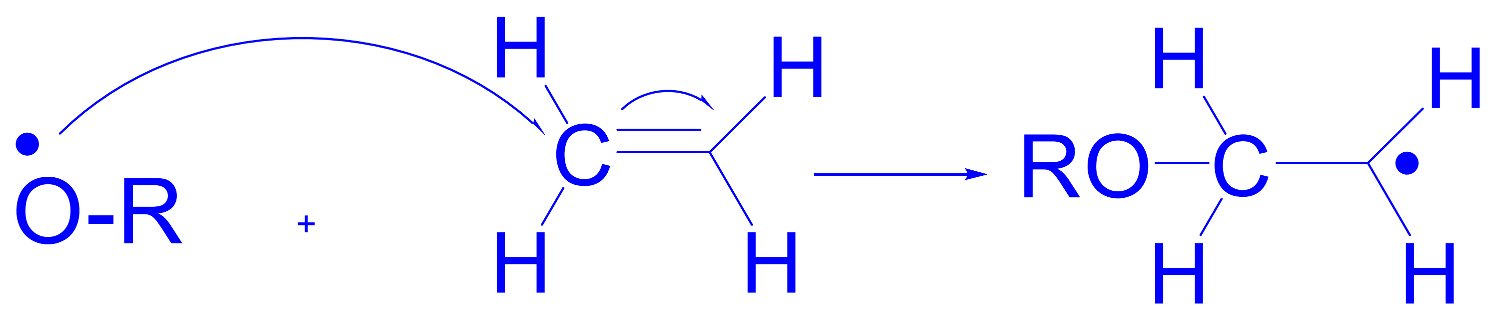
$$\text{R-O-O-R} \longrightarrow 2\,\dot{\text{O}}\text{-R}$$

A free-radical is a chemical component that contains a free electron that forms a covalent bond with an electron on another molecule.

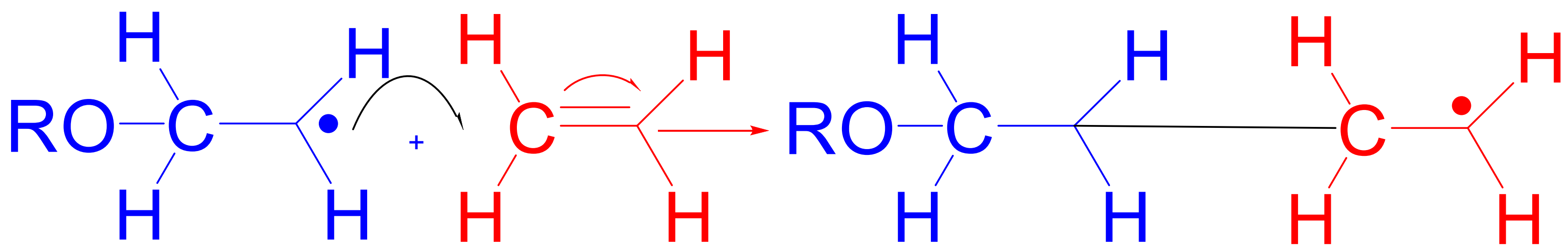
Polymerization Process (Mechanism)

Step 1: Initiation

The first step in the chain-reaction polymerization process, initiation, occurs when the free-radical catalyst reacts with a double bonded carbon monomer, beginning the polymer chain. The double carbon bond breaks apart, the monomer bonds to the free radical, and the free electron is transferred to the outside carbon atom in this reaction.



2) Propagation Step

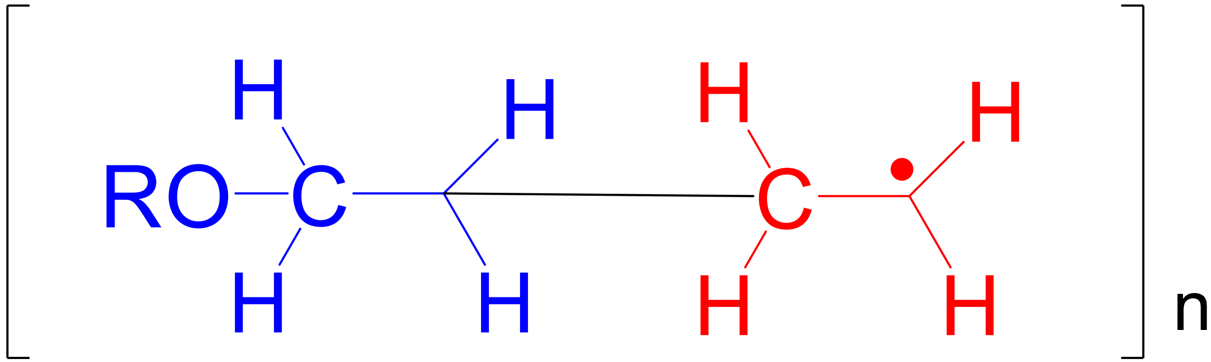


Propagation polymer chain

monomer

New polymer chain

The next step in the process, propagation, is a repetitive operation in which the physical chain of the polymer is formed. The double bond of successive monomers is opened up when the monomer is reacted to the reactive polymer chain. The free electron is successively passed down the line of the chain to the outside carbon atom



Termination

Termination occurs when another free radical (R-O·), left over from the original splitting of the organic peroxide, meets the end of the growing chain. This free-radical terminates the chain by linking with the last $CH_2\cdot$ component of the polymer chain. This reaction produces a complete polymer chain. Termination can also occur when two unfinished chains bond together.

Initiation

$$I \xrightarrow{k_d} 2R^{\bullet}$$

$$R^{\bullet} + M \xrightarrow{k_i} RM^{\bullet}$$

Propagation

$$\left.\begin{aligned}
RM^{\bullet} + M &\xrightarrow{k_p} RMM^{\bullet} \text{ or } RM_2{}^{\bullet} \\
RM_2{}^{\bullet} + M &\xrightarrow{k_p} RMMM^{\bullet} \text{ or } RM_3{}^{\bullet} \\
RM_3{}^{\bullet} + M &\xrightarrow{k_p} RMMMM^{\bullet} \text{ or } RM_4{}^{\bullet}
\end{aligned}\right\}$$

In general terms,

$$RM_{n-1}{}^{\bullet} + M \xrightarrow{k_p} RM_n{}^{\bullet}$$

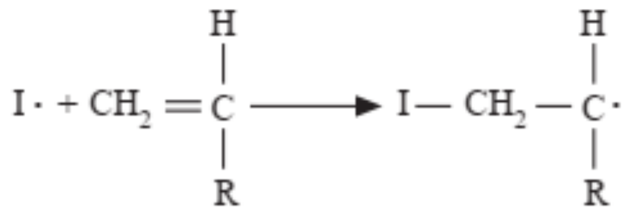
Termination

$$RM_n^{\bullet} + RM_m^{\bullet} \begin{cases} \xrightarrow{k_{tc}} RM_{(n+m)}R \\ \xrightarrow{k_{td}} RM_n + RM_m \end{cases}$$

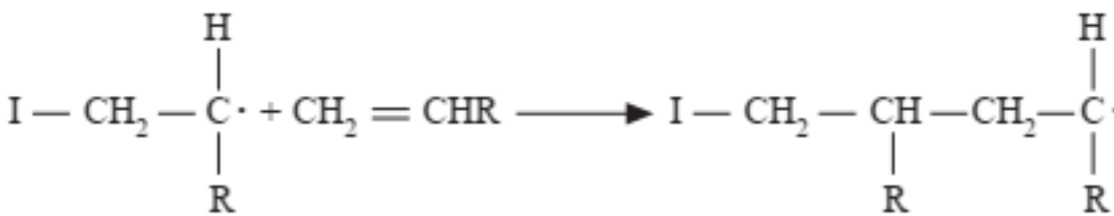
Formation of radicals

$$I\text{-}I \rightarrow 2I\cdot$$

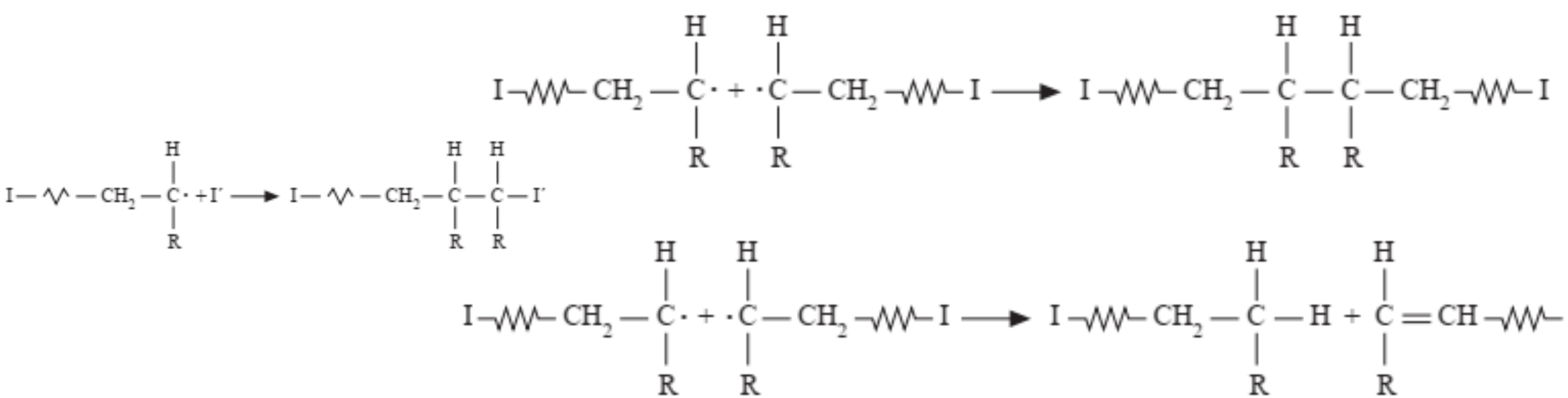
Addition of the initiator radical to a vinyl monomer molecule



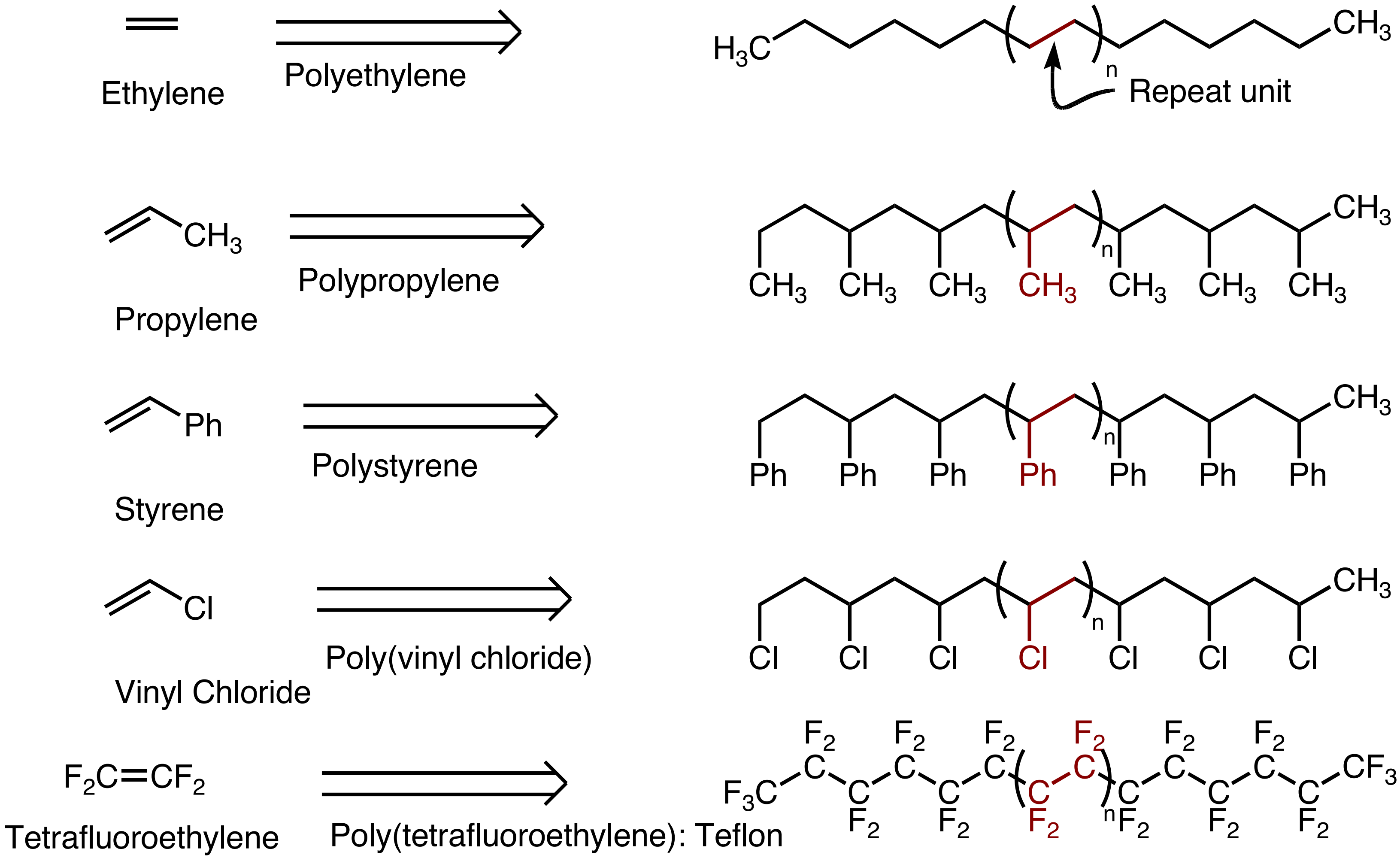
Initiated monomer adds other monomers



Termination

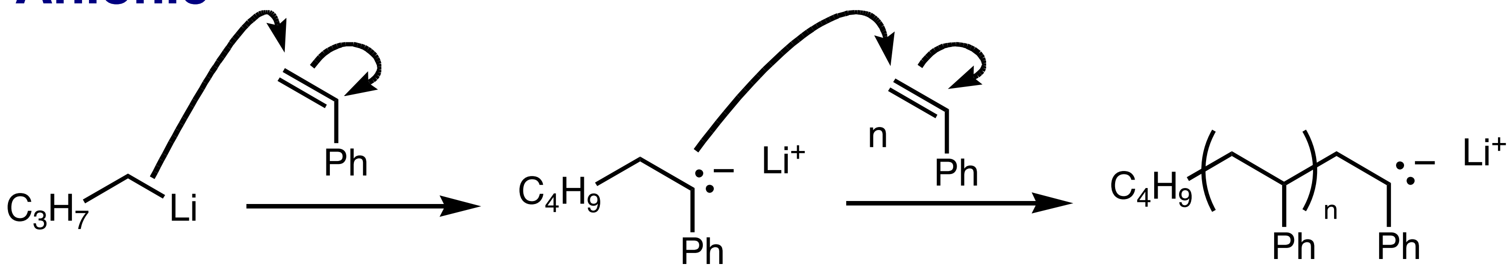


Monomer **Polymer**

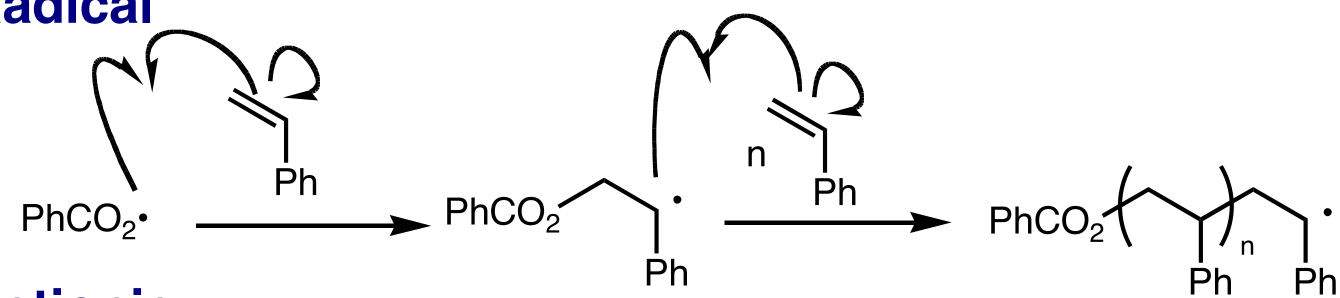


Types of Addition Polymerizations

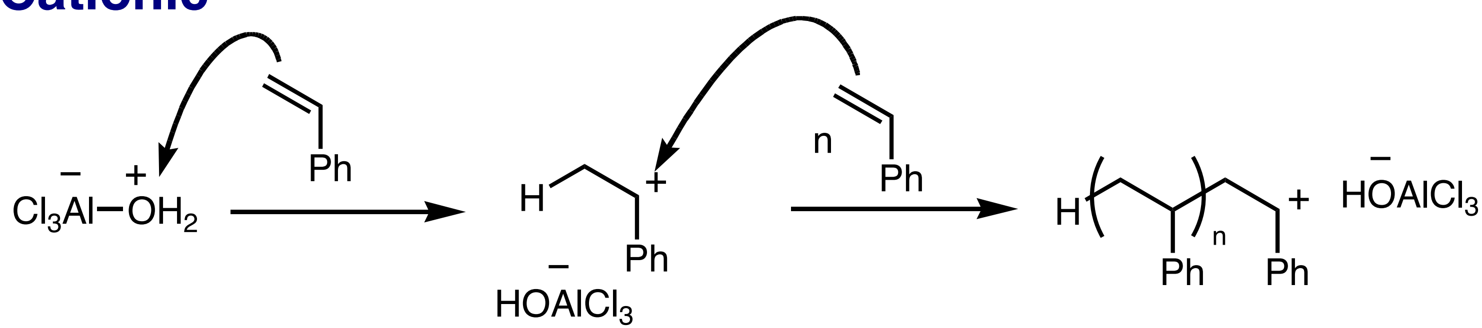
Anionic



Radical



Cationic

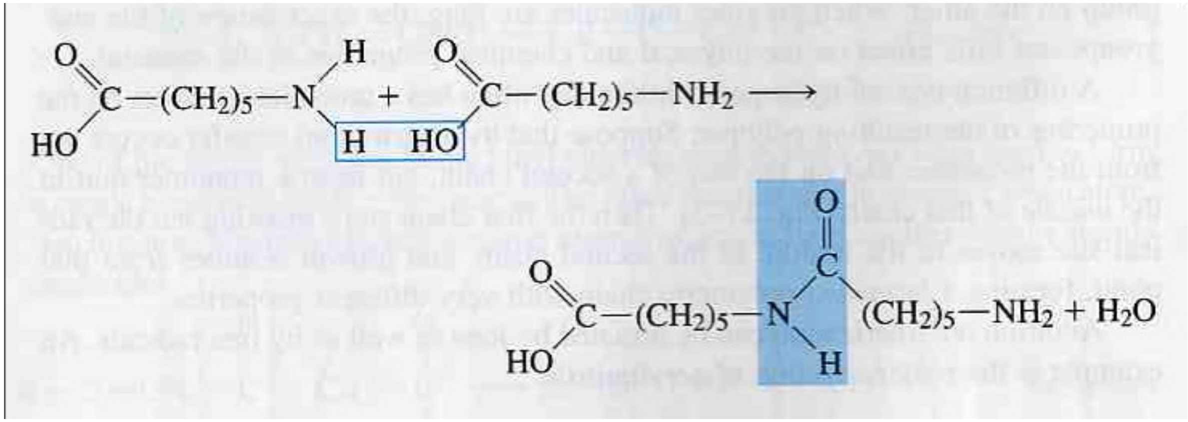


Condensation polymerization

Condensation polymerization: the polymer grows from monomers by splitting off a small molecule such as water or carbon dioxide.

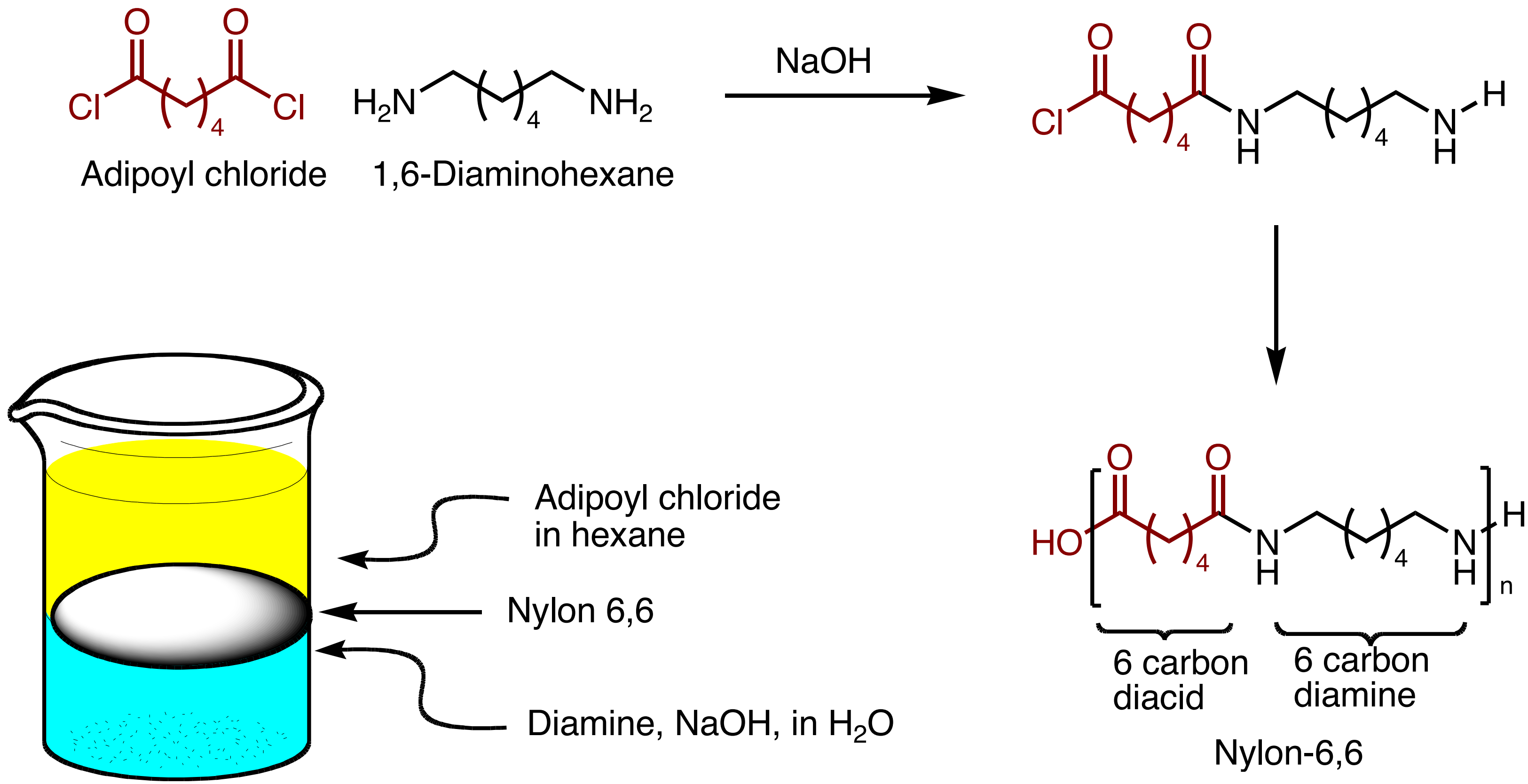
Example: formation of amide links and loss of water

Monomers



First unit of polymer + H_2O

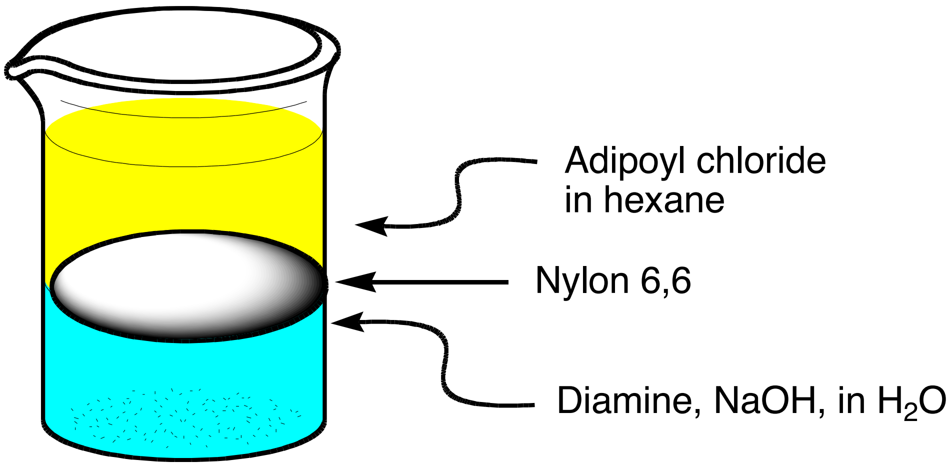
Nylon-6,6

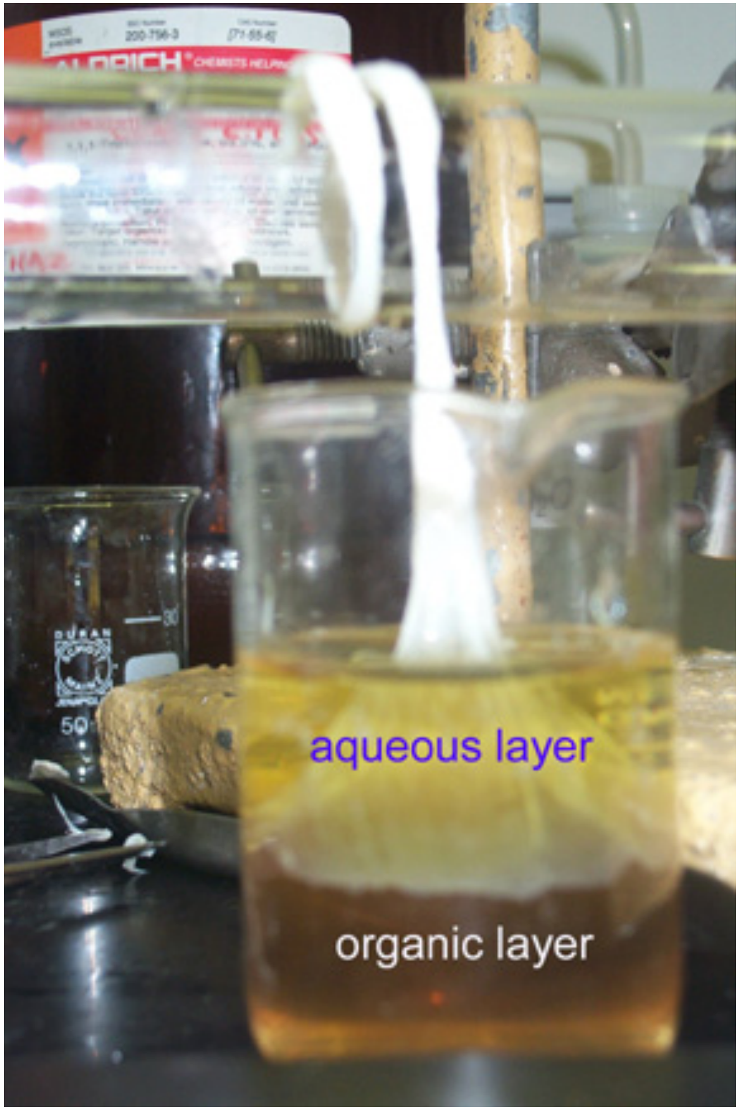


Cl
O
O
Cl
4
H_2N
4
NH_2
Adipoyl chloride
1,6-Diaminohexane
NaOH
Cl
O
O
4
N
H
4
N
H
H
Adipoyl chloride
in hexane
Nylon 6,6
Diamine, NaOH, in H_2O
HO
O
O
4
N
H
4
N
H
H
n
6 carbon
diacid
6 carbon
diamine
Nylon-6,6

Nylon-6,6

Since the reactants are in different phases, they can only react at the phase boundary. Once a layer of polymer forms, no more reaction occurs. Removing the polymer allows more reaction to occur.

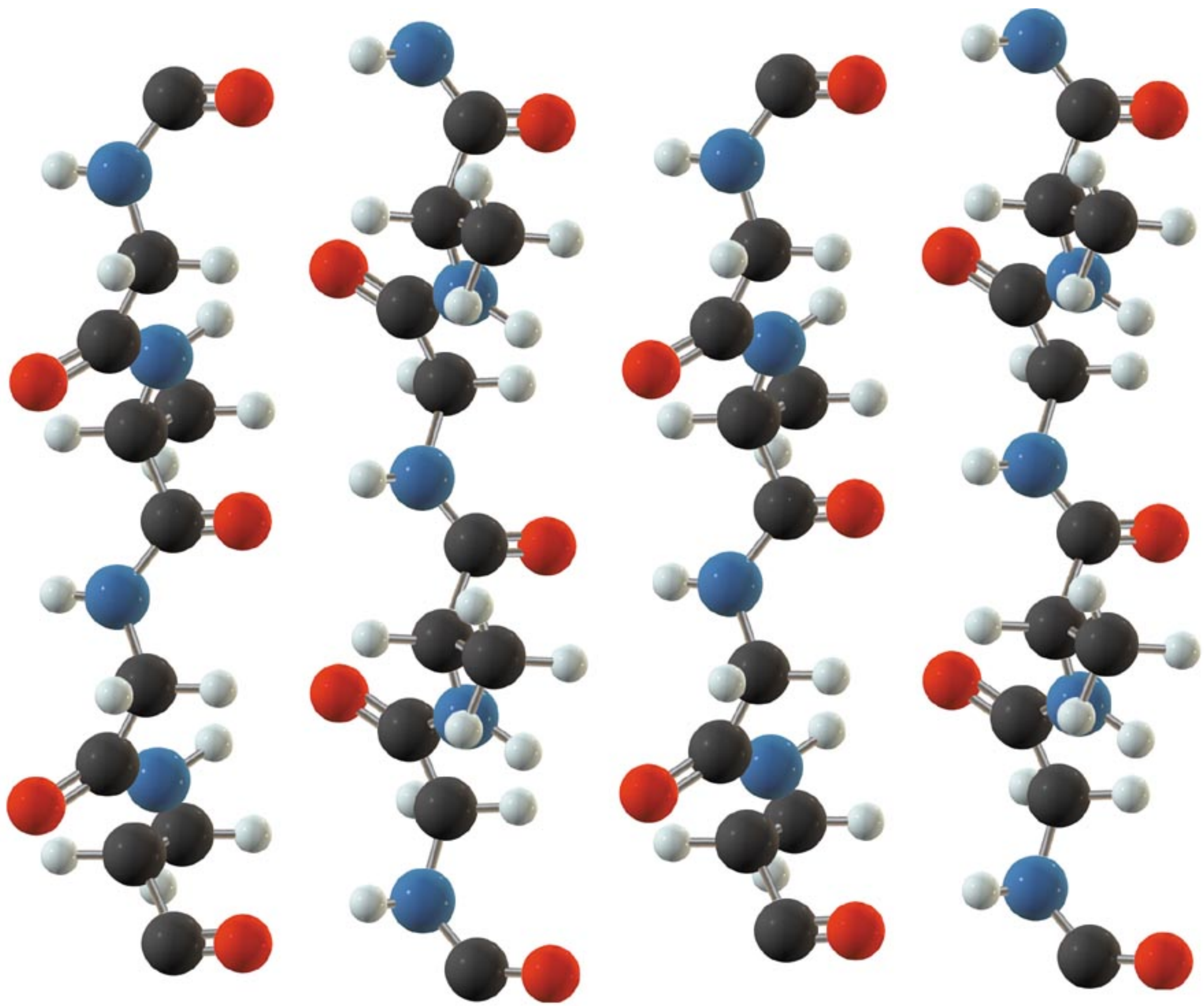




Hydrogen bonds between chains

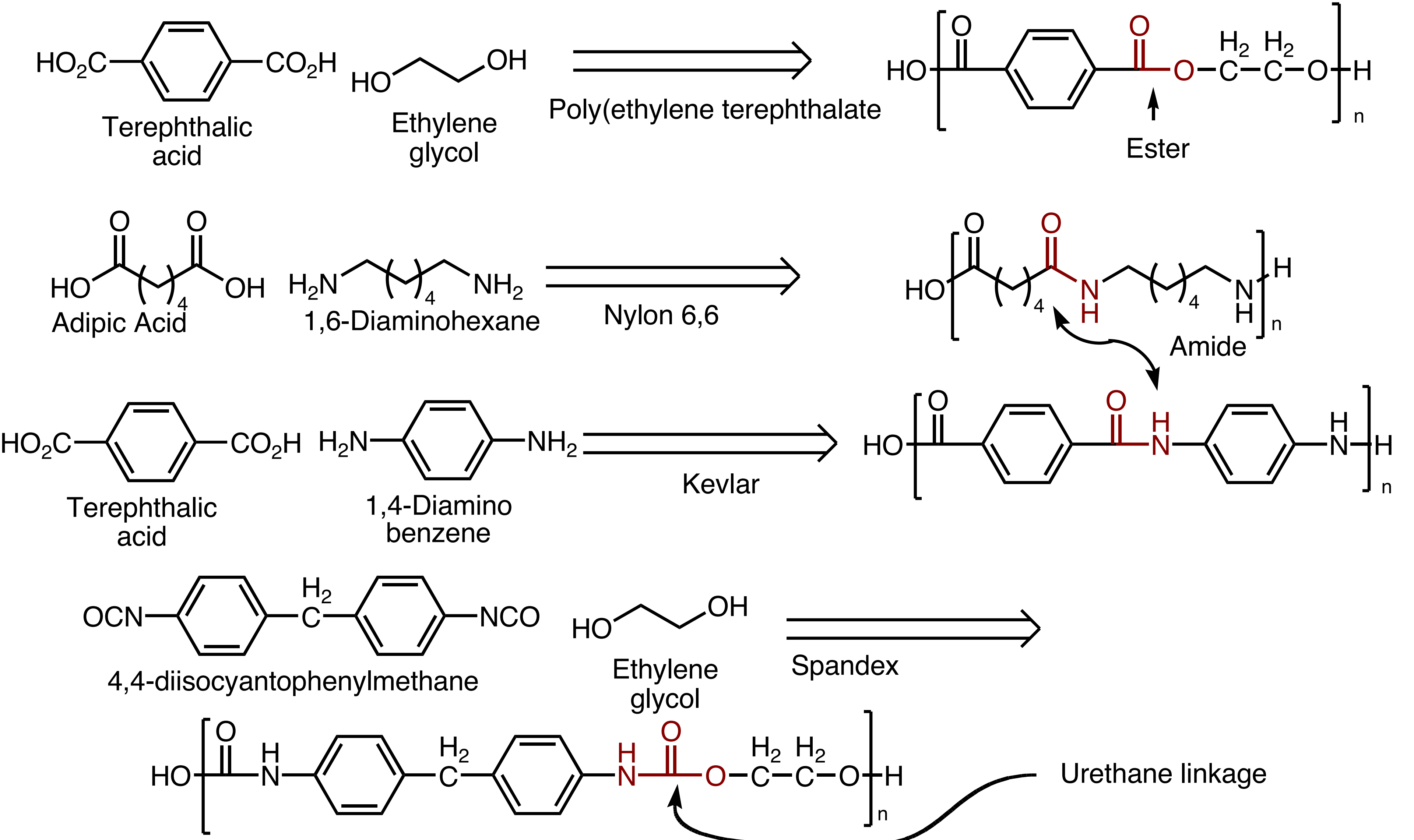
Supramolecular Structure of nylon

Intermolecular hydrogen bonds give nylon enormous tensile strength



Polyesters, Amides, and Urethanes

Monomer　　　　　　　　　　　　　　Polymer



In the 1967 movie, "The Graduate",

A businessman takes aside the baby-faced Dustin Hoffman and declares,

"I just want to say one word to you -- just one word -- 'plastics.' "

Range of Polymers

- Traditionally, the industry has produced two main types of synthetic polymer – plastics and rubbers.
- Plastics are (generally) rigid materials at service temperatures
- Rubbers are flexible, low modulus materials which exhibit long-range elasticity.

Polymer Classifications

Thermoplastics can be softened or melted by heat and reformed (molded) into another shape.
Most addition polymers are thermoplastics. The polymer chains are held together by weak interactions (noncovalent bonds) such as :??

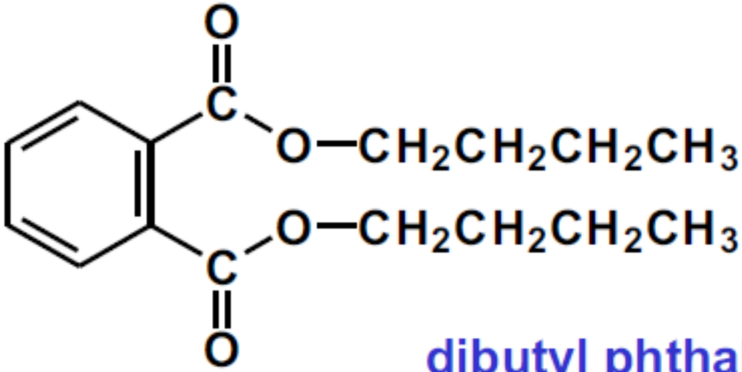
These interactions are disrupted by heating, allowing the chains to become independent of each other. Heating and reforming can be repeated indefinitely (if degradation doesn't occur). This allows recycling.

Most polymers of high molecular weight are quite rigid.

PLASTICIZERS

Can be softened and made flexible by adding plasticizers

The plasticizer separates the individual polymer chains from one another. It acts as a lubricant which reduces the attractions between the polymer chains.



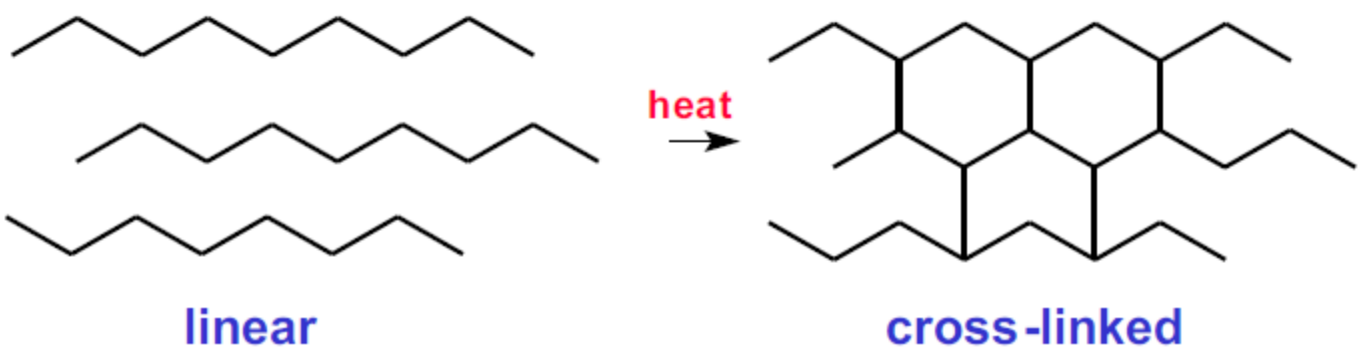
dibutyl phthalate

THERMOSET PLASTICS

Thermoset plastics melt initially, but on further heating they become permanently hardened.

Once formed, thermoset plastics cannot be remolded, and they cannot be recycled.

On heating, thermoset plastics become cross-linked (covalent bonds form between the chains). The cross-linked chains form a rigid network

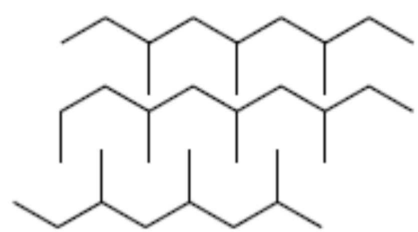


HIGH -DENSITY POLYMERS

Linear polymers with chains that can pack closely together. These polymers are often quite rigid.

LOW-DENSITY POLYMERS

Branched -chain polymers that cannot pack together as closely. There is often a degree of cross -linking. These polymers are often more flexible than high - density polymers.



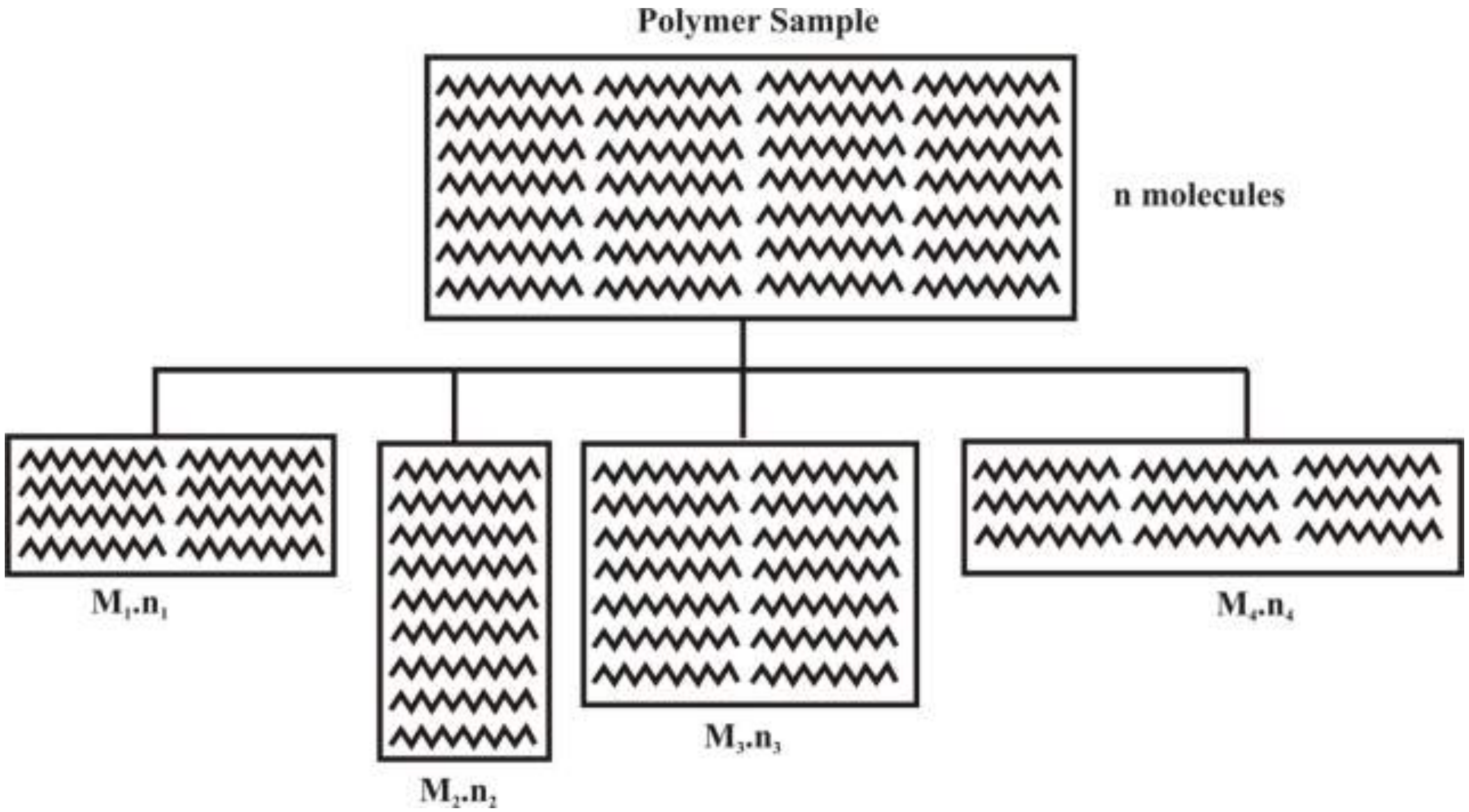
Molecular Weight

Molecular weight of Polymers

"Molecular weight of a polymer is defined as sum of the atomic weight of each of the atoms in the molecules, which is present in the polymer".

This distribution of molecular weights is caused by the statistical nature of the polymerization process e.g. methane (CH_4) molecules have the same molecular weight (16), but all polyethylene do not have the same molecular weight because the statistical distribution of molecular weight may be different for the different grade of the polyethylene and the degree of polymerization may also be different.

Generalization of Concept



Molecular Weight Averages

Let us assume that unit volume of a polymer sample contains a total of A molecules consisting of a_1 molecules with molecular weight M_1, a_2 molecules with molecular weight M_2, ... a_j molecules with molecular weight M_j. The arithmetic mean molecular weight M is then the total measured quantity divided by the total number of molecules:

$$\overline{M} = \frac{a_1M_1 + a_2M_2 + \cdots + a_jM_j}{a_1 + a_2 + \cdots + a_j}$$
$$= \frac{a_1M_1 + a_2M_2 + \cdots + a_jM_j}{A}$$
$$= \frac{a_1}{A}M_1 + \frac{a_2}{A}M_2 + \cdots + \frac{a_j}{A}M_j$$

The ratio a_i/A represents the proportion of molecules with molecular weight M_i. Denoting this proportion by f_i, the arithmetic mean molecular weight will be given by

$$\overline{M} = f_1M_1 + f_2M_2 + \cdots + f_jM_j = \sum_i f_iM_i$$

Number-average molecular weight

$$\boxed{Mn=\frac{\sum niMi}{\sum ni}=\frac{\sum wi}{\sum wi/Mi}}$$

Weight-average molecular weight

$$\boxed{Mw=\frac{\sum niMi^2}{\sum niMi}=\frac{\sum wiMi}{\sum wi}}$$

z-Average molecular weight

$$Mz=\frac{\sum niMi^3}{\sum niMi^2}=\frac{\sum wiMi^2}{\sum wiMi}$$

Where,

n = Moles of molecules (n_1 + n_2 + n_3 + ----------n_i) i.e. weight (w)/molecular weight (M)

w = Weight of individual molecules (w_1 + w_2 + w_3 + ---------w_i)

M = Molecular weight of each molecules

Number Average Weight (Mn)

The number average molecular weight is not too difficult to understand. It is just the total weight of all the polymer molecules in a sample, divided by the total number of polymer molecules in a sample.

Consider a polymer, which contains four molecular weight polymers in different numbers and weight and these

Polymer entity	Number of unit in each entity, n	Weight of each grams, M(g)	Total weight of each entity, W = nM(g)
Poly-1	2	10	20
Poly-2	4	20	80
Poly-3	6	100	600
Poly-4	3	250	750
Total	15	-	1450

Total number of polymer in containing each entity of poly-1, poly-2, poly-3 and poly-4 is = 15

Number of Poly-1 present in the polymer = 2

Number of fraction of poly-1 = 2/15

Similarly, Number of fraction of poly-2 = 4/15, Number of fraction of poly-3 = 6/15, Number of fraction of poly-4 = 3/15

Contribution made by poly-1 towards the average weight of polymer = number of fraction of each polymer x weight of each poly entity

Therefore, each poly contribution is

(2/15) x 10 =1.33g, (4/15) x 20 = 5.33g, (6/15) x 100 =40g, (3/15) x 250 = 50g

Summing up the contribution to get Number Average Molecular Weight=

1.33 + 5.33 + 40 + 50 = 96.66g

Generalization of Concept

Total number of molecules (n) id given by

$$n = n_1 + n_2 + n_3 + n_4 + ... = \sum n_i$$

Number fraction of each molecule is = $\dfrac{n_i}{\sum n_i}$

Number average weight contribution of each entity is = $\dfrac{n_i M_i}{\sum n_i}$

Number average weight molecular weight is

$$\frac{n_1 M_2}{\sum n_i} + \frac{n_2 M_2}{\sum n_i} + \frac{n_3 M_3}{\sum n_i} + \frac{n_4 M_4}{\sum n_i} ...... = \frac{\sum n_i M_i}{\sum n_i} = M_n$$

Weight Average Molecular Weight (M_w)

Total weight of each poly present in the polymer =1450g

Weight of poly-1 present in polymer = 20g

Weight fraction of poly-1 = 20/1450, Weight fraction of poly-2 = 80/1450, Weight fraction of poly-3 = 600/1450, Weight fraction of poly-4 = 750/1450

Contribution made by each poly towards average weight of polymer = weight fraction of poly-1x weight of each unit

For poly-1 (20/1450) x 10 = 0.14g

For poly-2 (80/1450) x 20 = 1.10g

For poly-3 (600/1450) x 100 = 41.38g

For poly-4 (750/1450) x 250 = 129.31g

Summing up the contribution made by each poly to get weight average molecular weight is

$$0.14 + 1.10 + 41.38 + 129.31 = 171.93g$$

Generalization of Concept

Total number of molecules (n) id given by

$$n = n_1 + n_2 + n_3 + n_4 + ... = \sum n_i$$

Total weight of the polymer is = $\sum N_i M_i$ W

Weight fraction of each molecule is = $\dfrac{n_1 M_1}{W} = \dfrac{n_1 M_1}{\sum n_1 M_1}$

Weight average weight contribution of each entity is = $\dfrac{n_1 M_1 M_1}{\sum n_i M_i} \dfrac{n_i M_i^{\ 2}}{\sum n_i M_i}$

Number average weight molecular weight is

$$\frac{n_1 M_1^{\ 2}}{\sum n_i M_i} + \frac{n_2 M_2^{\ 2}}{\sum n_i M_i} + \frac{n_3 M_3^{\ 2}}{\sum n_i M_i} + \frac{n_4 M_4^{\ 2}}{\sum n_i M_i} ...... = \frac{\sum n_i M_i^{\ 2}}{\sum n_i M_i} = M_w$$

For synthetic polymers Mw is greater than the Mn. If they are equal than they will consider as perfectly homogeneous. (Each molecule has same molecular weight).

Molecular weight and degree of polymerisation

Number of repeating unit in a polymer called as *degree of polymerisation* (DP). DP provides the indirect method of expressing the molecular weight and the relation is as follows;

$$M = DP \times m$$

Where, M is the molecular weight of polymer, DP is the degree of polymerisation and m is the molecular weight of the monomer

$$(DP)_n = \frac{\sum n_i (DP)_i}{\sum n_i} \text{ and } (DP)_w = \frac{\sum n_i (DP)_i^2}{\sum n_i (DP)_i}$$

Each of these averages can be related to the corresponding molecular weight average by the following two equations;

$$M_n = (DP)_n . m$$

$$M_w = (DP)_w . m$$

Influence of Molecular weight of Polymers

The influence of molecular weight on the bulk properties of polyolefin's, an increase in the molecular weight leads to

Increase in:

- Melt viscosity
- Impact strength

Lowers in:

- Hardness
- Stiffness
- Softening point
- Brittle point

High molecular weight polymer does not crystallize so easily as lower molecular weight material crystallizes due to chain entanglement and that reflect in bulk properties of the high molecular weight polymer.

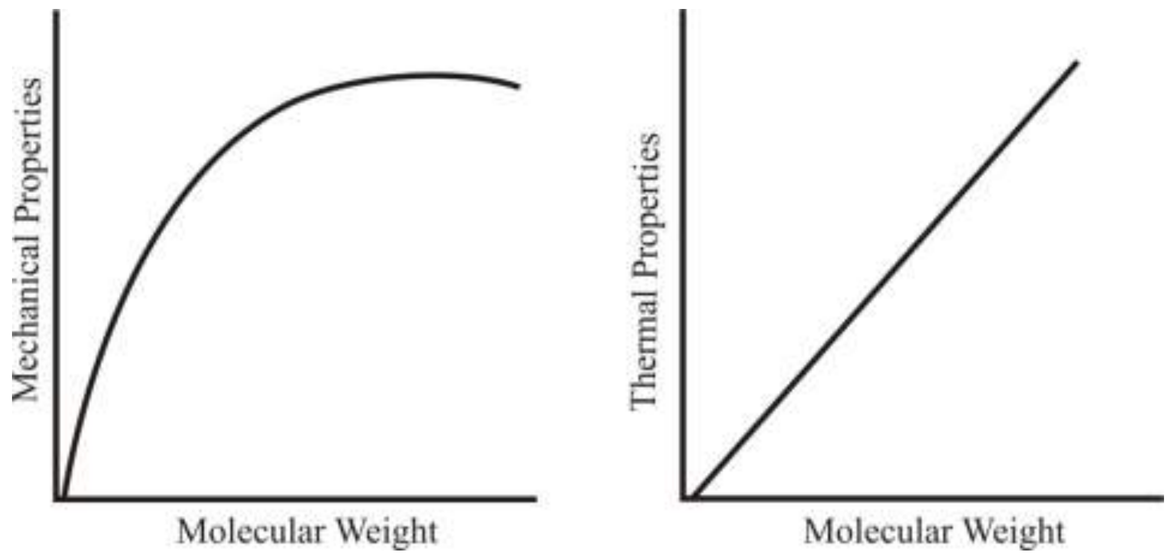
Effect of Molecular Weight on Properties of Polyolefins

Property	With Incease Molecular Weight
Melt Viscosity	↑
Melt Strength	↑
Melt Viscosity	↑
Toughness	↑
Abrasion Resistance	↑
Ultimate Strength	↑
Resistance to Creep	↑

Effect of Molecular Weight Distribution on Properties of Polyolefins

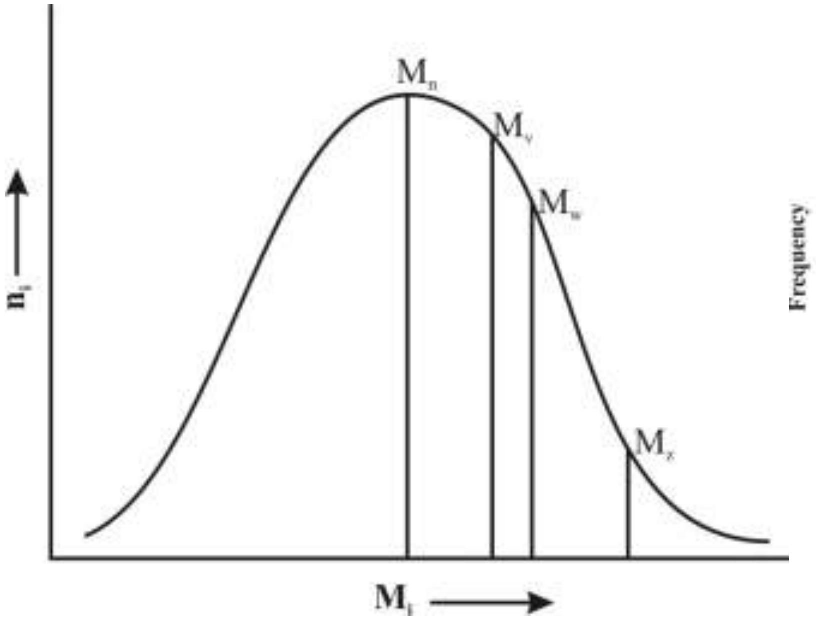
Property	Molecular Weight Distribution Incerasing
Melt Viscosity(High Shear)	↓
Surface Gloss	↓
Toughness	↓
Optical Properties	↓
Melt Strength (Low Shear)	↑

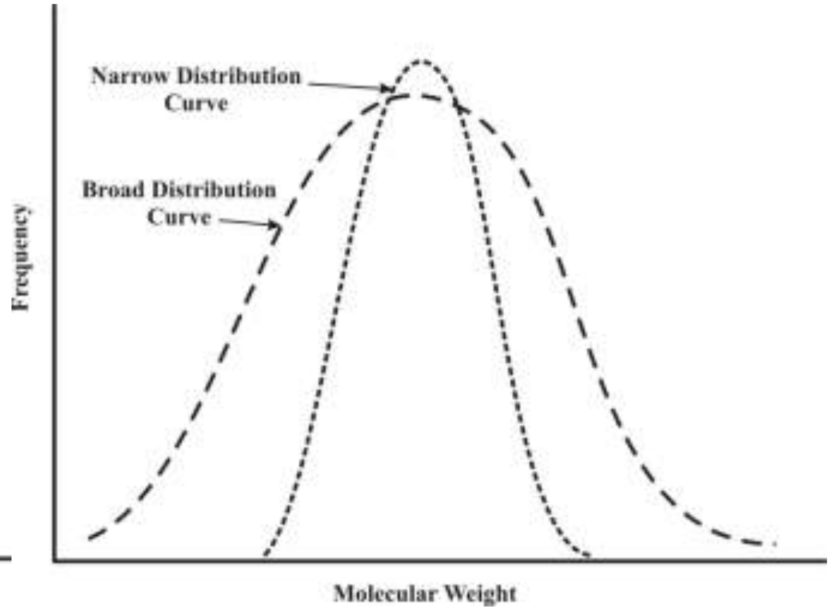
- A high molecular weight polymer increases the mechanical properties. Higher molecular weight implies longer polymer chains and a longer polymer chain implies more entanglement thereby they resist sliding over each other.
- Increasing the molecular weight and the chain length of the polymer increases impact strength.
- Thermal properties can also improved by increasing the molecular weight.



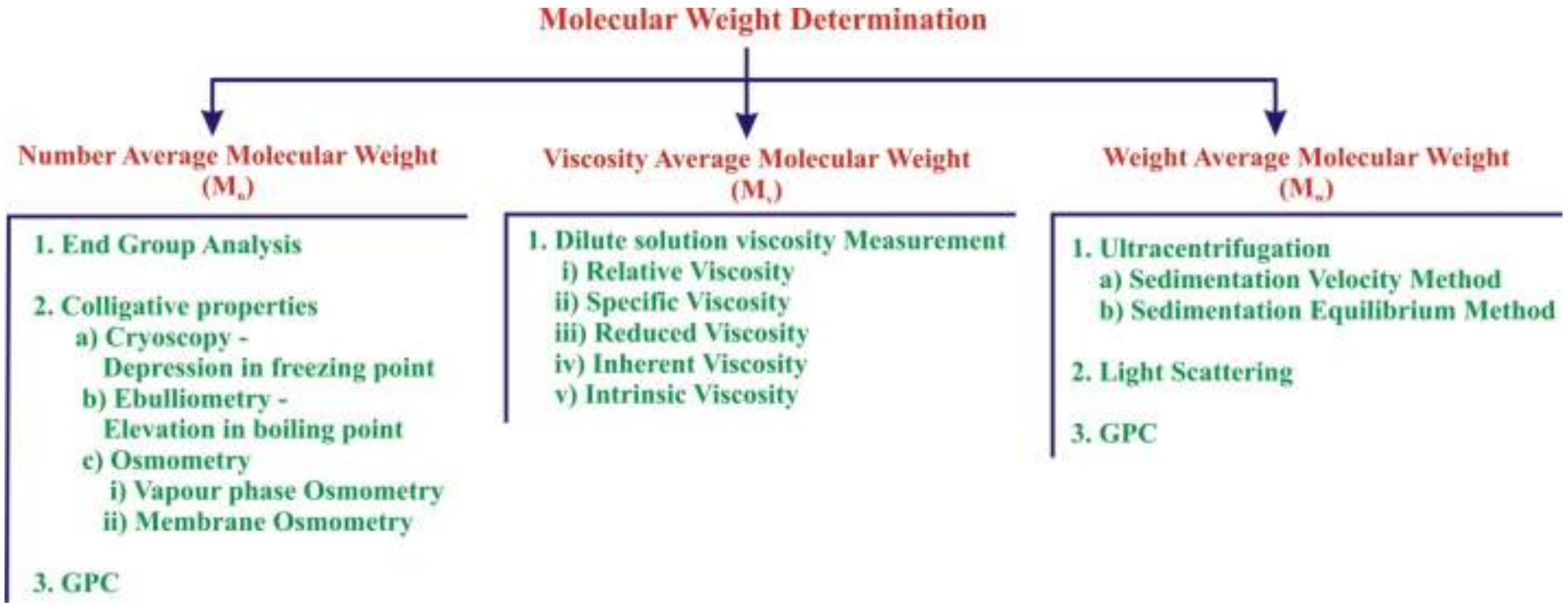
Polydispersity Index for Molecular weight of Polymers

Polydispersity is a very important parameter and it gives an idea of lowest and the highest molecular weight species as well as the distribution pattern of the intermediate molecular weight species. Plastics processing are affected by the molecular weight distribution.





Determination of Molecular Weight



END-GROUP ANALYSIS: Molecular-weight determination through group analysis requires that the polymer contain a known number of' determinable groups per molecule.

Requirements:

1)Quantitative determination of end groups

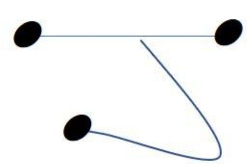
Amenable end groups

Concentration should be high enough

other group in the polymer should not interfere in the determination of end group

2) The no. of end groups for polymer chain (for each polymer chain how many)

For linear



Step growth polymers

Polyesters- OH, COOH

Polyamides-NH2, COOH

The methods become insensitive at high molecular weight, as the fraction of end groups becomes too small to be measured with precision.

Loss of precision often occurs at molecular weights above 25,000.

Me(equivalent mass) = m/n

m = coccentration of polymers

n= concentration of endgroups

$\boxed{M_n = M_e * f}$

f= no. of end groups/chain

Colligative properties

“Those properties that depend only on the concentration of solute molecules”

The colligative property methods are based on

- vapor-pressure lowering,
- boiling point elevation *(ebulliometry),*
- freezing-point depression *(cryoscopy),*
- and the osmotic pressure *(osmometry*

$$\lim_{c\to 0} \frac{\Delta T_b}{c} = \frac{RT^2}{\rho\, \Delta H_v} \frac{1}{\bar{M}_n}$$

$$\lim_{c\to 0} \frac{\Delta T_f}{c} = \frac{RT^2}{\rho\, \Delta H_f} \frac{1}{\bar{M}_n}$$

$$\lim_{c\to 0} \frac{\pi}{c} = \frac{RT}{\bar{M}_n}$$

where ΔT_b, ΔT_f, and Π are the boiling-point elevation, freezing-point depression, and osmotic pressure, respectively;

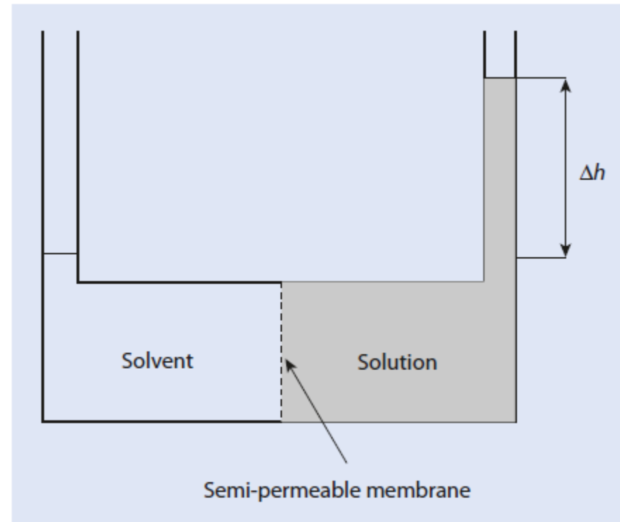
ρ is the density of the solvent;

ΔH_v, and ΔH_f are the enthalpies of vaporization and fusion, respectively, of the solvent per gram;

and c is the solute concentration in grams per cubic centimeter

Membrane Osmometry (absolute method)

"change in the osmotic pressure of a polymer solution as compared to the pure solvent".



$$\frac{\pi}{RTc} = A_1 + A_2 c + A_3 c^2 + \cdots$$

$$= \frac{1}{\overline{M}_n}\left(1 + \Gamma c + g\Gamma^2 c^2 + \cdots\right)$$

where r = *A2/A1 and g is a slowly varying function of the polymer-solvent interaction* with values near zero for poor solvents and values near 0..25 for good Solvents

In most cases the term in *c2 may be neglected; when dependence on c2 is* insignificant, it may be convenient to take g = 0.25 and write

Membranes for different solvents

Organic- cellulose, gel cellophane

Aqueous-cellulose acetate, nitrocellulose

Corrosive-glass

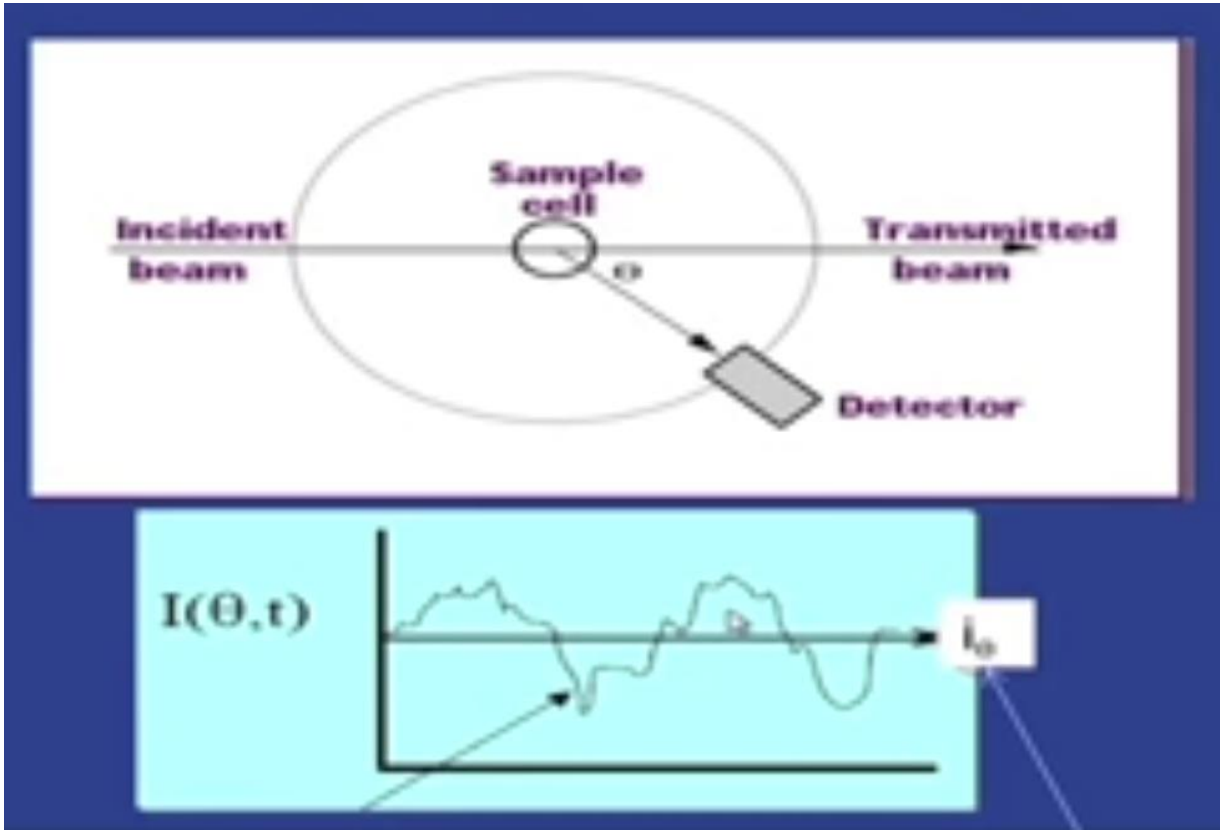
Membrane porosity

Must consider pore size and its distribution

Membrane conditioning

Should be stored in wet condition

Determination of M_w by Light scattering



Analysis of fluctuation in time
Dynamic light scattering
Hydrodynamic Radius

Analysis based on average intensity
Static light scattering
Absolute Mw, Ro^2, A^2

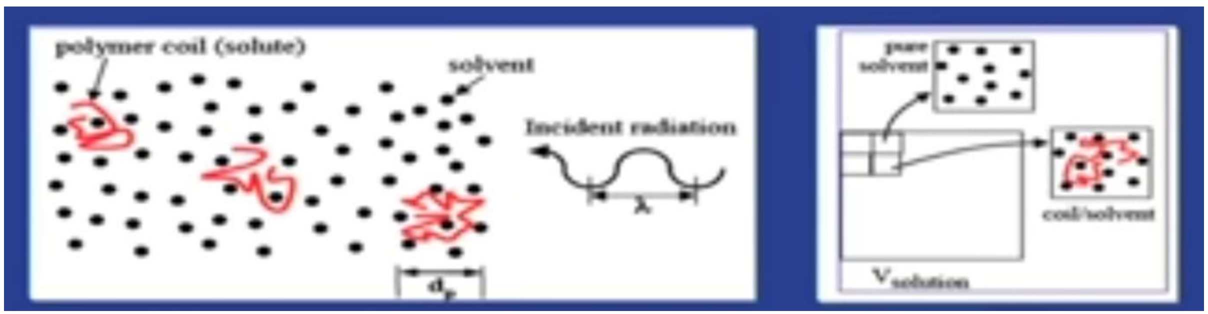
Scattering of light from polymer solution

MALS : Multi angle light scattering

SAXS: small angle X-Ray scattering

SANS: small angle Neutron scattering

Light scattering from dilute polymer solutions



Features in a binary component system

1). Each cell has on avaerage the same number of solvent molecules but the fluctuations in the solvent density will give rise to some weak scattering

2) Fluctuations in the concentration of solute molecules will give rise to scattering

$\Delta R = R_{solution} - R_{solvent}$

Following basic Rayleigh equation for light scattering and fluctuation theories of Einstein and Smouluchowski

$$\Delta R_\theta = \frac{2\pi^2 c\, n_0^2 \left(\frac{dn}{dc}\right)^2}{\lambda^4 N_A [\frac{1}{M} + 2A_2 c + 3A_3 c^2 + \ldots]}$$

$$\Delta R_\theta = R_{\theta,\,solution} - R_{\theta,\,solvent}$$

$$R_\theta = \frac{i_\theta r^2}{I_0 (1 + \cos^2 \theta)}$$

Rayleigh ratio

$$K = \frac{2\pi^2 n_0^2 \left(\frac{dn}{dc}\right)^2}{\lambda^4 N_A}$$

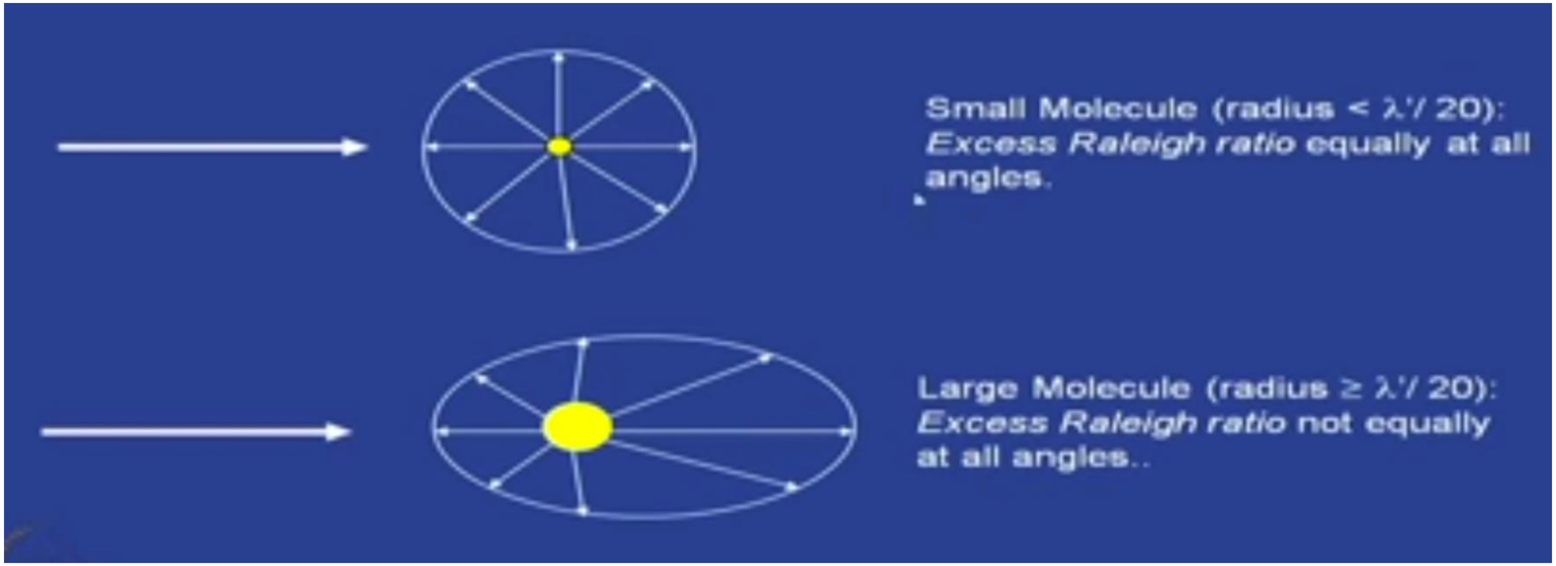
Optical constant

$$\frac{Kc}{\Delta R_\theta} = [\frac{1}{M} + 2A_2 c + 3A_3 c^2 + \ldots]$$

Basic priciples of light scattering from dilute polymer solutions

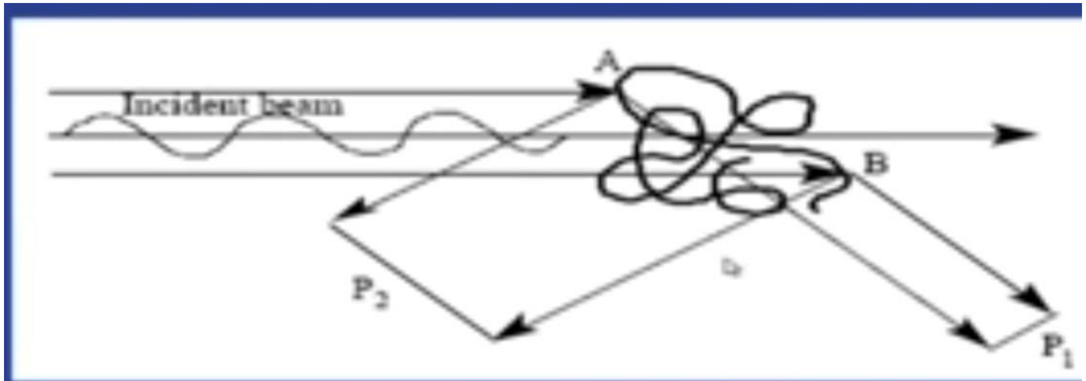
Principle 1: The amount of light scattered is directly proportional to the product of the polymer molar mass and concentration

$I_{scattered}$ proportional to $M_c(dn/dc)^2$



Light scattering angular dependance

Principle 2: The angular variation of the scattered light is directly related to the size of the molecule



$$I_{scattered} \propto P(\theta)$$

$$\frac{Kc}{\Delta R_\theta} = \frac{1}{P(\theta)}\left[\frac{1}{M} + 2A_2c + 3A_3c^2 + \ldots\right]$$

Following Debye and Guinier, for a monodisperse Gaussian Coil

$$P(\theta) = \left[\left[1 + \frac{16\pi^2 n_0^2 \sin^2(\theta/2)}{3\lambda^2}\langle s^2\rangle\right]\right]^{-1}$$

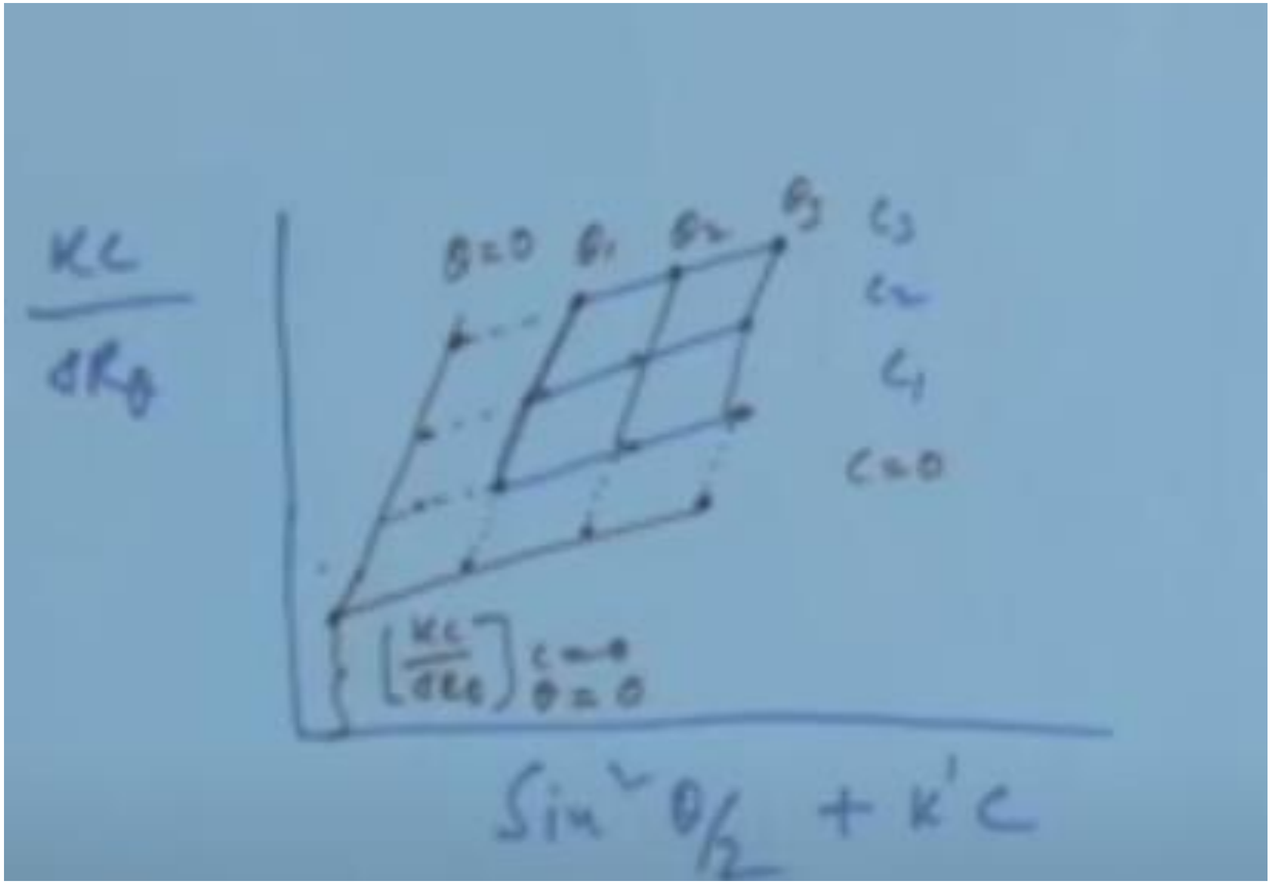
$$\frac{Kc}{\Delta R_\theta} = [\frac{1}{M} + 2A_2c + 3A_3c^2 + \ldots][1 + \frac{16\pi^2 n_0^2 \sin^2(\theta/2)}{3\lambda^2}\langle s^2 \rangle]$$

Valid for dilute solution and low angle

$$\frac{Kc}{\Delta R_\theta} = [\frac{1}{Mw} + 2A_2c + 3A_3c^2 + \ldots][1 + \frac{16\pi^2 n_0^2 \sin^2(\theta/2)}{3\lambda^2}\langle s^2 \rangle_z]$$

Zimm plot

How do you construct Zimm plot?



Why extrapolate

To eliminate intermolecular scattering effects: measure scattering from one molecule
To eliminate intramolecule scattering effects: to measure scattering at zero angle

Polymer: Amorphous and Crystalline State

Amorphous Polymer

- **Glass Transition**
- **Free Volume concept**
- **Factors affecting Glass Transition**

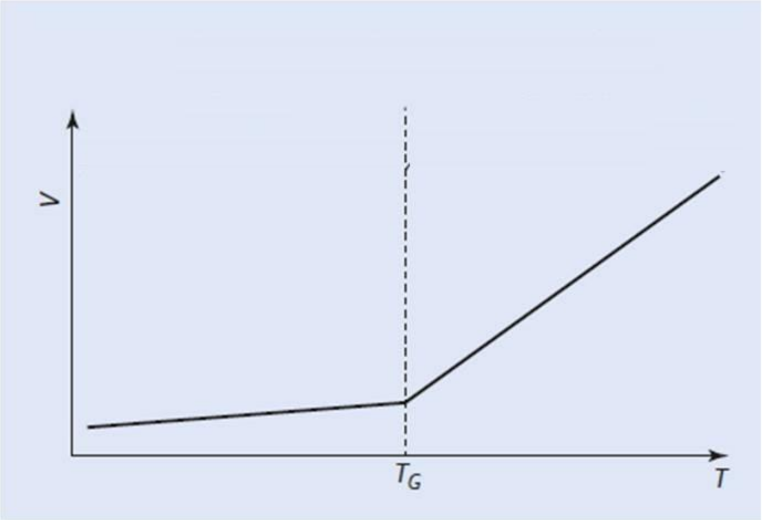
Polymer: Amorphous

Characteristics

- **No well-defined ordered arrangement of polymer molecules**
- **Example: Polystyrene, Polycarbonates etc.**
- **(Validated by SANS)**
- **It do not show any first order melting transitions.**
- **Response to thermal**

➢ **Transition of a polymer from a rubbery state to a hard glassy state**

➢ **Glass Transition Temperature (Tg)**



Change of Polymer Properties at Tg

➢ **Sharp increase in stiffness below Tg**

➢ **Abrupt change in heat capacity and TEC at Tg**

➢ **Change in slope of specific volume Vs T at T_g**

Free volume theory : Empty space between the molecules

$$V = V_0 + V_f$$

V_f = free volume

V_0 = Volume occupied by polymer molecules

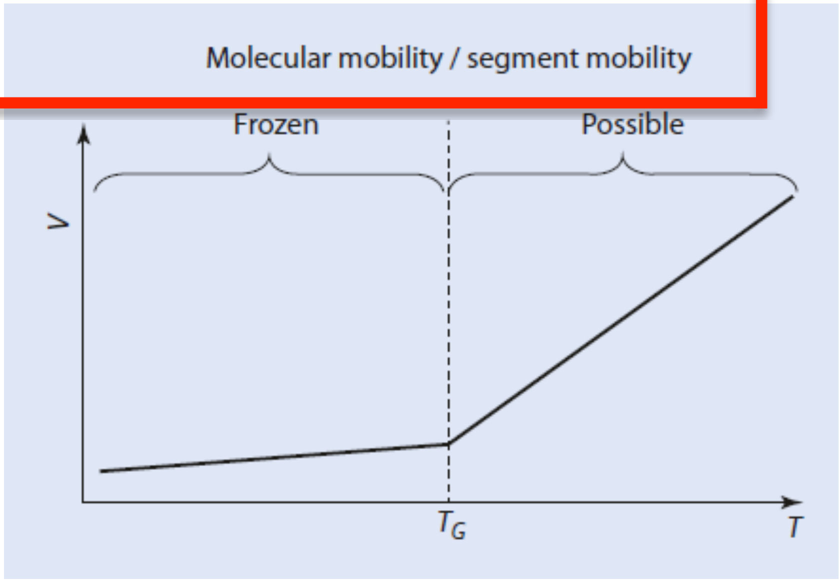
At temperatures below T_g, *movement of the chain* segments is not possible and the molecules cannot change their relative positions.

The only movements possible are vibrations at an atomic level.

If the temperature rises above T_g, *the* movement of individual segments becomes possible.. **This corresponds to Tg**

Fractional free volume = V_f/V

$= Vf/Vo+Vf$



Factors affecting T_g

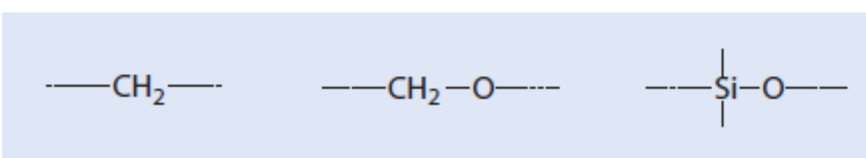
- **Chemical structure : Chain flexibility; nature of side groups**
- **Copolymerization and blending**
- **Film thickness**
- **Molecular architecture of the chain**

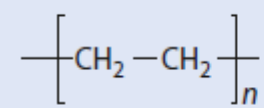
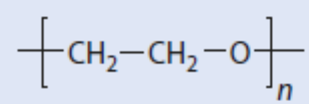
Chemical structure :

Chain flexibility: decided by nature of chemical groups constituting the main chain.

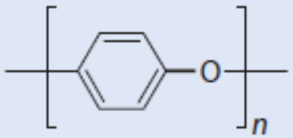
flexible chains are more mobile and lead to a lower glass transition temperature

Examples of flexible chain elements



	−93
	−67

Poly(P-Xylelene)- higher T_g

	+83

Nature of side groups : Generally increase T_g by restricting bond

Large bulky side groups cause chain stiffening, increases T_g (Polystyrene)

If side group is flexible lowers T_g

Polar substituent increases the glass transition temperature (PVC)

Table 6.3 Influence of polar substituents on the glass transition temperature (T_G) of $(CH_2–CHX)_n$

–X	T_G (°C)
$–CH_3$	–20
–Cl	+81
–CN	+105

Copolymerization and blending

For single phase or miscible polymer blend

$$T_g^{cop} = T_{g1}W_1 + T_{g2}W_2$$

W = weight fraction

Fox equation: reciprocal relationship

$$1/T_g^{cop} = W_1/T_{g1} + W_2/T_{g2}$$

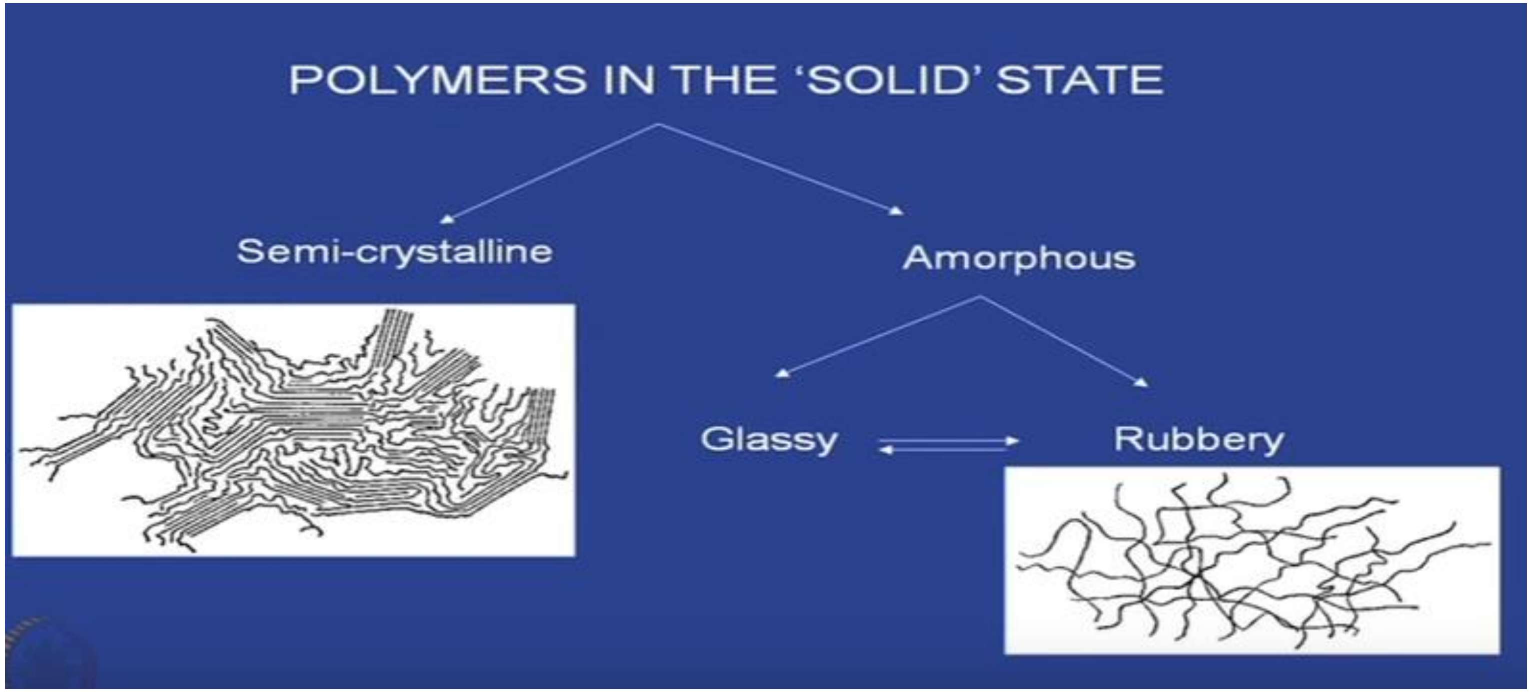
Molecular Architecture: Effect of molar mass on Tg

$$T_G = T_G^{\infty} - \frac{const.}{M_n}$$

T_G^{∞} Glass transition temperature for $M_n \to \infty$

Polymer Chemistry: *Sebastian Koltzenburg, Michael Maskos, Oskar Nuyken*

Textbook of Polymer Science: *FRED W. BILLMEYER, JR*



Crystalline state of polymers

Degree of crystallinity

Crystal structure

Crystallisation

Kinetics of crystallisation

Melting

Factors affecting Melting temperature

- **Polymers are semi-crystalline, rarely fully crystalline (crystallinity never equal to 100%)**
- **Consisting of coexisting of amorphous and crystalline regions** (Examples: Polyethylene, Polypropylene)
- **Polymer chains in the crystalline domains are arranged in an ordered fashion**
- Chains typically fold to form ordered lamellar structures.
- **Due to the presence of long chain structure of polymer molecule, their crystallisation is a complex process.**

Thermal transition

- Fully crystalline polymers exhibit melting transition but not glass transition
- Completely amorphous polymers show glass transition but not melting transition
- Semi crystalline polymers can show both glass and melting transition.

Degree of crystallinity

A measure of the amount of crystalline part in the sample

Density method

Volume fraction: Degree of crystallinity, ϕ_c

$$\phi_c = \frac{V_c}{V} = \frac{\rho - \rho_a}{\rho_c - \rho_a}$$

V_c = volume of crystalline part of sample

V=Total volume of the sample

$\boldsymbol{\rho}$=density of semicrystalline sample

$\boldsymbol{\rho}_c$=density of crystalline part of the sample

$\boldsymbol{\rho}_a$=density of amorphous part of sample

Mass fraction: Degree of crystallinity, X_c

$$X_c = \frac{W_c}{W} = \frac{\rho_c v_c}{\rho v}$$

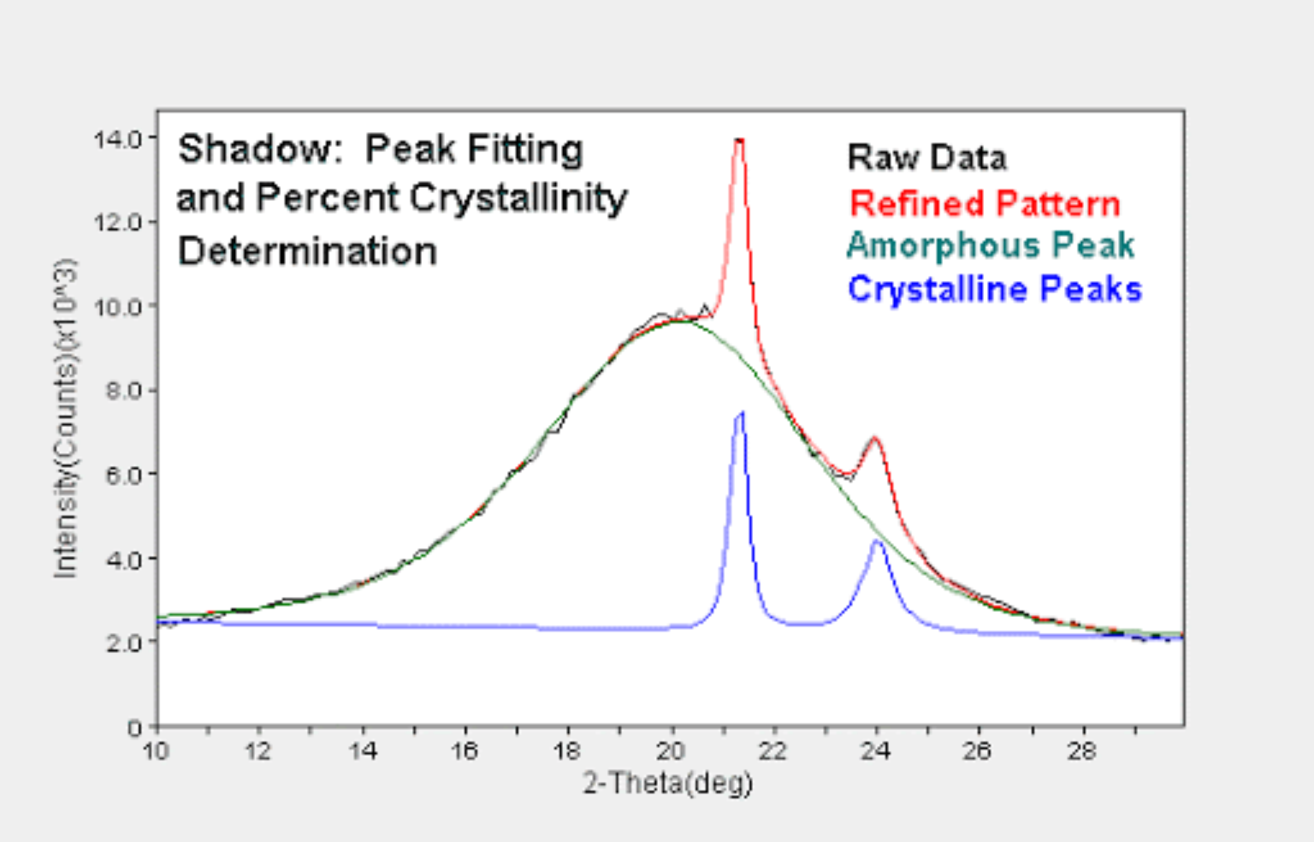
M_c = mass of crystalline part of sample

V=Total volume of the sample

Wide angle X-ray scattering

$$\boxed{X_c = A_c/(A_a+A_c)}$$

Aa = Area under amorphous halo
Ac = area under crystalline part



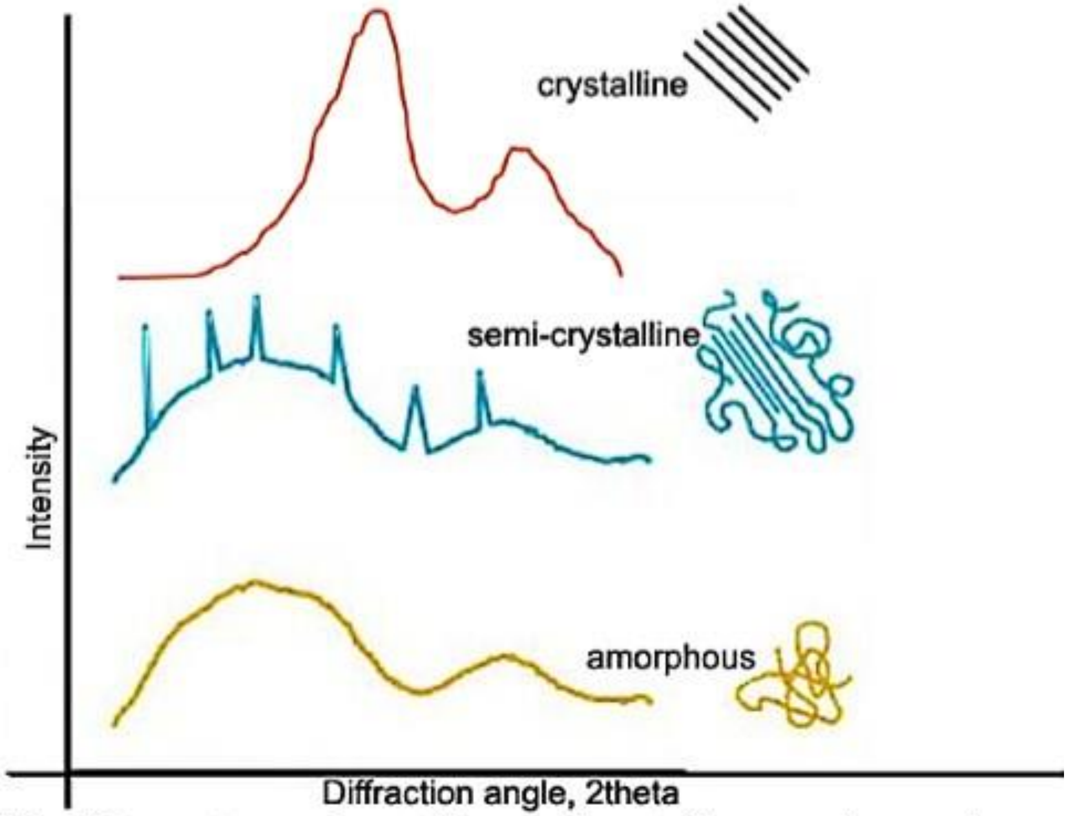


Fig: XRD spectarum of crystalline,semi-crystalline,amorphous polymer

Crystal structure

- Crystal structure is studied using XRD
- Unit cell, in polymer crystals contain several repeat units of the polymer chain. Ex: Polyethylene unit cell can be orthorhombic or monoclinic, isotactic polystyrene is trigonal etc.
- Polymer molecules pack into crystals as zig-zag or helics Ex: Polyethylene adopts Zig-Zag; isotactic polypropylene and polystyrene form helix

Factors determining crystallisation and crystal structure

- Linear chains
- Copolymerization
- Tacticity

Crystal Structure

Polymer crystals are typically lamellar –plate like

Lamellar thickness is typically 10nm (for solution grown crystal)

Polymer molecules fold multiple times between the lamellar structure

Crystal structure

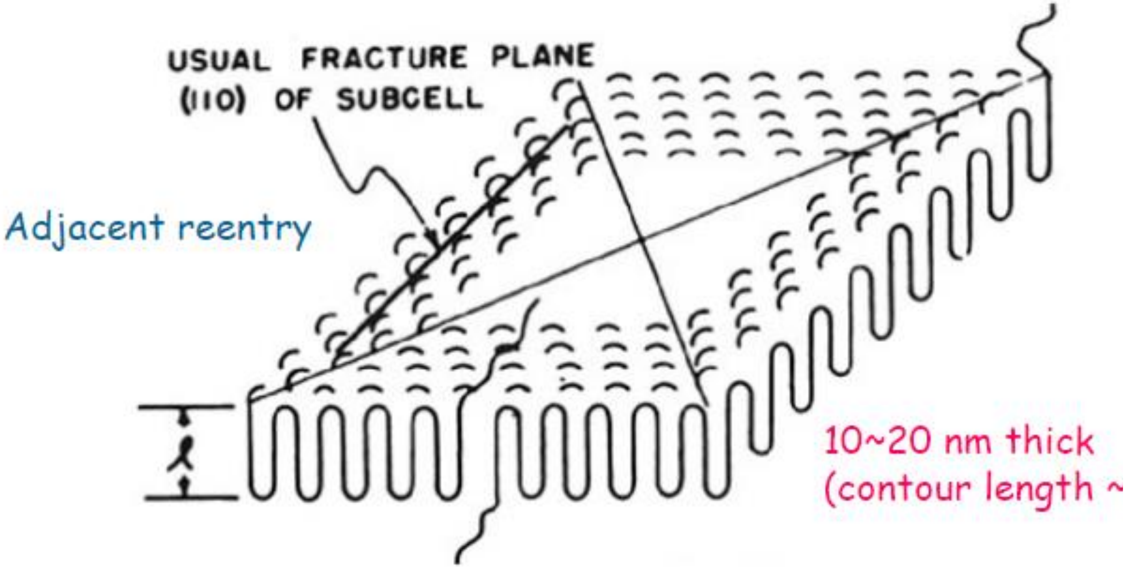
- Folds can be regular and tight or irregular and loose
- Concentrated solution and melt polymers polymer chains are entangled
- Crystallization can result in polymer chains being incorporated in more than one crystal
- In melts, chains are highly entangled, crystals are more regular

Spherulites:

In melt crystallization spherulites form by nucleation at different points and then grow

Consists of many lamellar crystals radiating from the centre

When viewed under polarised light it shows Maltese cross pattern



Spherulite

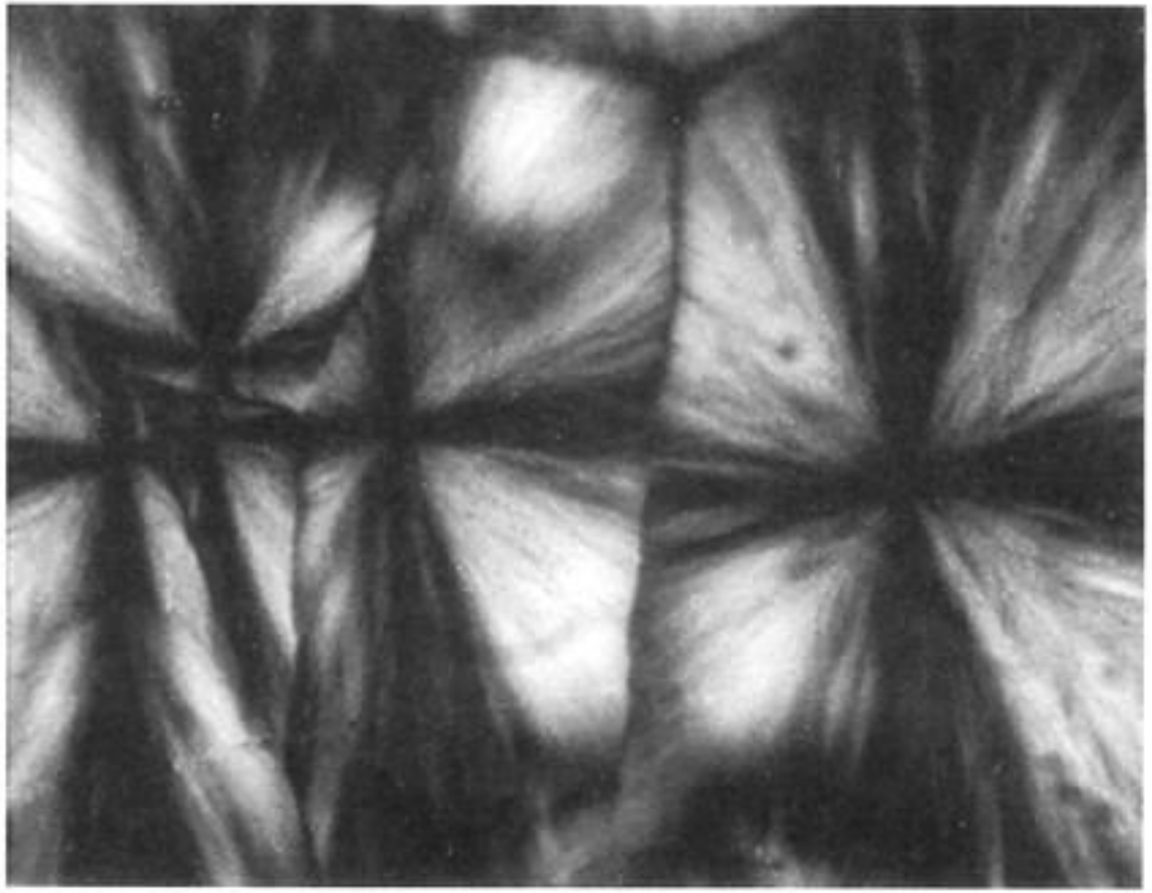


Figure 6.13 Spherulites of low-density polyethylene, observed through crossed polarizers. Note characteristic Maltese cross pattern (34).

Crystallization

Ordered structure from disordered state

Polymers can crystallize from solution or melt

When polymer melt is cooled below melting temperature

Tangled molecules in the melt tend to become aligned and form small ordered regions

The small crystal nuclei grow by addition of more chains

Crystallization takes place through nucleation and growth

Crystallization kinetics

Change in linear dimension of growing crystalline entities (at agiven Tc) is linear with time

$\boxed{R = vt}$

R= spherulite radius

Tc = tempearure of crystallisation

V = growth rate

Melting

- Takes place over a range of temperature
- Depends on crystallization temperature
- Depends on rate of cooling

Chemical Structure

Higher flexibility, lower Tm

Polar groups in main chain: raise Tm

Nature of side groups: Bulky group raise Tm

Molecular architecture: Chain ends and branches introduce defects in the crystals, hence lower Tm

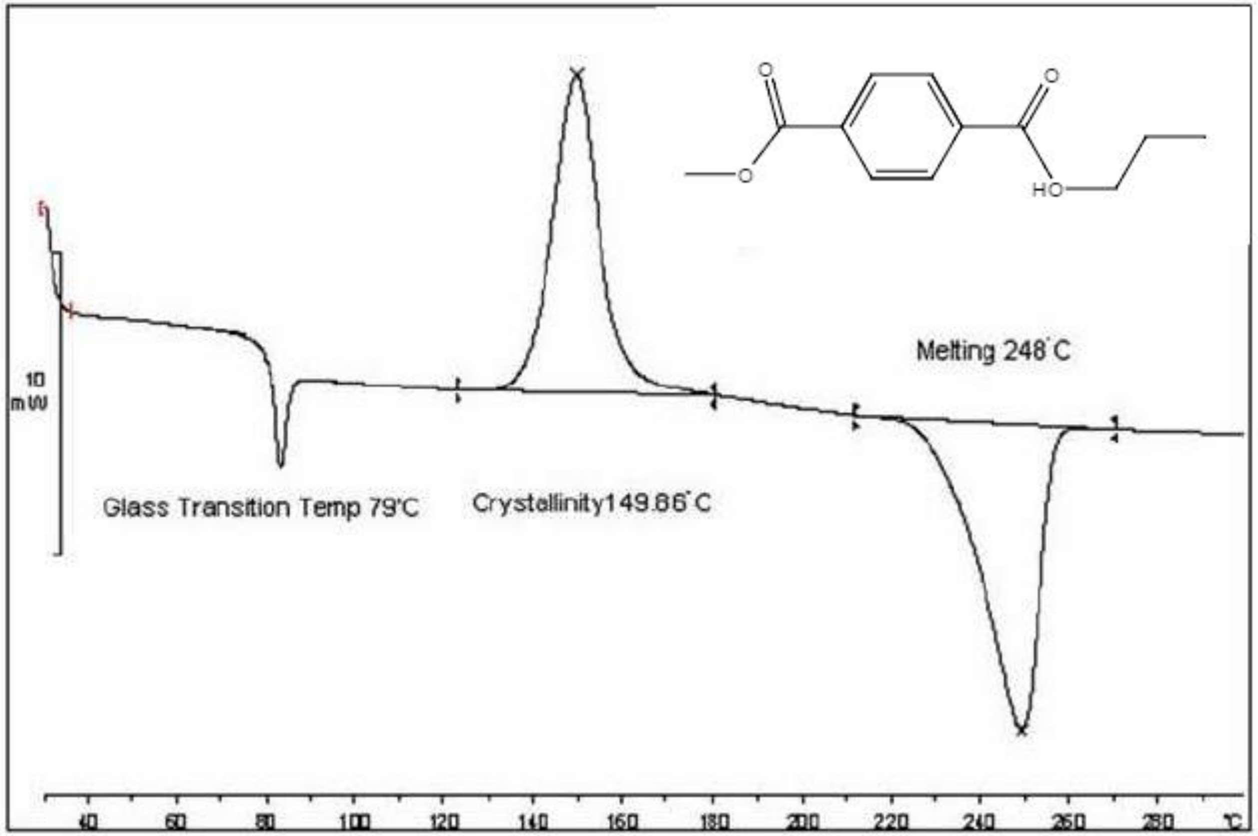
Low molar mass show low melting point; higher molar mass show higher melting point.

Amorphous polymer

- Heated from low T region, volume expands at constant rate.
- At Tg, rate of expansion increases abruptly. At same time change in physical behavior from hard brittle, glassy solid below Tg to a soft rubbery region above Tg.
- Further heating polymer changes from rubbery to viscous liquid

Semi-crystalline polymer

- Changes at Tg less drastic since changes are restricted to amorphous regions while crystalline zones are unaffected
- Between Tg and Tm semicrystalline polymer is composed of rigid crystallites dispersed in a rubbery amorphous matrix.
- At Tm, crystallites melt, leading to viscous state.

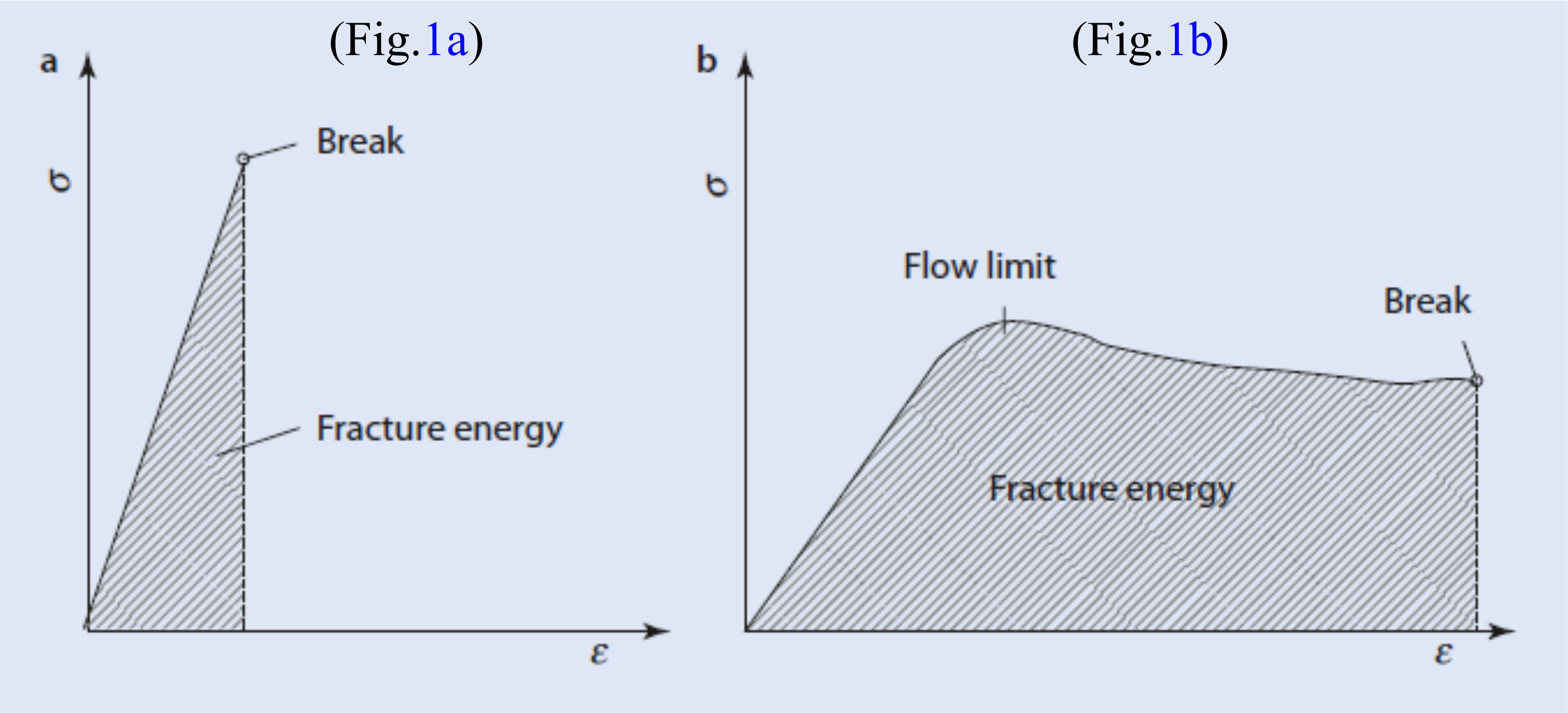


O
O
O
HO
10
mW
Melting 248°C
Glass Transition Temp 79°C
Crystallinity149.86°C
40
60
80
100
120
140
160
180
200
220
240
260
280
°C

Table 6.2 Selected crystallographic data (27)

Polymer	Crystal System, Space Group, Lattice Constants, and Number of Chains per Unit Cell[a]	Molecular Conformation	Crystal Density (g/cm^3)
Polyethylene $\text{-(}CH_2-CH_2\text{)-}_n$	Stable form, orthorhombic, $Pnam$-D_{2h}^{16}, $a = 7.417$ Å, $b = 4.945$ Å, $c = 2.547$ Å, $N = 2$	Planar zigzag (2/1)	1.00
	Metastable form, monoclinic, $C2/m-C_{2h}^{3}$, $a = 8.09$ Å, b (f.a.) $= 2.53$ Å, $c = 4.79$ Å, $\beta = 107.9°$, $N = 2$	Planar zigzag (2/1)	0.998
	High-pressure form, orthohexagonal (assumed), $a = 8.42$ Å, $b = 4.56$ Å, c (f.a.) has not been determined		
Polytetrafluoroethylene $\text{-(}CF_2-CF_2\text{)-}_n$	Below 19°C, pseudohexagonal (triclinic), $a' = b' = 5.59$ Å, $c = 16.88$ Å, $\gamma' = 119.3°$, $N = 1$	Helix (13/6) 163.5°	2.35
	Above 19°C, trigonal, $a = 5.66$ Å, $c = 19.50$ Å, $N = 1$	Helix (15/7) 165.8°	2.30
	High-pressure form I[12], orthorhombic, $Pnam$-D_{2h}^{16}, $a = 8.73$ Å, $b = 5.69$ Å, $c = 2.62$ Å, $N = 2$	Planar zigzag (2/1)	2.55
	High-pressure form II,[193] monoclinic, $B2/m$-C_{2h},[3] $a = 9.50$ Å, $b = 5.05$ Å, $c = 2.62$ Å, $\gamma = 105.5°$, $N = 2$	Planar zigzag (2/1)	2.74
it-Polypropylene $\text{-(}CH_2-CH(CH_3)\text{)-}_n$	α-Form, monoclinic, $C2/c$-C_{2h}^{6} or Cc-C_s^4, $a = 6.65$ Å, $b = 20.96$ Å, $c = 6.50$ Å, $\beta = 99°20'$, $N = 4$	Helix (3/1) $(TG)_3$	0.936
	β-Form, hexagonal, $a = 19.08$ Å, $c = 6.49$ Å, $N = 9$	Helix (3/1) $(TG)_3$	0.922
	γ-Form, trigonal, $P3_121$-D_3^4 or $P3_221$-D_3^6, $a = 6.38$ Å, $c = 6.33$ Å, $N = 1$	Helix (3/1) $(TG)_3$	0.939
it-Polystyrene $\text{-(}CH_2-CH(C_6H_5)\text{)-}_n$	Trigonal, $R3c$-C_{3v}^{6} or $R\bar{3}c$-D_{3d}^{6}, $a = 21.90$ Å, $c = 6.65$ Å, $N = 6$	Helix (3/1) $(TG)_3$	1.13

Tension–elongation curves



Viscoelastic behaviour: Most of the polymer show this kind of behaviour

Pure elastic response: stress is proportional to strain; Pure viscous response: stress proportional to the strain rate

Fig.1a : For brittle

Fig. 1b: For viscoelastic materials

The mechanical properties of polymers are not single-valued functions of the chemical nature of the macromolecules.

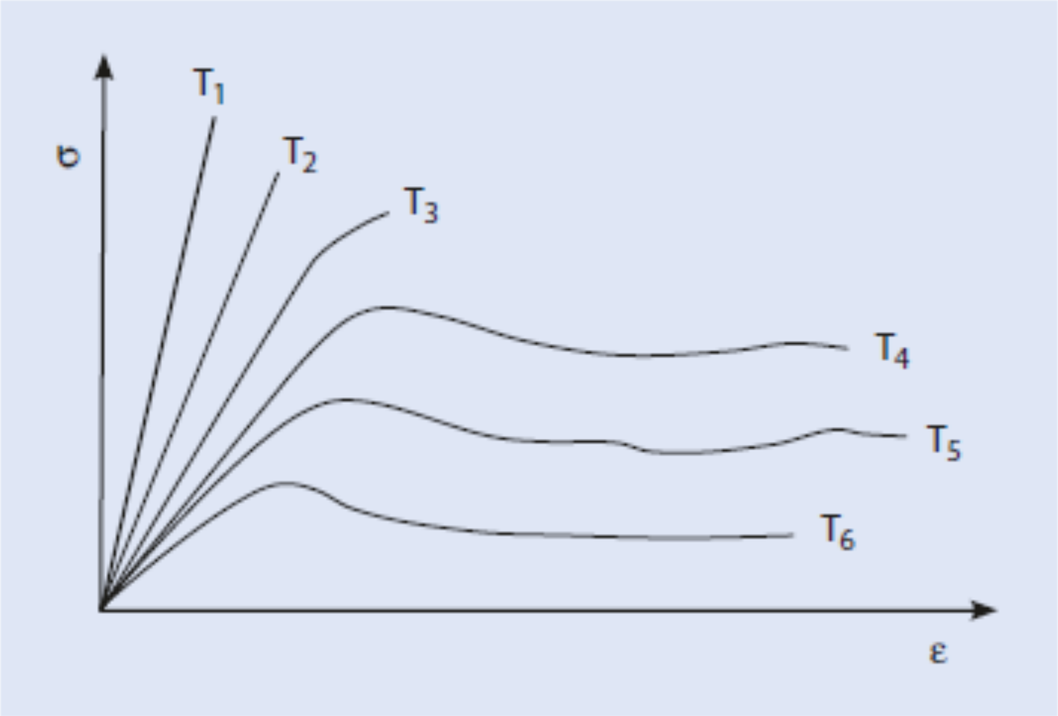
They will vary also with ***molecular weight, branching, cross-linking, crystallinity, plasticizers, fillers and other additives, orientation,*** and other consequences of processing history and sometimes with the thermal history of the particular sample.

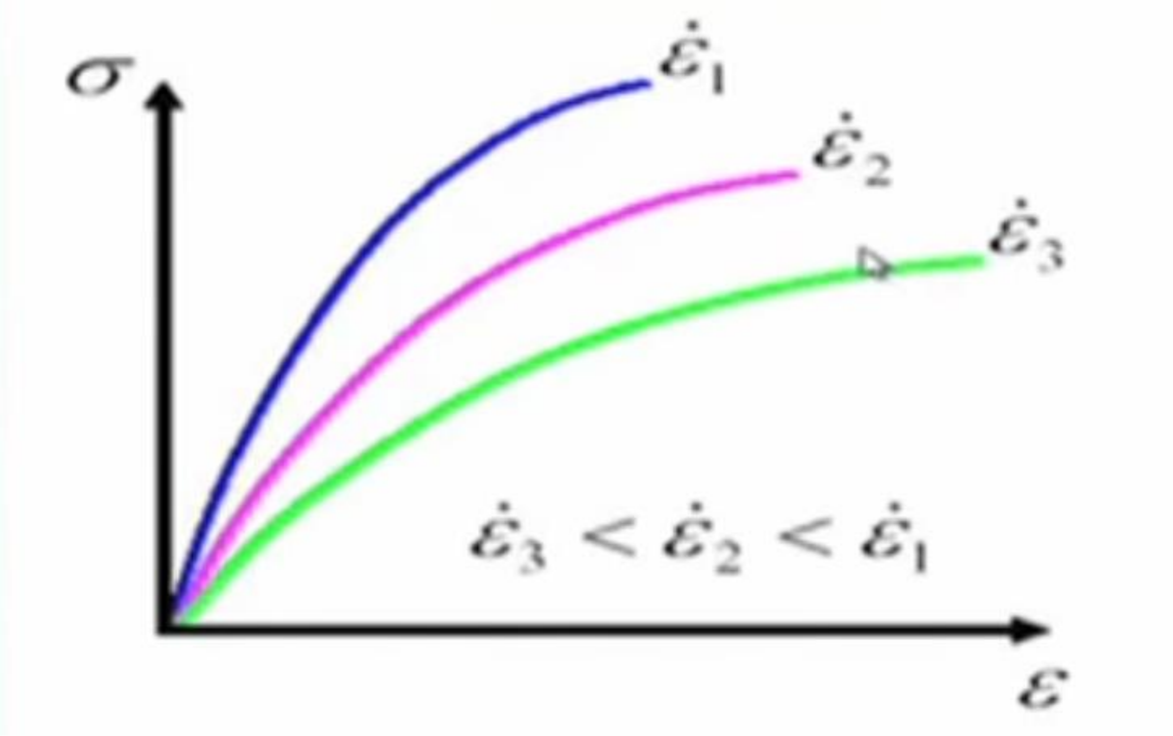
When all these variables are fixed for a particular specimen, it will still be observed that the properties of the material will depend strongly on the temperature and time of testing compared.

This dependence is a consequence of the viscoelastic nature of polymers.

•

Response with time and temperature dependent



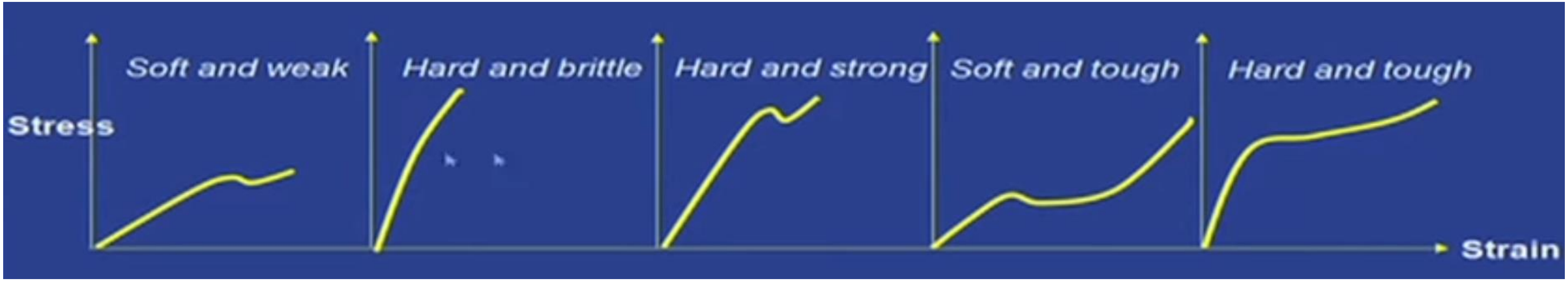


Stress strain curve behaviour on the temperature and time dependence

At short times higher strain rate at low Temperature : Brittle and rigid material (elastic response)

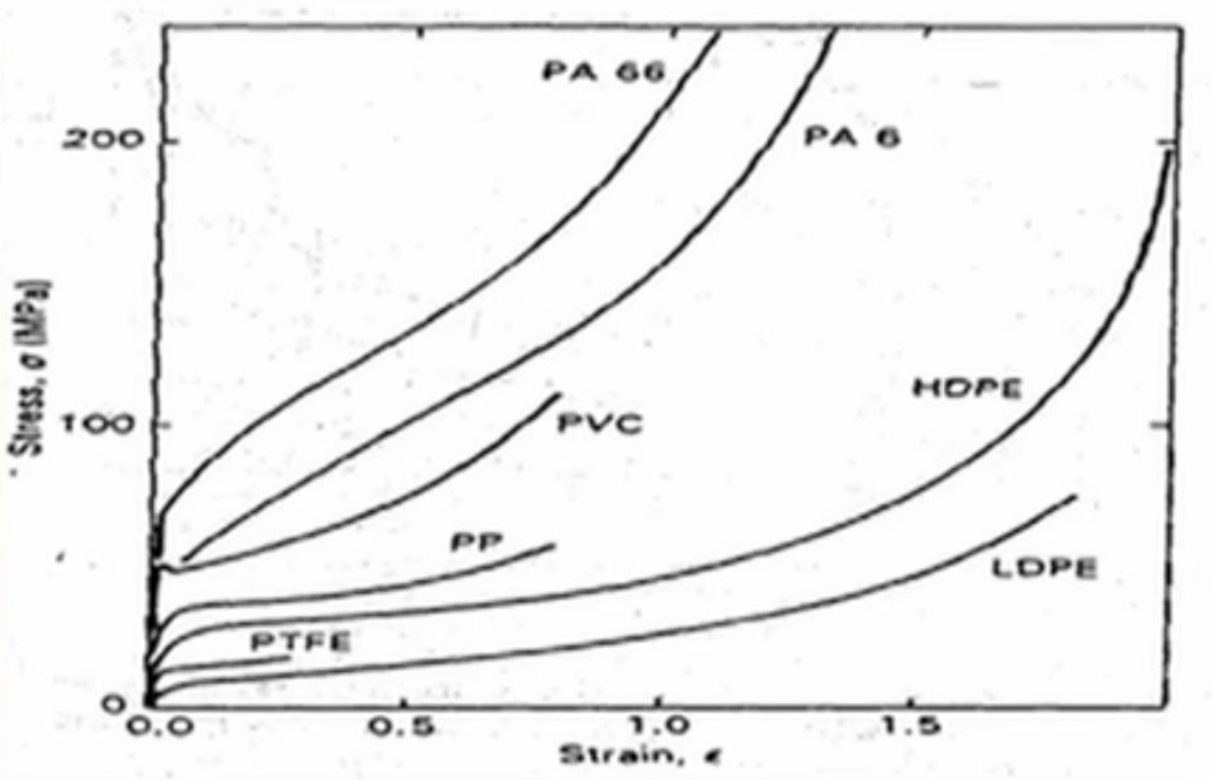
At long times strain rate slow, Temp. High: Tough and ductile material (viscous response)

Mechanical properties Different types of Stress strain curve



	Elastic modulus	Yield point	Tensile strength	Elongation at break	Ex.
Soft and weak	Low	low	low	moderate	Polymer gels
Hard and brittle	High	Low	High	Low	PS
High and strong	High	High	High	Moderate	PVC
Soft and tough	Low	Low	Moderate	High	rubbers
Hard and tough	High	high	high	high	Cellulose acetate, nylon

Stress strain curve for few common polymers



- Modulus of elasticity
  - highly elastic polymer ≈ 7 MPa
  - stiff polymer ≈ 4 GPa
  - metals ≈ 48 – 410 GPa
- Maximum tensile strength
  - Bulk polymers <~ 100 MPa
  - metal alloys ~ > 200 - 400 MPa
- Elongation
  - metals – rarely plastically elongate to more than 100%
  - polymers can elongate to 1000% (elastomers)

Mechanical properties : Structure Property relationship

Elastic modulus

Increases as **crystallinity increases**; **Decreases** as content of **soft disperse phase increases**

Increases as orientation increases; Increases as resistance to deformation of amorphous phase increases (glassy additive/fiber

Yield strength

Decreases with addition of solvent/plasticizer; Increases as crystallinity increases

Elongation at break

Elongation at break decreases with increasing MW (rubbery/crosslinked/semicrystalline polymers)

Decreases with increasing short chain branching density

In general

Flexural backbone: lower modulus and strength

Chain stiffness or backbone or bulky side group: increases modulus and strength

Increases as crystallinity increases modulus, strength

Creep Experiment

Pure elastic response:

Strain will develop instantaneously and remain constant

If stress is removed strain will instantaneously becomezero

Pure viscous response:

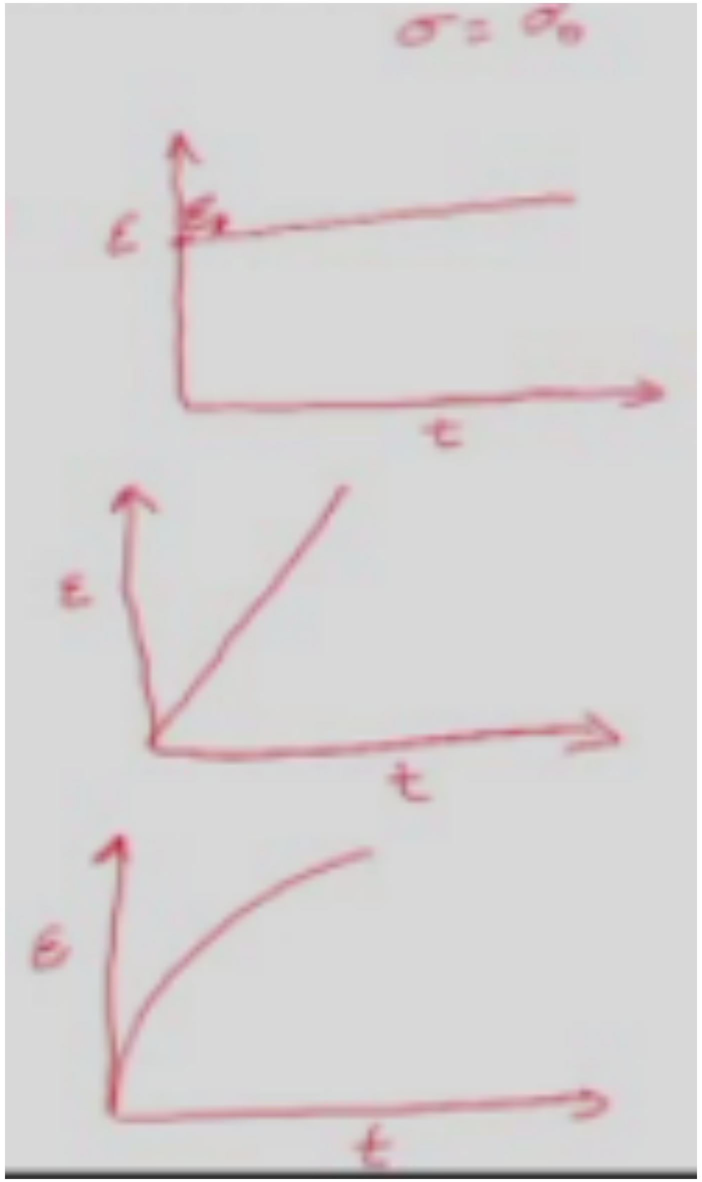
Strain will increase linearly as long as stress is applied

If stress is removed strain will not decay

Viscoelastic response:

Strain will increase as long as the stress is applied

If stress is removed, strain will decat but not becomes zero



Stress Relaxation Experiment:

A constant stress is applied and the stress is monitored with time

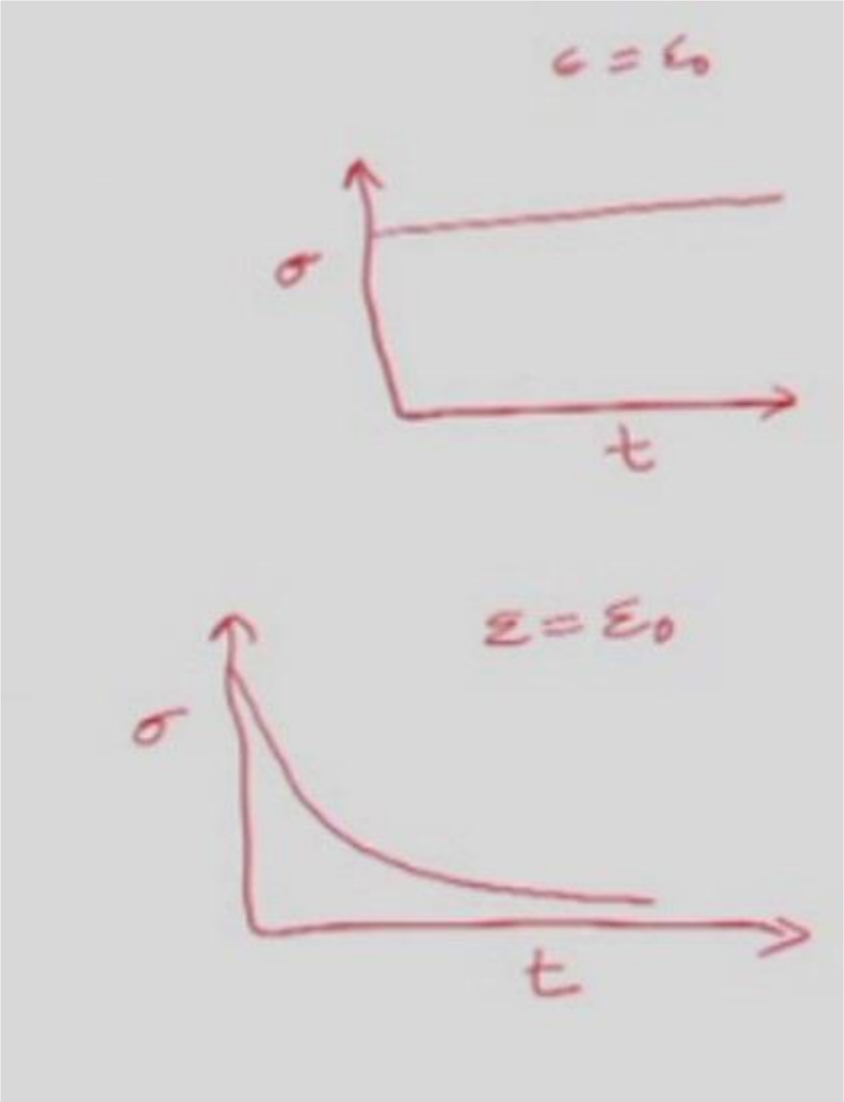
Pure elastic response

Stress will remain constant witht ime

Strain is removed stress will instantaneously become zero

Viscoelastic response

Stress will decay with time



CONDUCTING POLYMERS

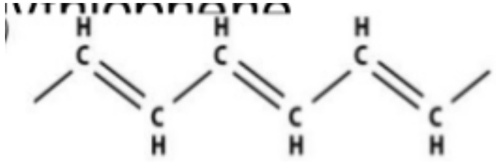
THE BEGINNING OF SEMICONDUCTING POLYMERS

Alan G. MacDiarmid
University of Pennsylvania,
Philadelphia, USA

Hideki Shirakawa
University of Tsukuba,
Japan

Alan J. Heeger
University of California
at Santa Barbara, USA

Nobel Prize in Chemistry in 2000



- Generally polymers are very good insulators and poor conductors
- There are certain polymers which conduct electricity due to the presence of conjugated multiple bonds along the polymer chin backbone or ring structure. This conductivity is because of delocalised mobile electrons
- ***Definition: The polymers that are capable of conducting eletrcicity due to the presence of conjugated system of delocalised electrons are called conjugated polymers***
- ***Two condition has to met 1) conjugated double bonds 2) dopants***
- They are unique as possible substitute for metallic conductors and semiconductors
- Exhibit highly reversible redox behaviour and the unusual combination of properties of metals and plastics
- They have unique conduction mechanism and good environmental stability in presence of oxygen and water
- These polymers could be highly promising for many technological uses because of their chemical versatility, stability, processability and low cost.

MATERIALS	CONDUCTIVITY	Sp. Gr.
Ag	10^5	10.5
Cu	$6x10^5$	8.9
Al	$4x10^5$	2.7
Polyacetylene	$1.5x10^5$	1.0
Polythiophene	10^4	1.0
Polypyrrole	600	
Carbon fiber	500	
polyaniline	10	
Poly-p-Phenylene	500	

Conductivities of conductive polymers with selected dopants

Polymer	Doping materials	Conductivity (s/cm)
Polyacetylene	I_2, Br, Li, Na, AsF_5	10^4
Polypyrrole	BF_4^-, ClO_4^-, Tosylate	500-7.5×10^3
Polythiophene	BF_4^-, ClO_4^-, tosylate, $FeCl_4^-$	10^3
Poly (3-alkylthiophene)	BF_4^-, ClO_4^-, $FeCl_4^-$	$10^3 - 10^4$
Polyphenylenesulfide	AsF_5	500
Polyphenylene-vinylene	AsF_5	10^4
Polyphenylene	AsF_5, Li, K	10^3
Polyisothianaphthalene	BF_4, ClO_4^-,	50
Polyaniline	HCL	200

Synthesis of conductive polymers

- Chemical polymerization
- Electrochemical polymerization
- Photochemical polymerization
- Concentrated emulsion polymerization
- Solid state polymerization
- Plasa polymerization
- Pyrolysis,
- Soluble precursor polymer preparation

Synthesis of conductive polymers: Poly aniline (PANI)

Produced as bulk powder, cast films or fibers

First discovered in the 19th century, also known as Aniline black

Found in one of the three idealised states

Leucoemeraldine ,emeraldine and (Per)nigraniline

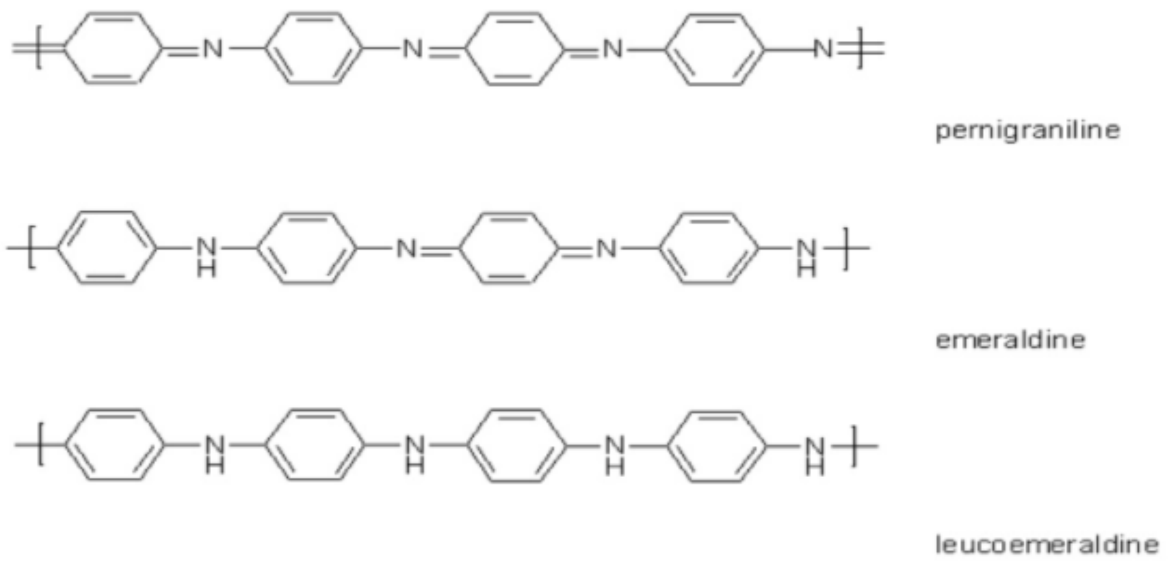
Why polyaniline

High sensitive

Rapid flexible

Easy in synthesis

Structure



Synthesis of conductive polymers: Poly aniline (PANI)

- Dissolve 5.17g of aniline hydrochloric acid
- Dissolve 11.42g of APS in 100ml water and add to monomer solution drop wise with constant stirring
- Stir the resultant solution for 1 h and keep for overnight
- Filter, wash with distilled water, 1(M) HCl, DI water, diethylether, 1(M) NH_4OH and acetone and dry in vacuum oven at 60C for 6 h
- Keep the polymer in ammonia solution for overnight to get the emeraldine base form of the polymer

Techniques of conductive polymer film fabrication

- Solution casting
- Spin casting
- Vacuum deposition

Nature of dopants and doping

The conductivity of a polymer can be increased several fold by doping it with an oxidative/reductive species or by donor/acceptor radicals.

Electrical conductivity of polymers, although primarily depends on doping are influenced by many factors, viz

- Synthesis method resulting in different structures
- Processing of polymers
- Degree of crystallinity
- Temperature

Stability and processability

- Conducting polymer exhibit
- Poor thermal and environmental stability
- Insoluble and infusible (due to extend chain configuration)
- Processing is poor

Due to the delocalisation of electron in the extended chain structure probably higher intermolecular cohesion occurs. This restricts the solubility in suitable solvent and causes degradation of polymers before melting.

Doping techniques

- Gaseous doping
- Solution doping
- Electrochemical doping
- Self doping
- Ion exchange doping

Doping agents or dopants are either strong reducing agent or strong oxidising agents.

They may be neutral molecules and compounds or inorganic salt forming ions, organic dopants or polymeric dopants.

Mechanism of doping in polymers

Since dopants are strong oxidising or reducing agents on doping positive or negative charge carriers are developed in the polymers

Polymer + Dopant ---------- $(\text{Dopant polymer})^{+}$ + Dopant^{-} (oxidation)

Polymer + Dopant ----------$(\text{Doped Polymer})^{-}$ + Dopant^{+}(Reduction)

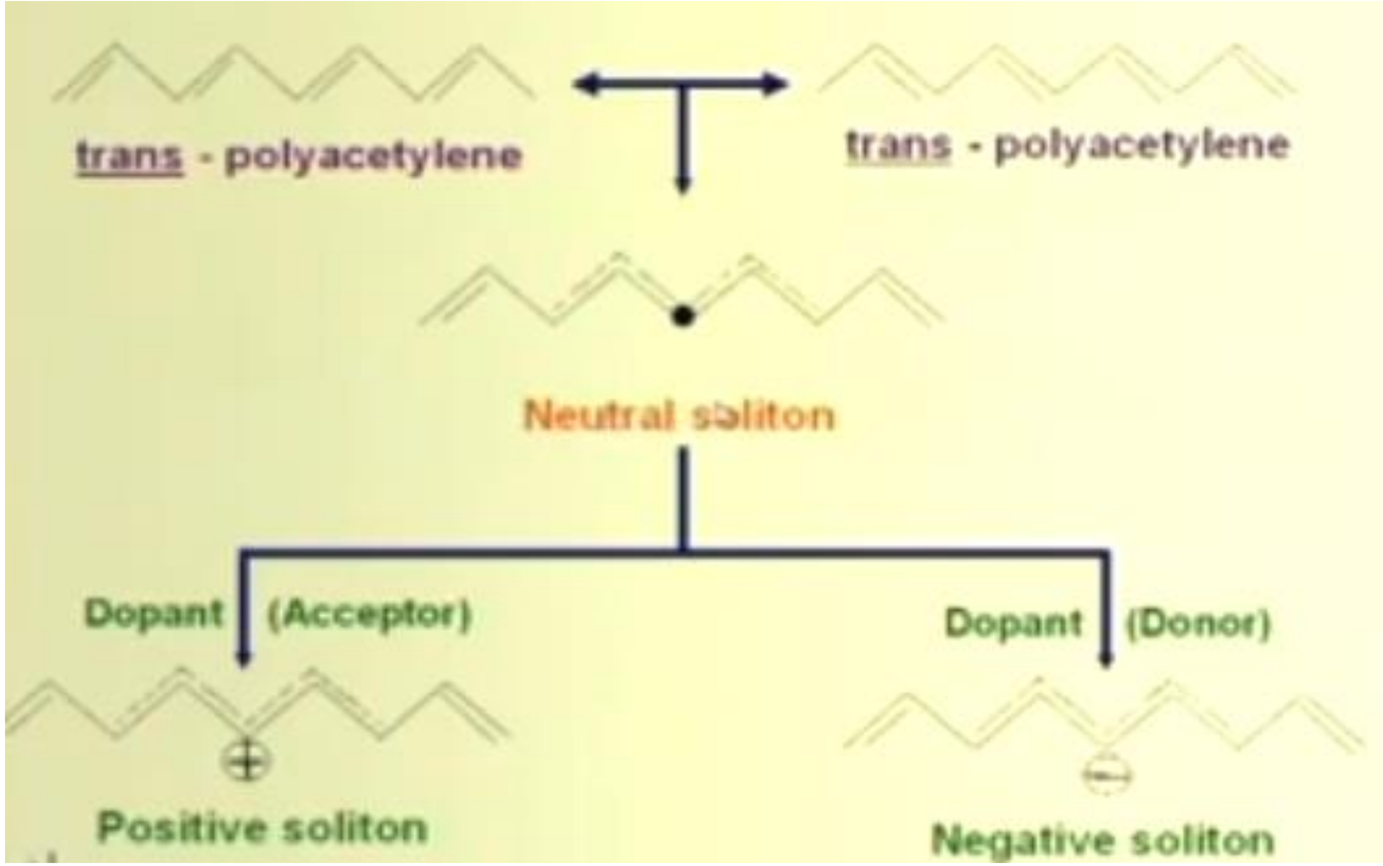
In doped polymers charged solution are formed

Solution ---- charged defect with no spin

A reducing donor type, dopants introduce an electron to the polymer chain which couples with a neutral defect resulting in a negative solution with zero spin

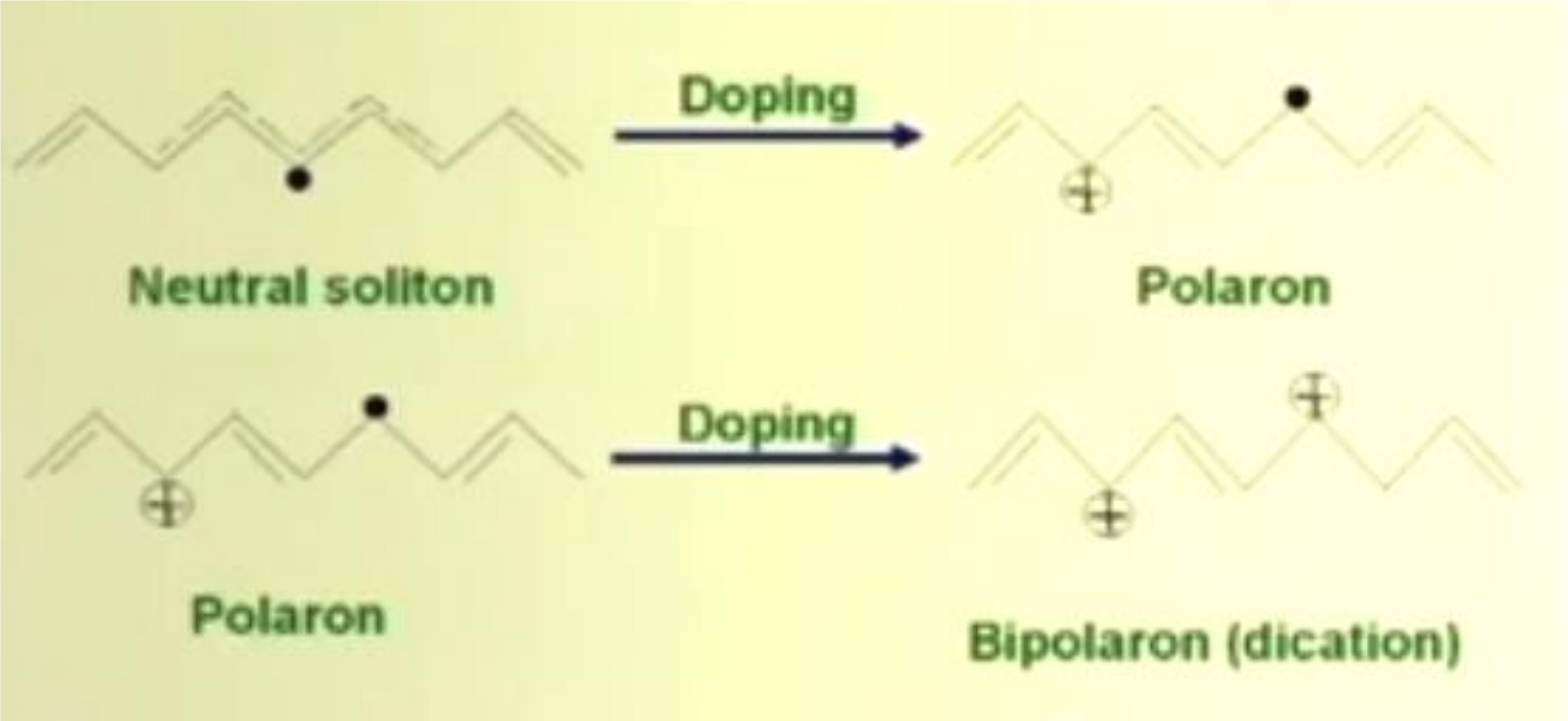
Similarly on oxidising dopant abstracts an electron from the polymer chain and a positive spinless soliton is formed

Sequence of electric events occurring in the polymer chain at a very low doping level



For increased doping level formation of polaron takes place

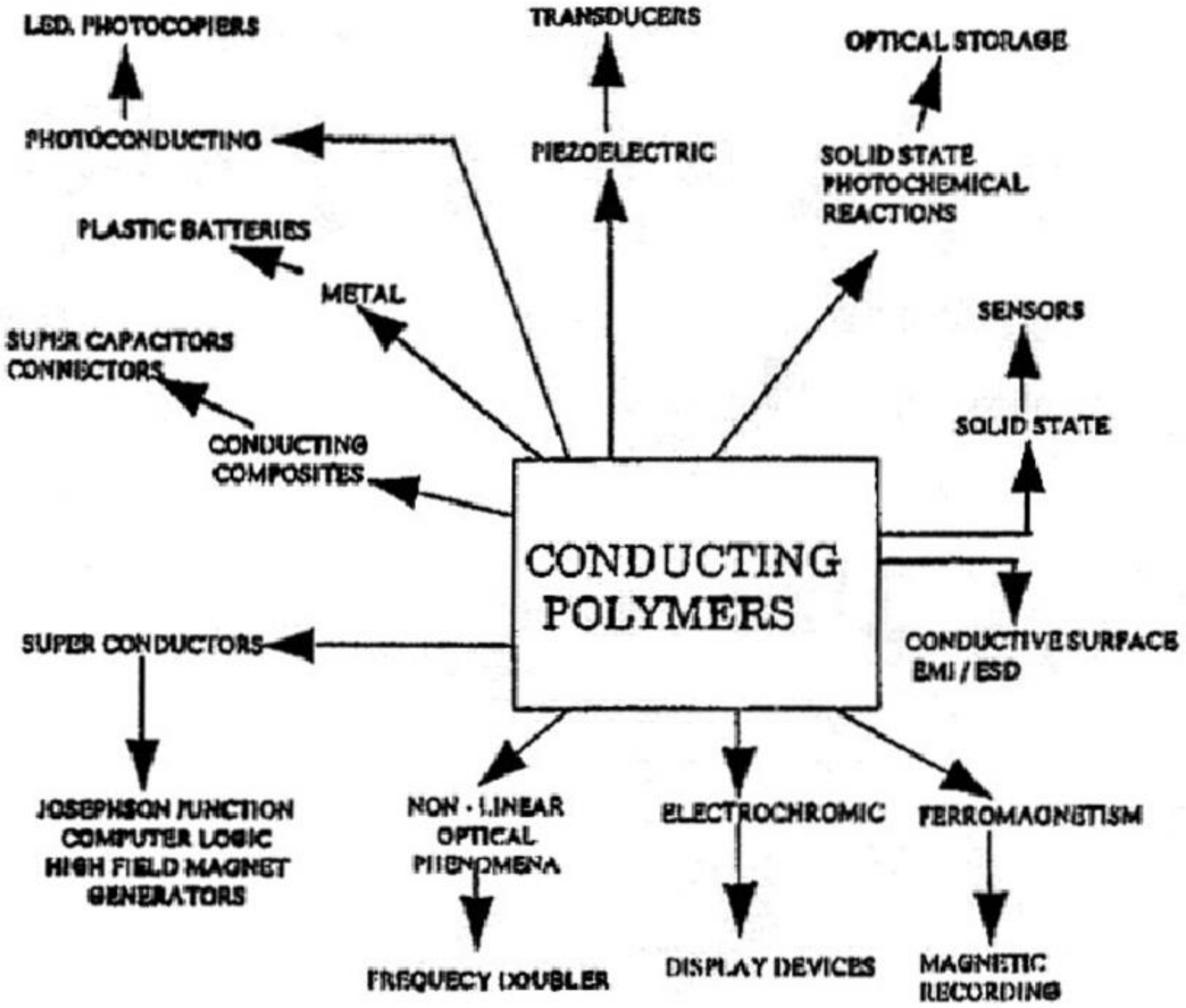
Polaron and bipolaron are the major charge carriers in conducting polymers ex., polyparaphenylene, polypyrrole, polythiophene



SMART WINDOWS

Shield for computer screen
against electromagnetic
"smart" windows
radiation

Smart Windows


LED, PHOTOCOPIERS
TRANSDUCERS
OPTICAL STORAGE
PHOTOCONDUCTING
PIEZOELECTRIC
SOLID STATE
PHOTOCHEMICAL
REACTIONS
PLASTIC BATTERIES
METAL
SENSORS
SUPER CAPACITORS
CONNECTORS
SOLID STATE
CONDUCTING
COMPOSITES
CONDUCTING
POLYMERS
SUPER CONDUCTORS
CONDUCTIVE SURFACE
EMI / ESD
JOSEPHSON JUNCTION
COMPUTER LOGIC
HIGH FIELD MAGNET
GENERATORS
NON - LINEAR
OPTICAL
PHENOMENA
ELECTROCHROMIC
FERROMAGNETISM
FREQUECY DOUBLER
DISPLAY DEVICES
MAGNETIC
RECORDING

Plastics :

Plastics are usually synthetic or semi synthetic organic compounds of very high molecular mass and can be mouled into solid objects of numerous shapes and sizes

Plastics generally include a main chain organic link, side linked molecular groups and some organic and inrorganic blends added as additives, plasticizers, fillers etc

History of plastics

- A new semi-synthetic plastic was unveiled by Alexander Parkes at the 1862 Great International Exhibition in London, England
- Parkesine, was an organic material consisting of cellulose nitrate and a solvent
- Parkesine could be heated formed while it retained its shape when cooled and was moulded into products such as buttons, combs, picture frames and knife candles.

- The first synthetic plastic was discovered in 1907, Dr. Leo H. Bakeland, reacted Phenol and Formaldeyde under pressure using Hexamethylenetetramine as a catalyst for the reaction
- The result was a thermosetting Phenolic Plastic, he named BAKELITE which was electrically resistant, chemically stable, heat resistant, rigid moisture and weather resistant.

Plastic Production process

Additives

Fillers, Plasticizers, Stabilizers, colorants, lubricants, flame retardants

Monomers → Heat, pressure, catalysis → Polymers

Process

Polymerisation condensation or addition

Base polymer + Additives

↓

Compounding (Blender or mixer)

↓

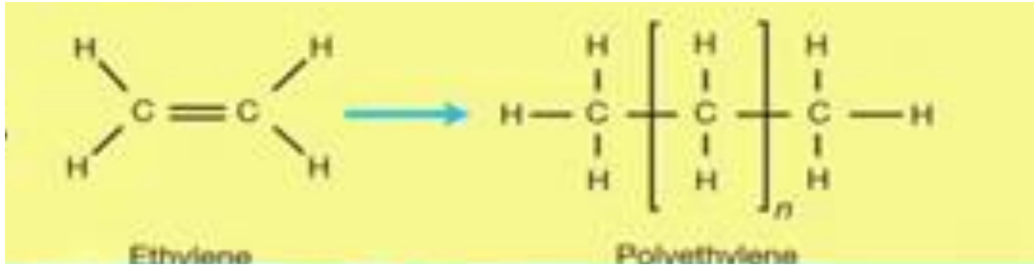
Polymer compound

↓

Processing

↓

Finished product

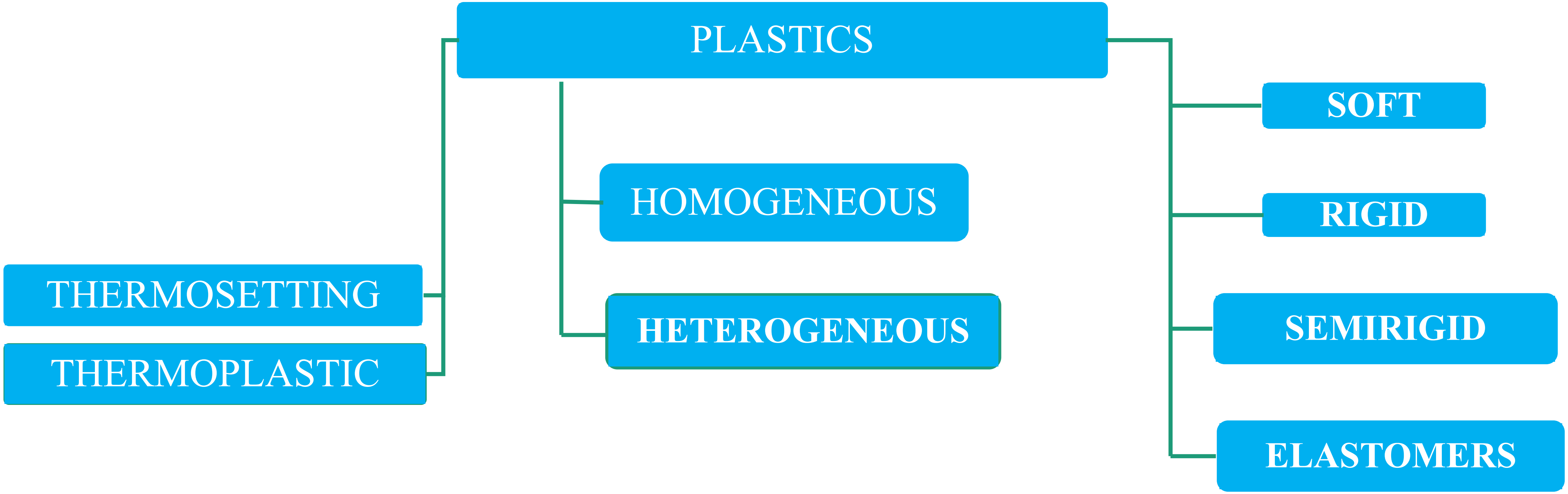


Plastic products

- Fillers: 20-100
- Plasticizers: 30-100
- Stabilizers: 0.02-10.0
- Colouring agents: 2-5
- Lubricants: 1-2
- Flow parameters: 2-15
- Crosslinking agents: 1-5

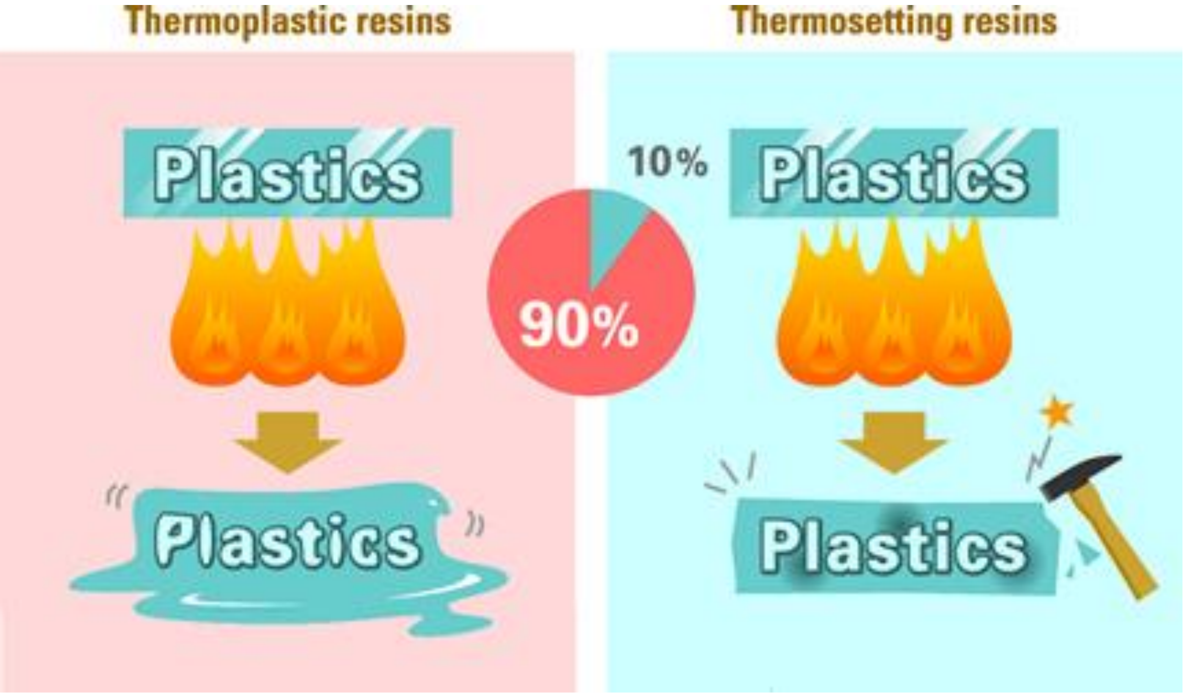
Characteristics of Plastics: **Every type of plastic have very distinct characteristics, but most plastics have the following general attributes**

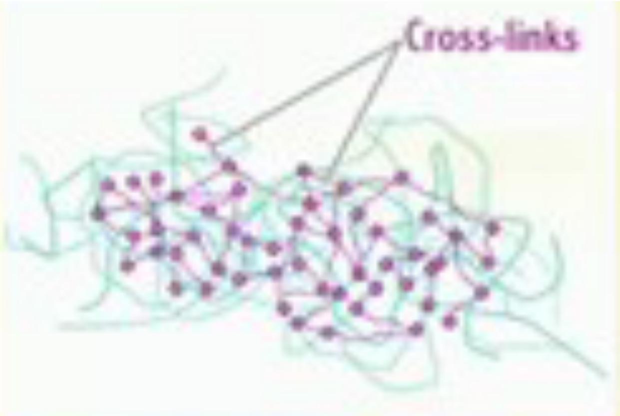
- Can be very resistant to chemical and corrosion
- Can be both thermal and electrical insulators
- Have very high strength to weight ratio
- Can be highly durable, resistant to water
- Plastics are materials with a seemingly limitless range of characteristics and colours and are easy to manufacture.


PLASTICS
THERMOSETTING
THERMOPLASTIC
HOMOGENEOUS
HETEROGENEOUS
SOFT
RIGID
SEMIRIGID
ELASTOMERS

PLASTICS

THERMOPLASTICS | THERMOSETS





Thermoplastics variety softens by heat and harness when cooled down. It can be used by remoulding as may times as required

Thermosetting plastics can not be reused. This variety requires a great pressure and momentary heat during moulding which harden on cooling.

Thermoplastic properties

- It may melt before passing to a gaseous state
- Allow plastic deformation when it is heated
- Chemical compositions do not change on heating
- They are brittle and glossy
- They are soluble in certain solvents
- Good resistance to creep

Examples

- **Polyethylene**
- **Polypropylene**
- **Polystyrene**
- **Acrylics, Teflon**
- **Polycarbonate, nylon, Acrylonitrile butadiene styrene**

Thermoplastic pros

- Highly recyclable
- Aesthetically superior finish
- High impact resistance
- Remoulding/reshaping capabilities
- Chemical resistant
- Hard crystalline or rubbery surface
- Eco-friendly manufacturing

Thermoplastic Cons

- Generally more expensive than thermoset
- Can melt if heated

Thermoset properties

- Soluble in alcohol and certain organic solvents, when they are in thermoplastic stage. This property is utilised for making paints and vanishes from these plastics
- They undergo irreversible chemical process
- They are durable, strong and hard
- These are available in a variety of beautiful colours
- They are mainly used in engineering application of products.

Thermoset pros

- More resistant to high temperature than thermoplastic
- High flexible design
- Thick to thin wall capabilities
- Excellent aesthetic appearance
- High levels of dimensional stability
- Cost effective

Cons

- Cannot be recycled
- More difficult to surface
- Cannot be remoulded or reshaped

Examples

- **Epoxies, polyurethane, Phenolics, unsaturated polyesters, silicones**

Based on structure

Homogeneous

- Plastics are composed only of hydrocarbon atoms and they exhibit a homogeneous structure

Examples

- Polyethylene
- Polypropylene
- Polystyrene

Heterogeneous

- These plastics are composed of the chain containing carbon, hydrogen, oxygen, nitrogen and other elements and they exhibit a heterogeneous structure

Examples

- Polytetraflouroethylene
- Polyamides or nylons
- Polyvinylchloride
- Acrylonitrile butadiene

Based on their physical and chemical properties

Rigid Plastics

These plastics have a high modulus of elasticity and they retain their shape under exterior stresses applied at normal or moderately increased temperature

Ex:HDPE tubes, PP cups, PET Pallets

Semirigid plastics

These plastics have a medium modulus of elasticity and the elongation under pressure completely disappears when pressure is removed

Ex:LDPE films, flexible ducts, PVC sheets

Soft Plastics

These plastics have a low modulus of elasticity and the elongation under pressure disappears when the pressure is removed

Ex: Children`s toy, Rattles, Fishing baits

Elastomers

These plastics are soft and elastic materials with a low modulus of elasticity. They deform considerably under load at room temperature and return to their original shape, when the load is released. The extensions can range up to ten times their original dimensions

Polyisoprene, Polybutadiene, Polyisobutylene

Code 1: PET (polyethylene terephthalate)

Code 2: HDPE (high density polyethylene)

Code 3: PVC (polyvinylchloride)

Code 4: LDPE (low density polyethylene)

Code 5: PP (polypropylene)

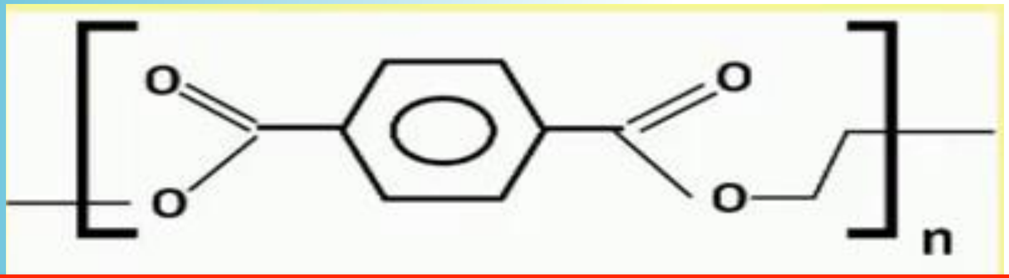
Code 6: PS (polystyrene)

Code 7: Other plastics including polycarbonate, acrylic, liquid crystal polymers (LCP) and nylon.

Polyethylene terephthalate (PET)

Introduced by J.Rex Whinfiled and James T.Dickson in 1940, this plastic is one of the most commonly used on the planet. It took another 30 years before it was used for crystal clear beverage bottles, produced by Coca-cola and Pepsi.

Monomer of PET



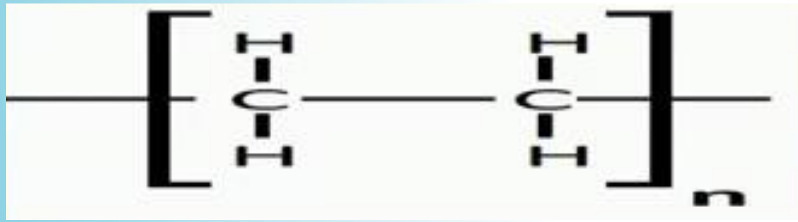
- PET is an excellent moisture and water barrier material
- Oxygen permeability
- Light weight
- Semi rigid
- Robust and impact resistant
- hygroscopic in nature

High Density polyethylene (HDPE)

In 19953, Karl Ziegler and Erhard Holzkamp used catalysts and low pressure to createhigh density polyethylene. It was first used for pipes in storm sewers, drains and culverts. Today this plastic is used for a wide variety of products.

It is the most widely commonly recycled plastic because it will not break under exposure to extreme heat or cold

Monomer of HDPE



- High strength to density ratio; Density can range from 930 to 970 Kg/m3
- It is harder and can withstand somewhat higher temperature (120deg C)
- The physical properties of HDPE can vary depending on the molding process that is used to manufacture a specific sample
- Strong and a dimensionally stable material

Polyvinyl chloride

PVC is one of the oldest synthetic materials in industrial production. It was actually discovered on accident twice, once in 1838 by French physicist Henri victor Regnault and again in 1872 by German chemist Eugin Baumann. On both occasions, these men found it inside vinyl chloride flasks left exposed to sunlight

It has been called the poison plastic because it contains numerous toxins and is harmful to our health and environment

Low density polyethylene (LDPE)

LDPE was the first polyethylene to be produced, making it the godfather of the material. It has less mass than HDPE.

Monomer of LDPE

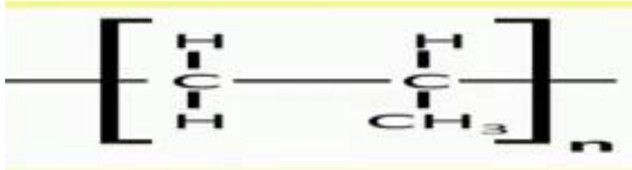
- Density range of 0.917-0.930 g/cm3
- Can withstand temperature of 80deg C continuously and 90 deg C for a short time
- Not reactive at room temperatures, except by strong oxidisingagents and some solvent cause swelling
- Strong and dimensionally stable material
- High resilience

Polypropylene (PP)

J.Paul Hogan and Robert L.Banks of Phillips Petroleum company discovered polypropylene in 1951.

At the time, they were simply trying to convert propylene into gasoline, but instead discovered a new catalytic process for making plastic.

Monomer



- Rigid and opaque
- Flexible and low density
- Electrical and abrasion resistance
- Good dimensional stability at high temperature and humidity
- Excellent chemical resistance
- Weathering resistance

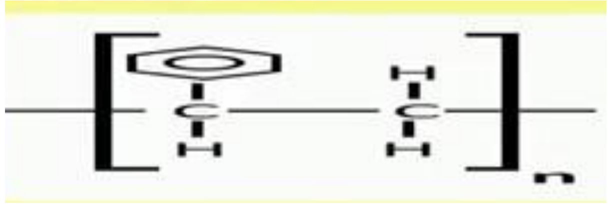
Polystyrene or Styrofoam (PS)

In 1839, German Eduard Simon accidentally came across polystyrene while preparing medication. He isolated a substance from natural resin and didn't realise what he had discovered.

It took German chemist Herman Staudinger to research the polymer and expand on its uses.

Lightweight and easy to form into plastic materials, Breaks effortlessly; More harmful to the environment

Monomer



Rigid, clear

Inexpensive resin

Good strain and abrasion resistance

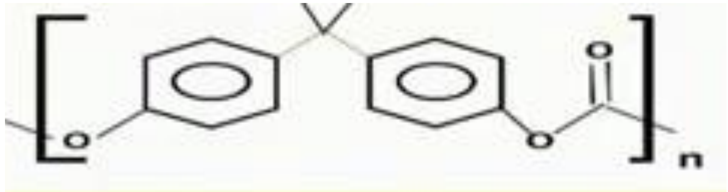
Transparent or can be colored with colorants

- **Other plastics**

Polycarbonate, polyactide, acrlylic, styrene etc…

POLYCABONATES

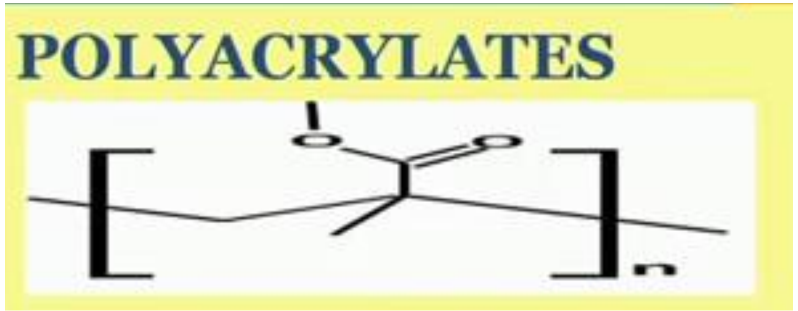
Monomer



- High temperature resistance
- Flame retardant
- Excellent ductility
- Self lubricating
- UV resistant with use of UV stabilizers
- Glass like visibility and transparency

POLYACRYLATES

Monomer



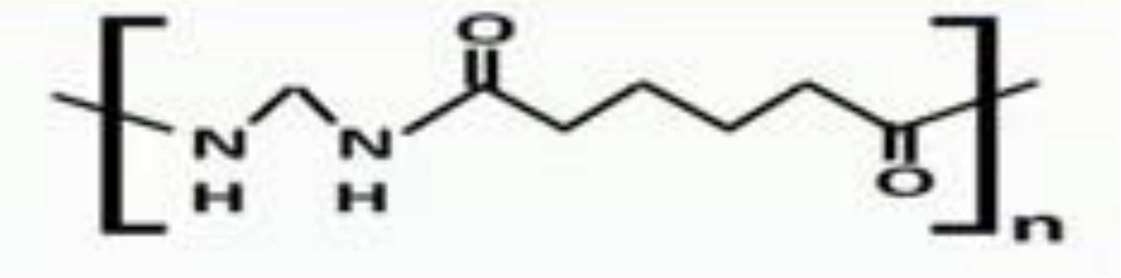
- Optical clarity
- Excellent outdoor material
- Very good tensile strength, flexural strength,
- easy handling and processing
- good impact resistance
- Glass like visibility and transparency

TEFLON (PTFE)

- Chemically inert
- Low friction coefficient
- High temperature resistant
- Excellent lubricant excellent dielectric properties
- High bulk resistivity

POLYAMIDES

- High mechanical strength
- Stiffness, thermal stability
- Tough at low temperature
- Excellent resistance to chemicals
- Low friction, easy processing
- durability



1 PETE **LDPETE:** Polyethylene terephthalate ethylene, used for soft drink, juice, water, detergent, cleaner and peanut butter containers.

2 HDPE **HDLDPE:** High-density polyethylene, used in opaque plastic milk and water jugs, bleach, detergent and shampoo bottles and some plastic bags.

3 V **PVC or V:** Polyvinyl chloride, used for cling wrap, some plastic squeeze bottles, cooking oil and peanut butter jars, detergent and window cleaner bottles.

4 LDPE **LDLDPE:** Low density polyethylene, used in grocery store bags, most plastic wraps and some bottles.

5 PP **PP:** Polypropylene, used in most deli soup, syrup and yogurt containers, straws and other clouded plastic containers, including baby bottles.

6 PS **PS:** Polystyrene, used in Styrofoam food trays, egg cartons, disposable cups and bowls, carryout containers and opaque plastic cutlery.

7 OTHER **Other:** Usually polycarbonate, used in most plastic baby bottles, 5-gallon water bottles, "sport" water bottles, metal food can liners, clear plastic "sippy" cups and some clear plastic cutlery.
New bio-based plastics may also be labeled #7.

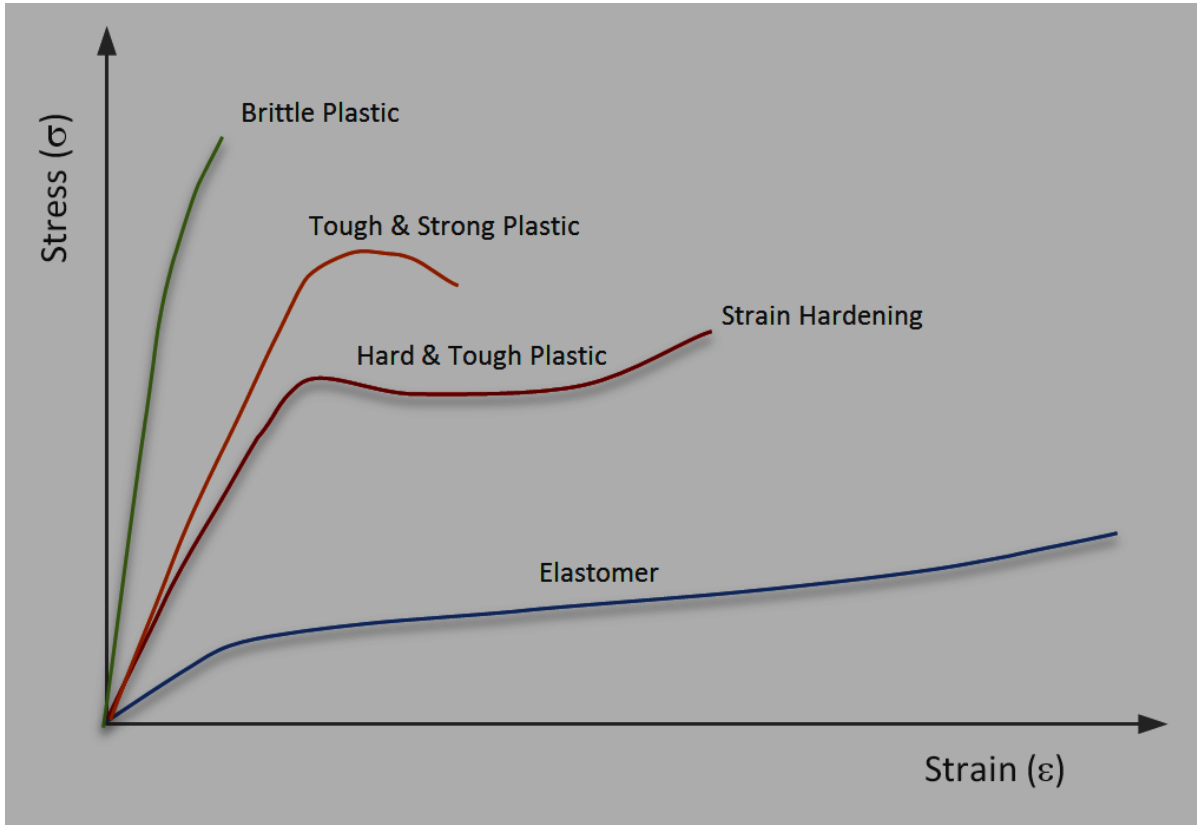
Elastomer

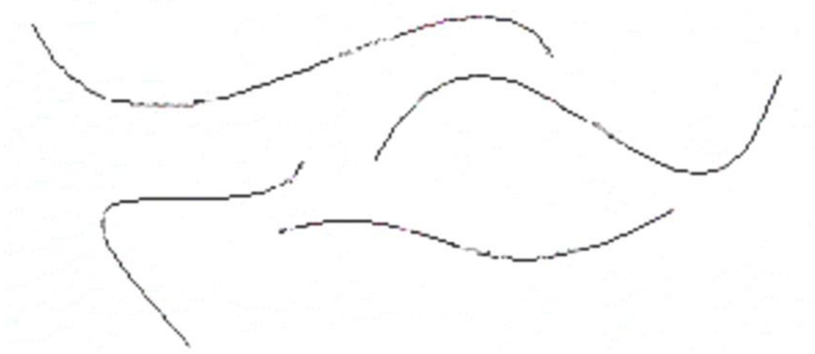
- An elastomer is a polymer with the property of viscoelasticity (colloquially "elasticity"), generally having notably low Young's modulus and high yield strain compared with other materials.
- They can be easily stretched & rapidly recover their original dimensions when applied stress is released
- Large reversible extensions (several hundred percent)
- Low intermolecular forces allow for flexibility
- An Elastomer is a polymeric material that has elongation greater than 100% and a significant amount of resilience
- Highly amorphous materials, High randomly orientated structure
- Rubberoid material

Rubber can be classified as natural rubber and synthetic rubber

Natural rubber is a product of tree HaveaBrasiliensis while synthetic rubber is a elastomer derived from petrochemical feedstock product. [Polyisoprene Styrene Butadiene Rubber (SBR) Chloroprene, Polybutadiene Nitrile rubber, Butyle rubber, Silicones, Ethylene propylene diene monomer (EPDM), etc]

Raw rubber vs Elastomer

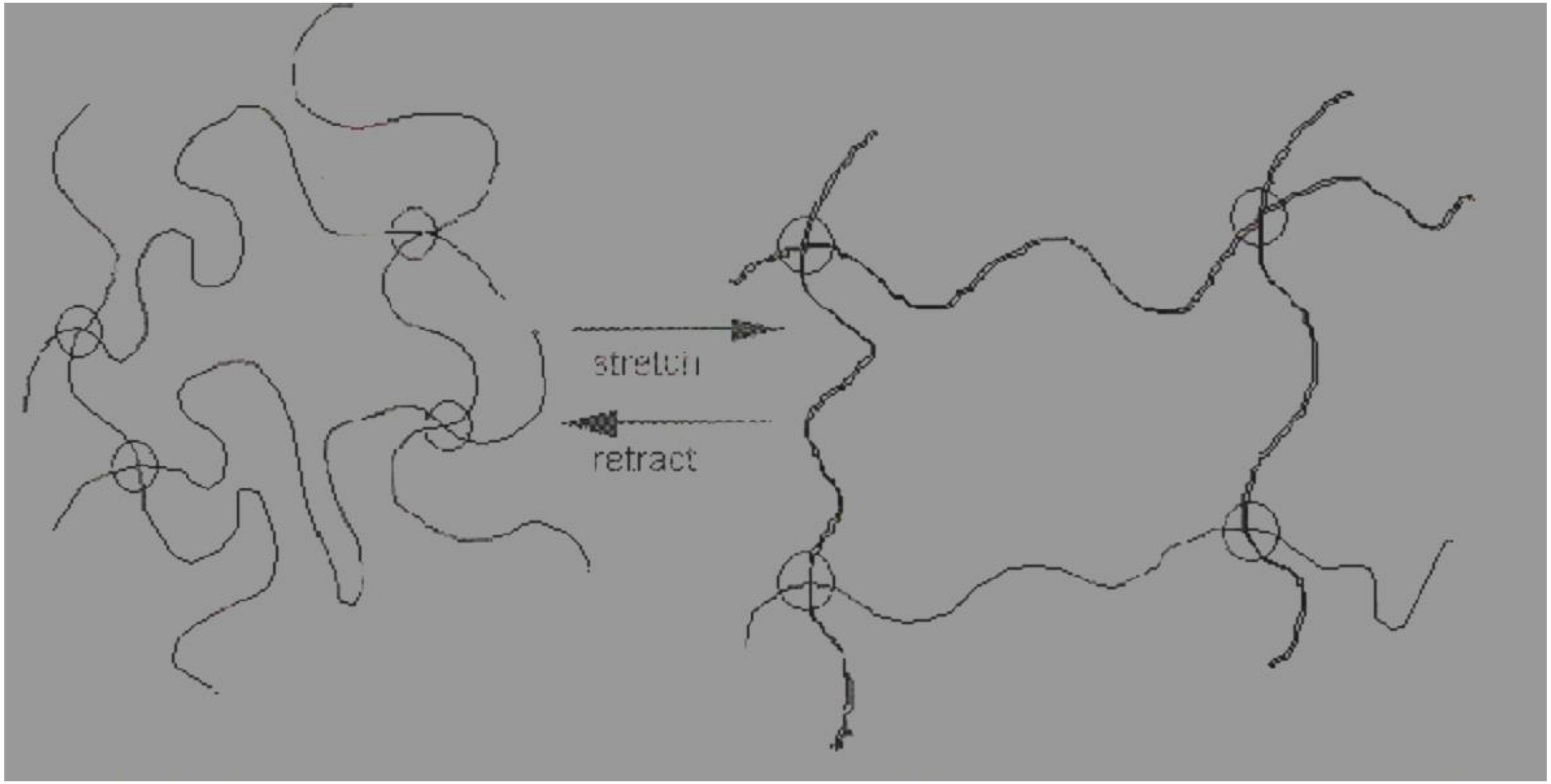




Raw rubber – no crosslinking

Elastomer – crosslinking

Function of crosslinks



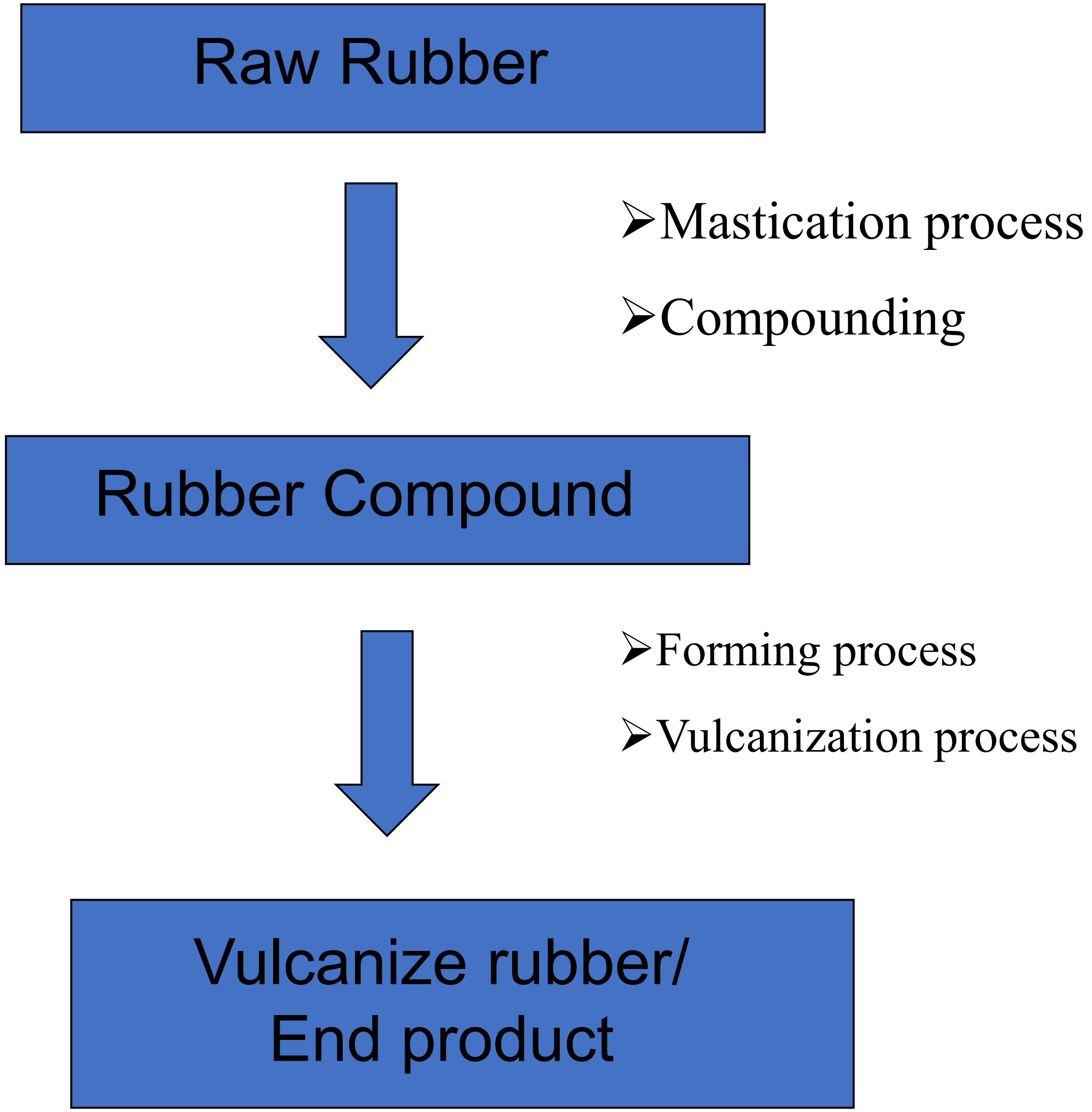
Rubber processing

Compunding

- Incorporation of all ingredients uniformly to the base polymer to form a semi homogeneous mass known as mix of Rubber compound

Major factors

Price, processing, properties



Composition of rubber products Parts per hundred rubber (pphr)	
General class	**Actual components**
Base Polymer	NR, BR, SBR, IIR, CR(cholorophene rubber), EPDM
Fillers	Used for reinforced or modified the mechanical properties and also to reduced the cost Carbon black, Clays, whiting silica fillers,
Processing aids	To ease the processing, to modify the specific properties and also as extender Petroleum oils, waxes
Curatives	Cross linking agents, organic accelerators, organic & inorganic activators, promoters and retarders
Antiageing compounds	Organic compounds as antioxidants and antiozonants
Accelerator	To increase vulcanization process and reduced the time of vulcanization
Activator	To increased the accelerator efficiency
Miscellaneous additives	Coloring agents, abrasives, antistatic compounds etc

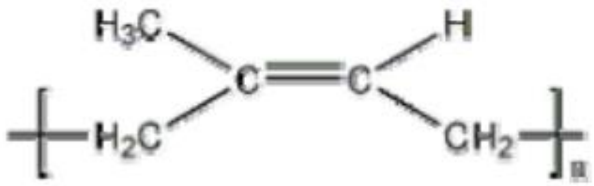
Natural Rubber (NR) cis-1,4-polyisoprene

Commercially significant source of natural rubber is Hevea Brasiliensis.

The biggest producer countries of natural rubber are Thailand, Indonesia, Malaysia and India.

Natural rubber also contains a few percent of non-rubber constituents such as resins, proteins, sugars and fatty acids, which can function as weak antioxidants and accelerators in the natural rubber.

Natural rubber i ur, but also peroxides and isocyanates can be used.



Cis-1,4-polyisoprene.

Applications: • tyres (60 - 70%) • tubes, conveyor belts and V-belts • coatings • gaskets • latex products • footwear • adhesives

Cis-1,4 polyisoprene

Properties of natural rubber

Molecular weight – 1.15x 106; 3.5 x 105

Molecular weight distribution broad

Hightack green & green/gum strength

Good processing and Crystallizing rubber

Vulcanisation properties of rubber

- Tensile strength : 17 to 24 Mpa
- Tensile strength filled with C-black : 24 to 32 Mpa
- Adhesion and wear resistance: Below 25 deg C better than SBR above 35 deg c SBR is better
- Skid resistance: Moderate and improve with OENR
- Dynamic properties: High resilience, at high strain fatigue life superior to that of SBR, compression set and creep : poorer than SBR
- Ageing: poor but improved antioxidants or antiozodants

Advantages of NR:

good processability

excellent elastic properties

good tensile strength

high elongation

good tear resistance

good wear resistance

little dissipation factor - low heat build-up in dynamic stress • excellent cold resistance • good electrical insulator

high resistance to water and acids (not to oxidizing acids)

Disadvantages of NR: • poor weather and ozone resistance • restricted high temperature resistance (short-time maximum temperature 100°C) • swelling in oils and fuels: low oil and fuel resistance • unsuitable for use with organic liquids in general (even though vulcanization considerably improves swelling resistance), the major exception being low molecular weight alcohols

Synthetic rubber

Homo and Co polymers and other difunctional polymers

Poly addition or polycondensation reactions

Why SR?

As a substitute for natural rubber

To meet the demands for some special properties

Ex: Processability

Oil and solvent resistance

Low temperature solubility

Heat resistant properties

Better dynamic properties

Abrasion resistance

Environmental stability

Synthetic rubber

GENERAL PURPOSE AND NON OIL RESISTANT RUBBERS

Styrene butadiene

Polyisoprene rubber

Polybutadiene rubber

Isobutylene-iso prene (butyl) rubber

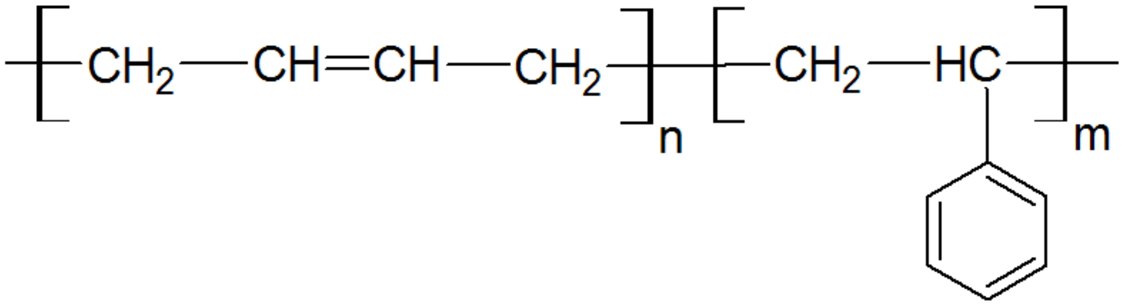
Ethylene propylene rubbers

Thermoplastic rubbers

Styrene butadiene

Emulsion SBR: Hot process (50 deg C) Cold process 95-10 deg C)

Solution SBR: Better grades with controlled microstructur



Properties Styrene butadiene

Properties of emulsion polymerization and solvent polymerization.

	Emulsion - SBR	*Solvent-SBR*
Molar mass M_n, g/mol	145000	200000
Molar mass M_w, g/mol	651000	420000
M_w / M_n	4.5	2.1
Styrene content, %	23.5	18
Cis-1,4-content, %	18	35
Trans-1,4-content, %	65	54
1,2-content, %	17	11
Glass transition temperature T_g, °C	- 50.6	- 69.7

Type designation according to numeric code:

• 10xx hot polymer without filler

• 12xx solution - SBR

• 15xx cold polymer without filler

• 16xx cold polymer, carbon black master batch

• 17xx hot polymer, oil-extended

• 18xx cold polymer, carbon black/oil master batch

• 19xx emulsion- resin- master batch

• 'xx' indicates viscosity, coagulant, content of styrene

Advantages of SBR rubbers:

• good abrasion and aging resistance

• good elasticity

• low price

Applications:

car tyres (blended with BR, IR and NR)

• footwear

conveyor belts, hoses

toys

• moulded rubber goods

• sponge and foamed products

• waterproof materials, belting

• adhesives

Comparison between the properties of SBR and NR.

5 = excellent, 4 = very good, 3 = good, 2 = fair, 1 = poor

	Styrenebutadiene-rubber SBR	*Natural rubber NR*
Hardness, °IRH	40....90	30...90
Tensile strength at break, *N/mm²*	7....25	7...28
Elongation at break, *%*	100...600	100...700
Operating temperature range		
- maximum, °C	100	80
- minimum, °C	- 45	- 55
Elasticity	5	5
Resistance:		
- weather and ozone	1...2	1...2
- abrasion	4	4...5
- radiation	2...3	2...3

SBR needs more reinforcement than natural rubber to achieve good tensile and tear strength and durability. SBR also has lower resilience than NR.

Fillers

Reinforcing : Carbon black, Silica

Non-Reinforcing: CaCo3, TiO2, clays and related minerals such as china clays, soft clays, hard clays, calcined clays etc, MgCO3, talc

Processing aids

Peptizers

Pepton 22 Dio-benzamido phenyl disulfide

Pepton 44 Activated Dio-benzamido phenyl disulfide

Renacit IV Zn Salt of pentachlorothiophenol

Renacit VII Pentachlorothiopheno with activating and dispersing additives

Accelerators: Aldehyde amines, Thiazoles, Sulfenamides, dithiocarbamates, tetramethylthiuram disulfide (TMTD, dithiodimorpholine (DTDM, Thiuram sulfides, Xanthates

Accelerator Modifiers- Activators

Metal Oxides: ZnO, hydrated lime, Litharge, Red lead, MgO, alkali carbonates and metal hydroxide

Organic acids: fatty acids

Alkaline substances: Ammonia, diethanol and triethanol amine, amine salts, reclaim rubber etc.

Retarders - PVI

N-Hexo cyclohexyl thiophthalimide

Pthalic anhydride

N-nitrosodiphenyl amine

Antidegradants

Antioxidants : Hindered Phenol, amino phenol, hydroquinone, phasphite,Diphenyl amine, Napthayl amine, Phenylene diamine

Antiozonants: Dialkyl phenylene diamine, Nickel dibutyldithiocarbonate

Waxes

Vulcanization

Batch curing methods

Autoclave or steam pan

Gas curing

Oven curing

Water curing

Cold curing

	Material	Chemical Group	Generally Resistant to	Generally Attacked by
NR, IR	Natural rubber, Isoprene	Polyisoprene	Most moderate wet or dry chemicals, organic acids, alcohols, ketones, aldehydes	Ozone, strong acids, fats, oils, greases, most hydrocarbons
SBR, BR	Butadiene, Styrene Butadiene	Styrene, Butadiene Copolymer, Polybutadiene	Similar to natural rubber	Similar to natural rubber
IIR	Butyl	Isobutylene, Isoprene, polymer	Water and steam	Petroleum solvents, coal, tar, solvents, aromatic hydrocarbons
EPM, EPDM	Ethylene Propylene	Ethylene Propylene copolymer and terpolymer	Water, steam and brake fluids	Mineral oils and solvents, aromatic hydrocarbons
NBR	Nitrile	Butadiene, Acrylonitrile copolymer	Many hydrocarbons, fats, oils, greases, hydraulic fluids, chemicals	Ozone, ketones, esters, aldehydes, chlorinated and nitro hydrocarbons
HNBR	Hydrogenated nitrile	Butadiene, Acrylonitrile copolymer	Similar to NBR but with improved chemical resistance and higher service temperature	Ozone, ketones, esters, aldehydes, chlorinated and nitro hydrocarbons

Processing of polymers

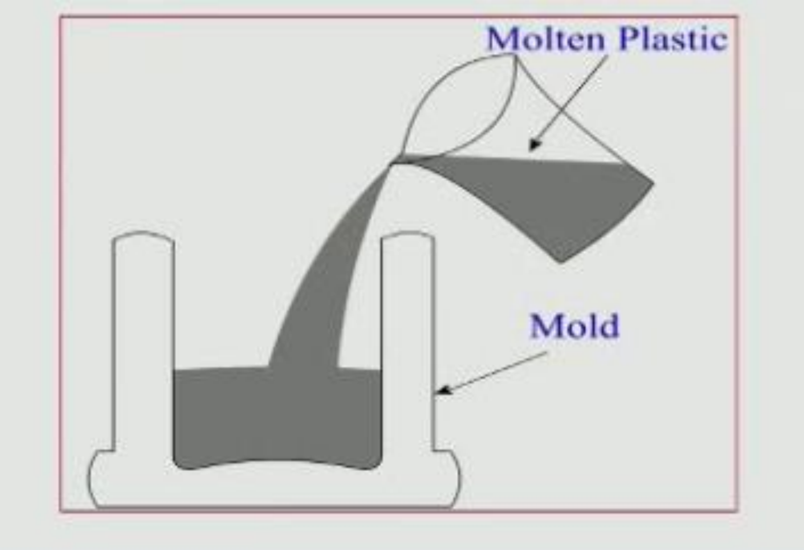
- Polymers are synthetic and semi-synthetic organic solids and can be easily processed
- The processing stages of polymers are heating, forming and cooling in continuous or repeated cycles
- The manufacturing processes of polymer products depends on the shape of the final product and the types of plastic material being used

Processing techniques for polymers

Casting	Compression molding
Rotational molding	Extrusion
Injection molding	Thermoforming
Transfer molding	Blow molding

Casting

- Casting is the process of manufacturing a plastic part into a mold
- Plastic material is melted and its poured into the mold cavity
- As plastic cools, it will solidify into the desired shape and part is ejected.



Processing Parameters

- Meting temperature of the polymer
- Pouring temperature
- Cooling rate

Adv:

- Low initial investment cost
- Flexible parts can be made
- Dimensional stability
- High production rate

Dis. Adv:

- Complex shape cant be produced
- This is more labor intensive

Application:

- Customized toys like designer toys, model of objects like trins, aircraft, boat and ships

THERMOFORMING PROCESS

It is a plastic processing technique in which the thermoplastic sheets are formed with the application of heat and pressure in a mold

Different types

- **Vacuum forming**
- **Pressure forming**
- **Matched die forming**

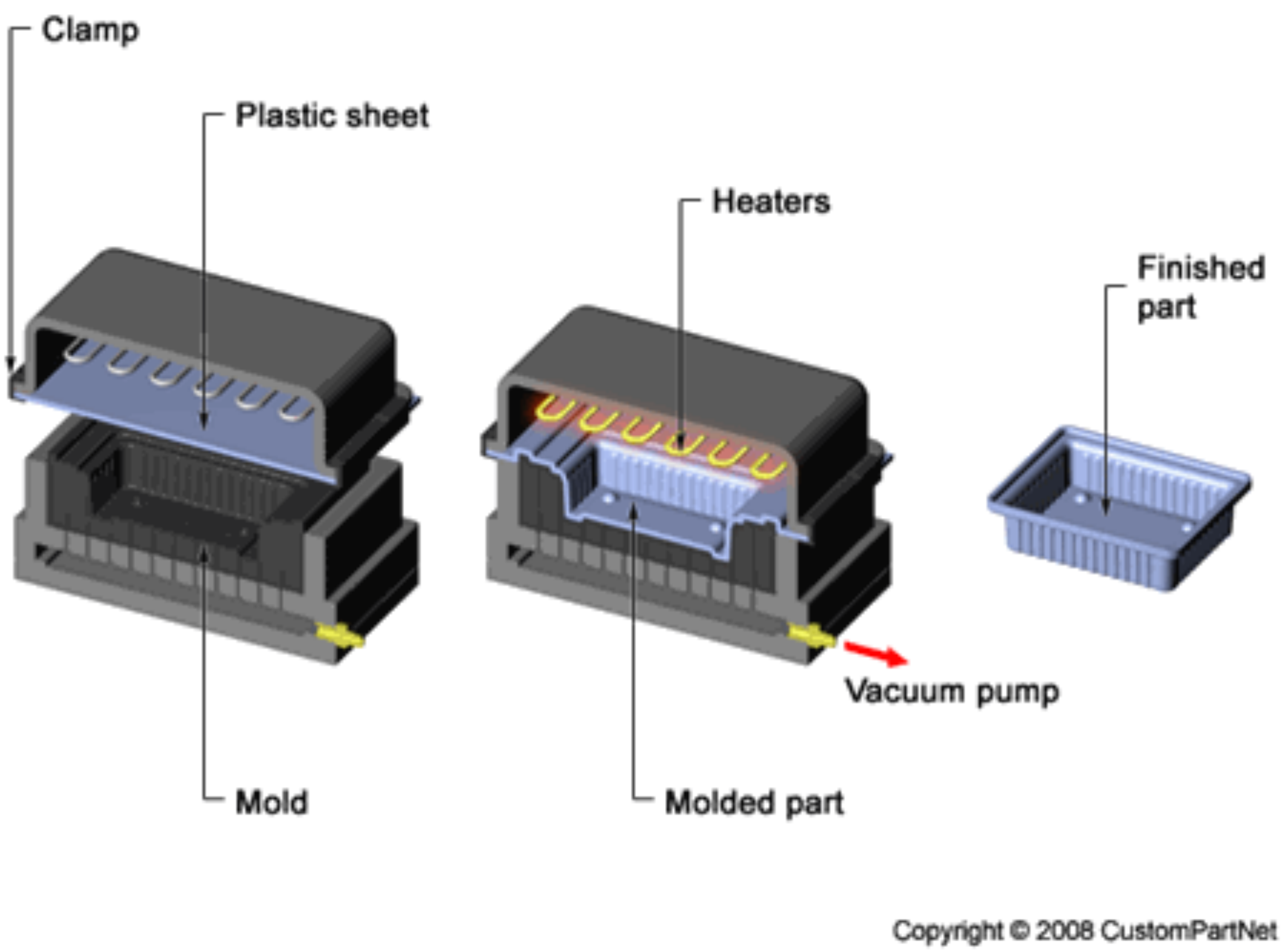
Vacuum forming

(1) A flat plastic sheet is softened by heating

(2) **soften sheet is placed over the mold cavity**

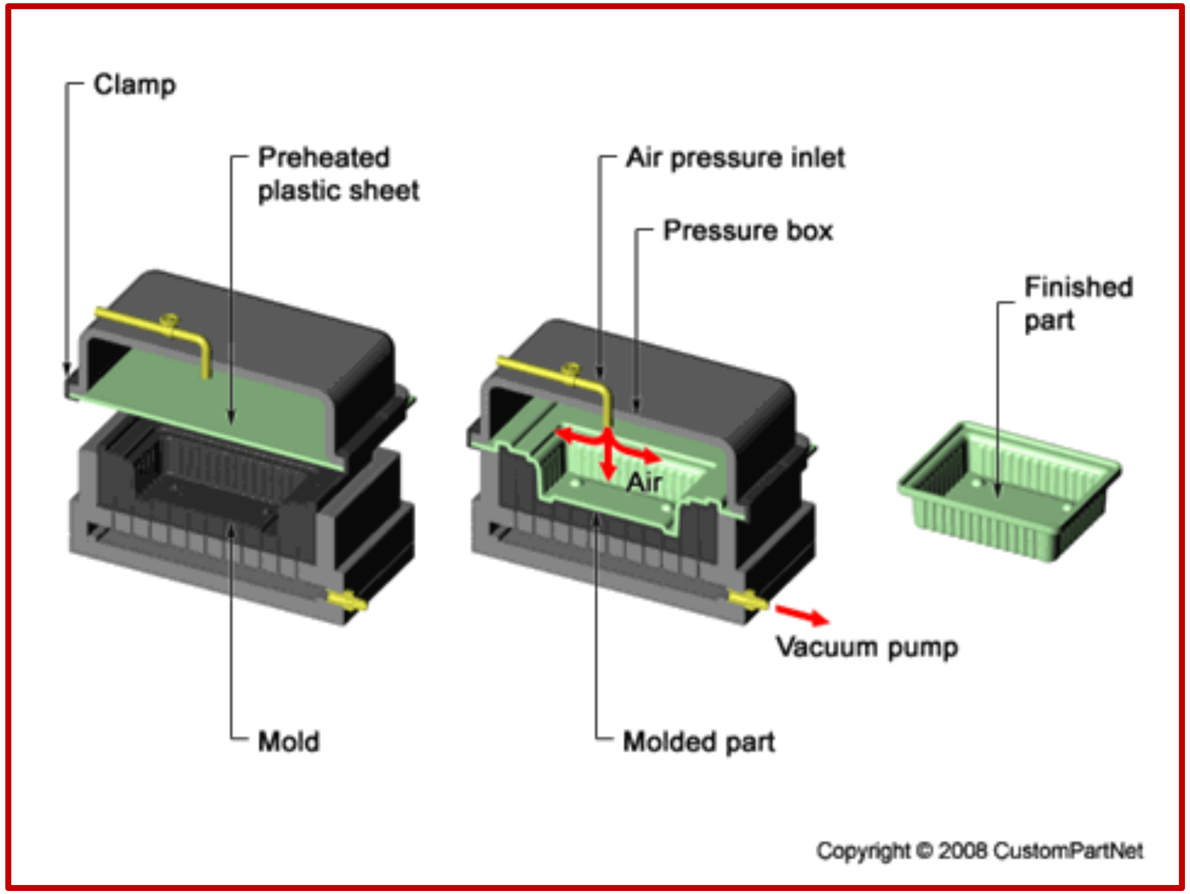
(3) *a vaccum draws the sheet into the cavity*

(4) **after cooling and solidification, part is removed and subsequently trimmed**



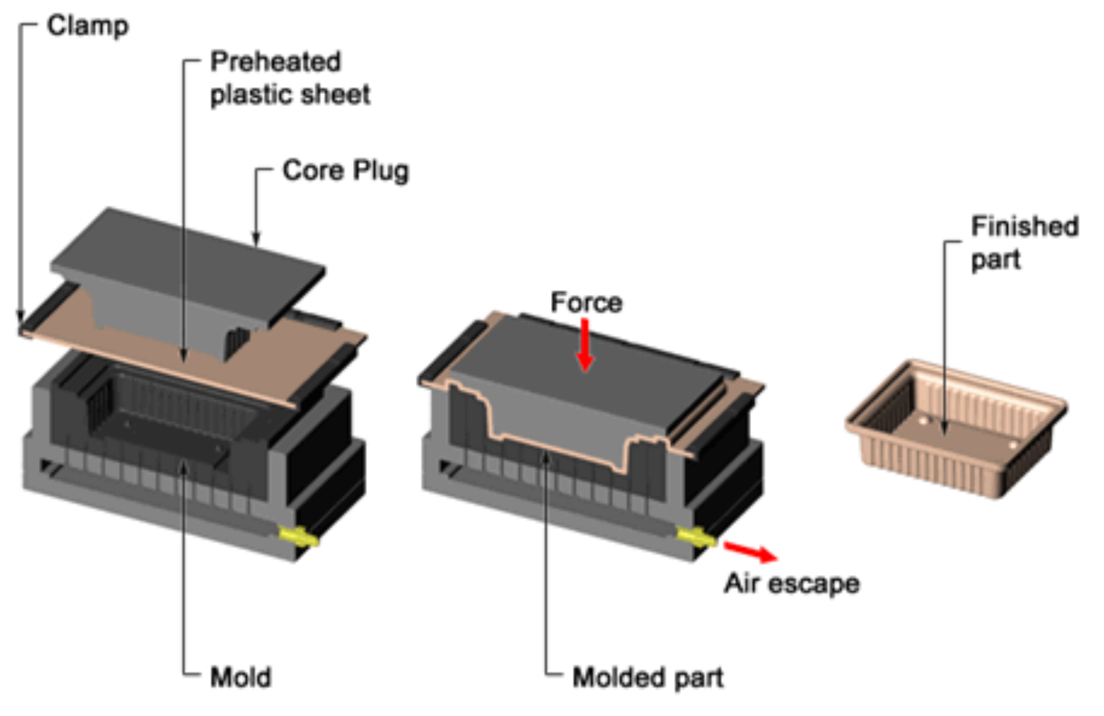
Pressure forming

- The pressure forming is closely related to vacuum forming
- The air pressure required is much higher as compared to the vacuum forming
- The preheated plastic sheet is placed on the mold surface and then air pressure is applied quickly above the sheet
- The high pressure is developed in between the softened sheet and the pressure box
- Due to high pressure, the preheated plastic sheet can be deformed into the mold cavity in a fraction of a second
- The formed part thereby solidifies and is ejected from the mold cavity



Matched die forming

- Matched die forming is also called mechanical forming
- Mold consists of two parts die and punch
- TP sheet is heated with the application of heat until it softens
- Preheated sheet is placed into the die and through punch pressure is applied on the sheet
- Air in between the die and sheet is evacuated by using vacuum pump and sheet conforms to the mold shape
- Formed part is cooled and ejected from th emold.



Materials used

Different types of thermoplastics which can be processed using thermoforming process are

- LDPE, PMMA
- HDPE
- PP
- PS
- ABS

ADV:

- Design flexibility
- Rapid prototype development
- Low initial setup costs
- Low production cost
- Less thermal stresses
- Good dimensional stability

DISADV:

- Poor surface finish
- parts may have nonuniform wall thickness
- All parts need to be trimmed
- Very thick plastic sheets can not formed

Applications

- Food packaging
- Automotive parts
- Trays
- Aircraft windscreens

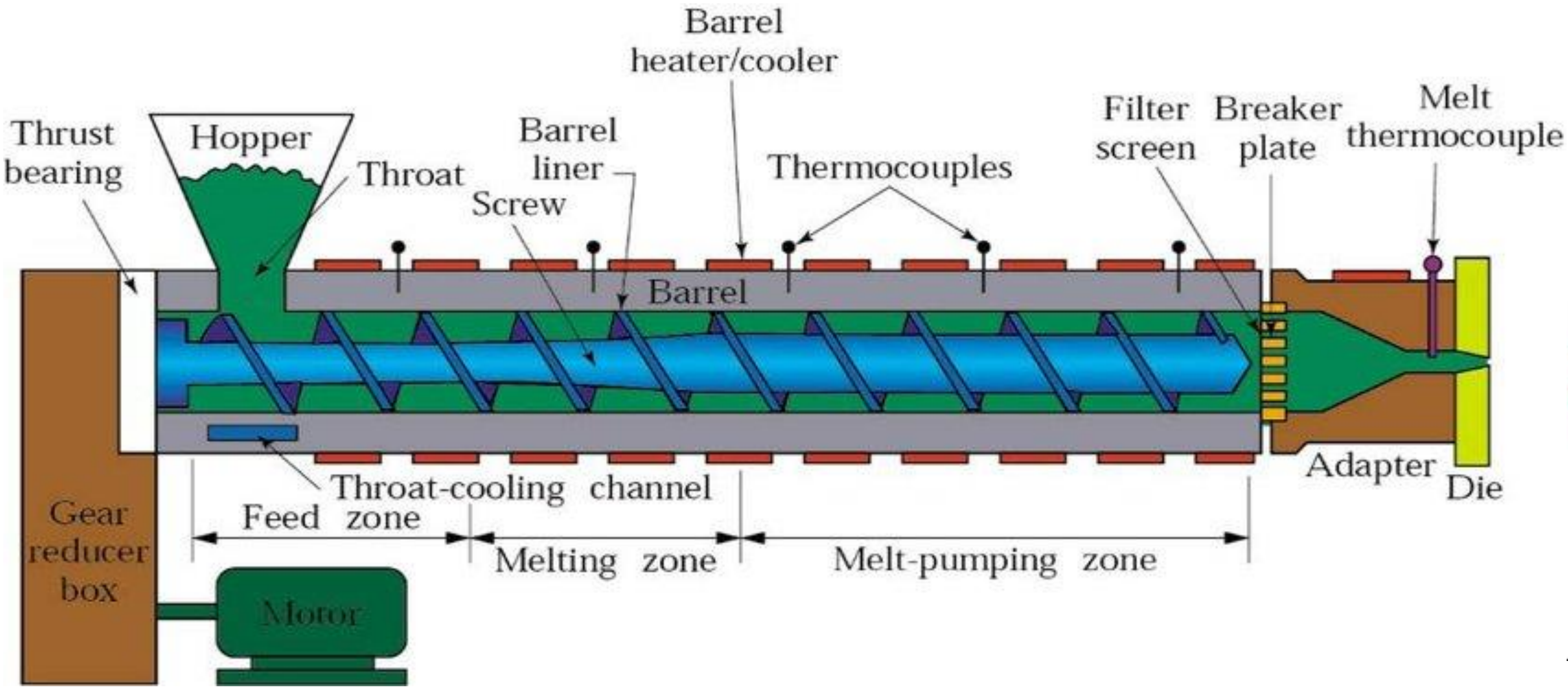
Automotive Headlight Covers

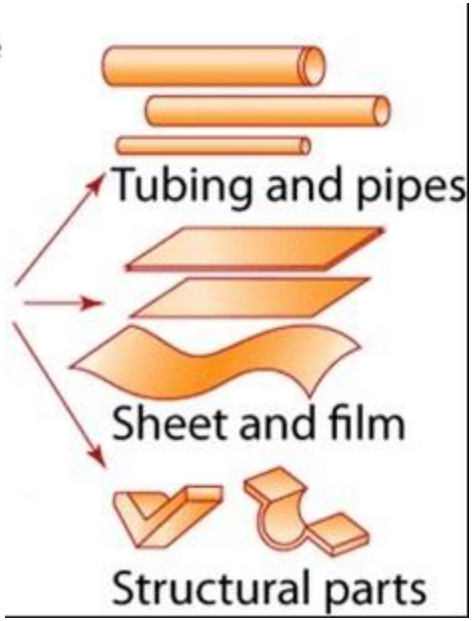
Custom Exterior Door Panels

Aircraft windscreens-PMMA

Extrusion

- High volume manufacturing process
- Plastic material is melted with the application of heat and extrude through die into a desired shape
- A cylindrical rotating screw is placed inside the barrel which forces out molten plastic material through a die.





Extrusion process

- Plastic material in the form of pellets or granules is gravity fed from atop mounted hopper into the barrel of the extruder.
- Additives such as colorants and UV inhibitors can be mixed n the hopper.
- Plastic material enters through the feed throat and comes into contact with the rotating screw.
- The rotating screw pushes the plastics beads forward into the barrel which is heated by using heating elements upto melting temperature of the plastic.

- The plastic material is completely melted in the melting zone.
- A thermostat is used to maintain the inside temperature of the barrel.
- The overheating of plastics should be minimized which may cause degradation in the material properties.
- A cooling fan or water cooling system is used to maintain the temperature of the barrel during the process.
- At the front o the barrel, the molten plastic leaves the screw and travels through a screen pack to remove any contaminants in the molten plastic.
- The screens are reinforced by a breaker plate.

- The back pressure gives uniform melting and proper mixing of the molten plastic material into the barrel.
- After passing through the breaker plate molten plastic enters into die.
- An uneven flow of molten plastic would produce unwanted stress in the plastic product.
- These stresses can cause warping after solidification of molten plastic.
- Plastics are very good thermal insulators and therefore it is very difficult to cool quickly.
- The cooling of plastic product is achieved by pulling through a set of cooling rolls.

SCREW DESIGN

- feeding mechanism
- uniform melting and mixing of the plastic (melting zone)
- push the molten material through the die (metering zone)

PROCESS PARAMETERS

- **Melting temperature of plastic**
- Speed of the screw
- Pressure
- **Type of die used**
- **Cooling medium**

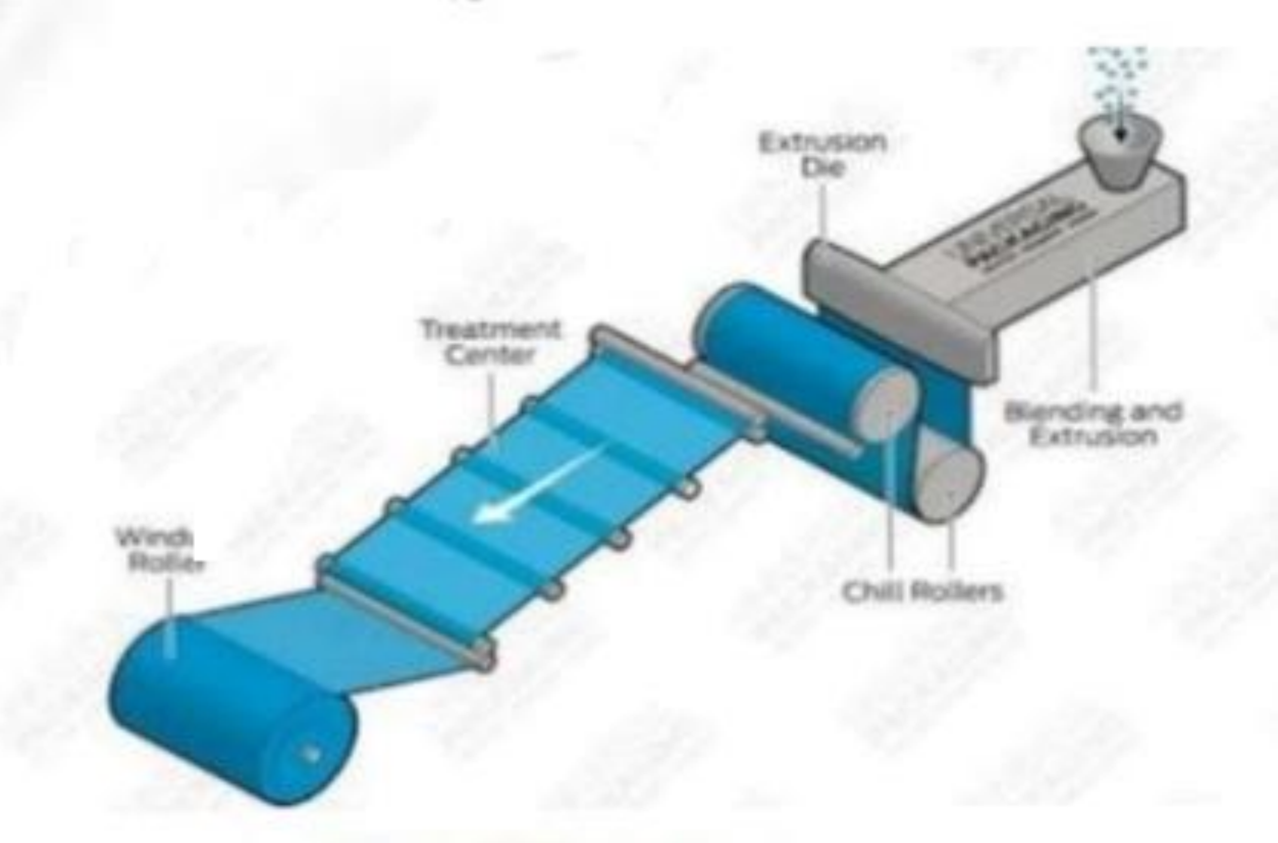
Types of extrusion processes

- Sheet/ film
- Blow film
- Over jacketing
- Tubing
- Co-extrusion
- Extrusion coating
- Compound extrusion

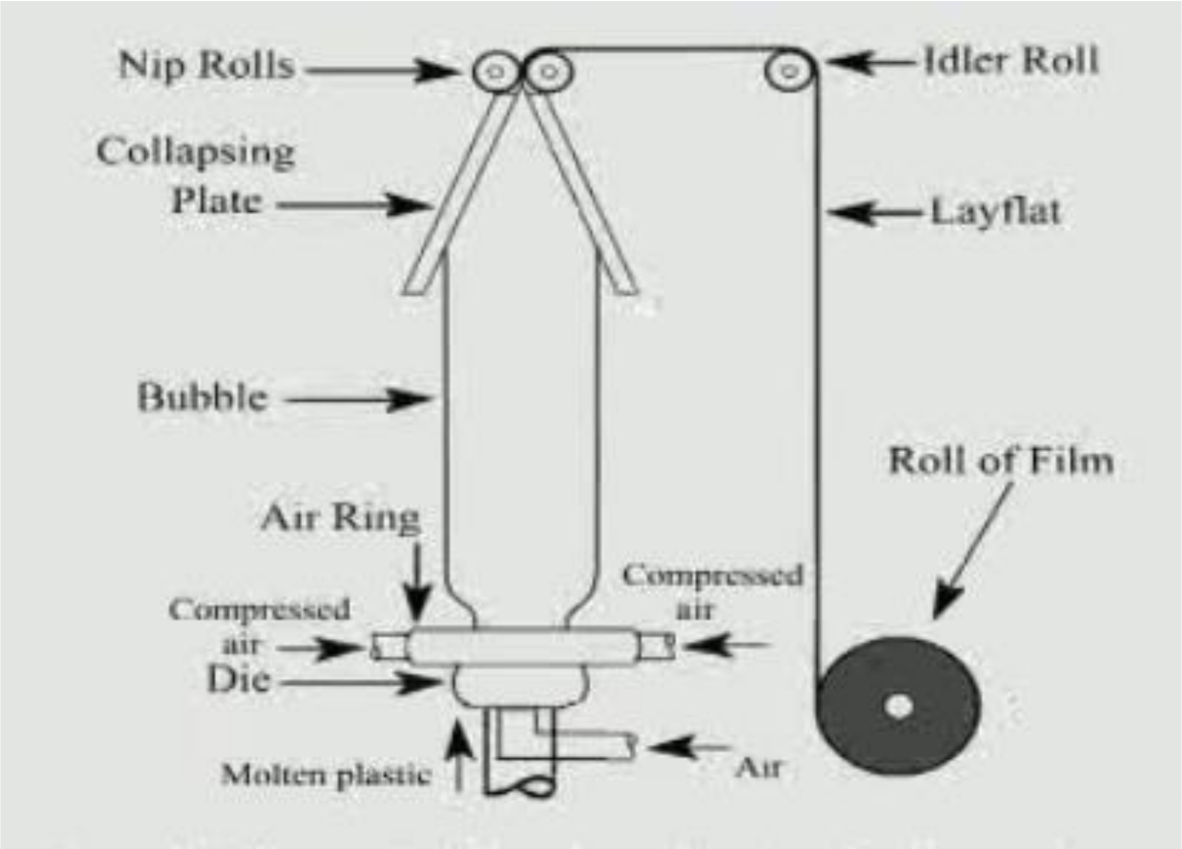
https://www.ptonline.com/knowledgecenter/profile-extrusion/profile-extrusion-fundamentals/history-and-fundamentals-of-extrusion

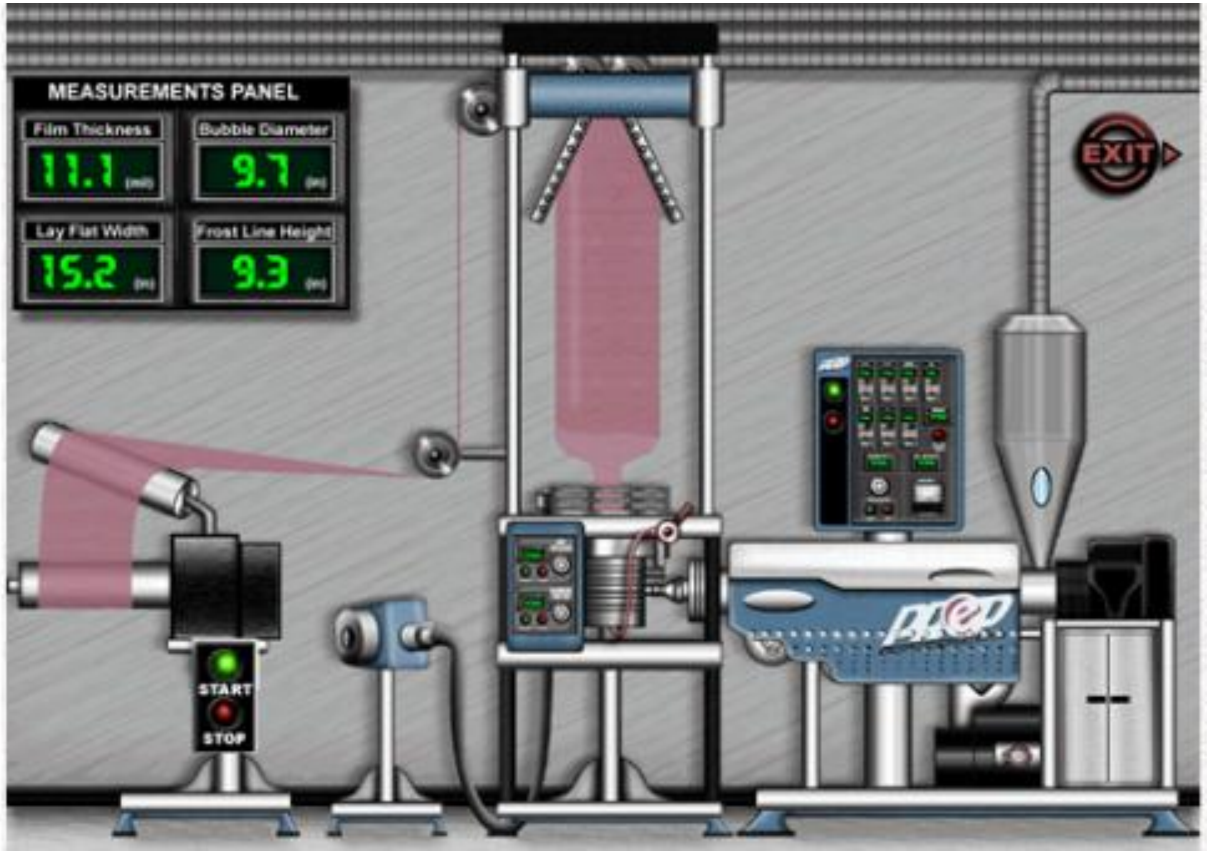
Sheet/ film extrusion

- The molten plastic is extruded through a flat die
- The cooling rolls are used t o determine the thickness of sheet/film and its surface texture.
- Thickness: 0.2 – 5 mm
- Generally polystyrene plastic is used as a raw material in the sheet extrusion process.



Blown film extrusion

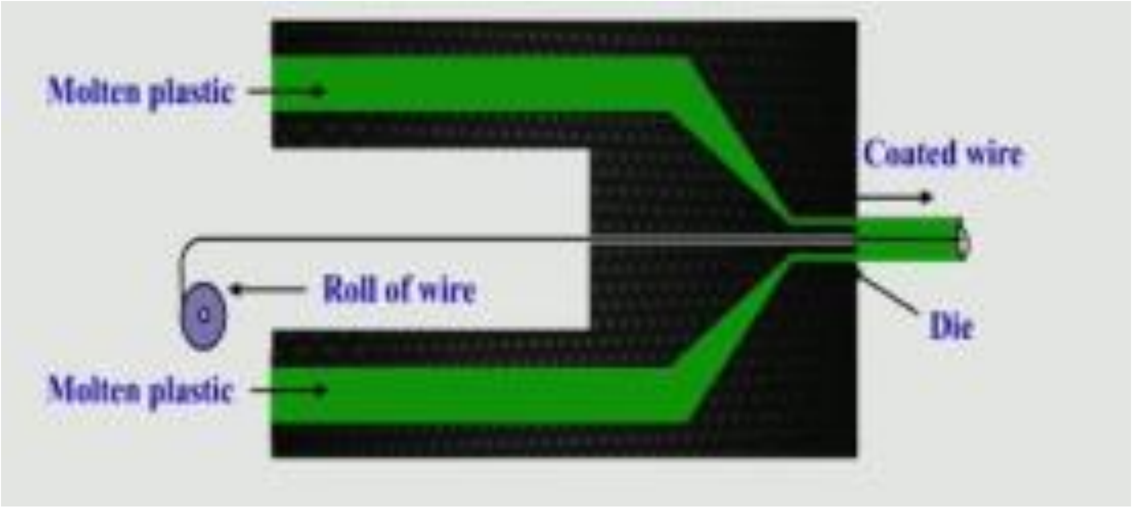




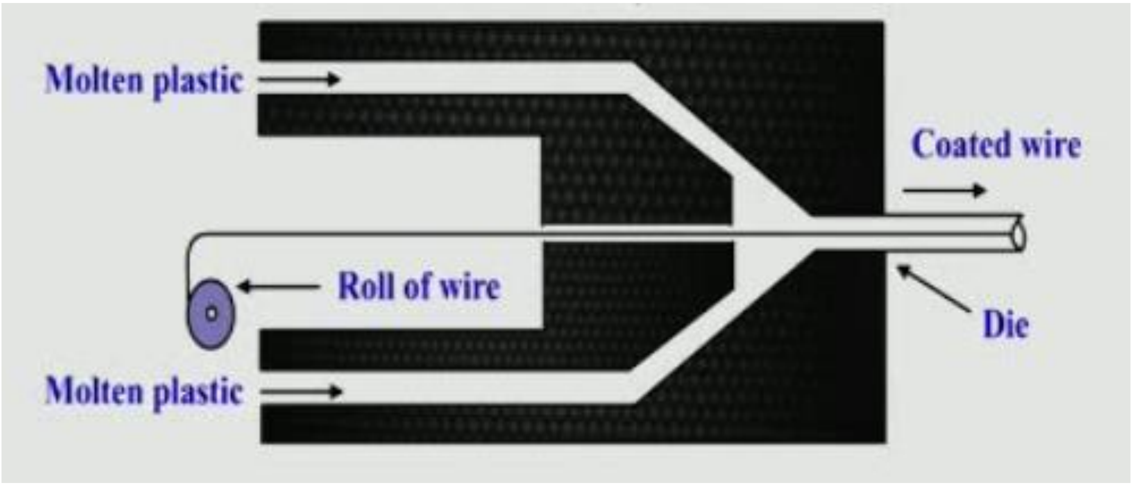
Blown film extrusion

- The die is like a vertical cylinder with a circular profile.
- The molten plastic is pulled upwards from the die by a pair of nip rollers.
- The compressed air is used to inflate the tube around the die.
- In the center of the die there is an air inlet from which compressed air can be forced into the center of the circular profile and hence tube creating a bubble.
- The bubbles are collapsed with the help of collapsing plate.
- The nip rolls flatten the bubble and then passes through the lay flat.
- The wall thickness of the film can be controlled by changing the speed of the nip rollers.
- The lay flat can be spooled in the form of roll or cut into desired shapes.

Jacketing coating

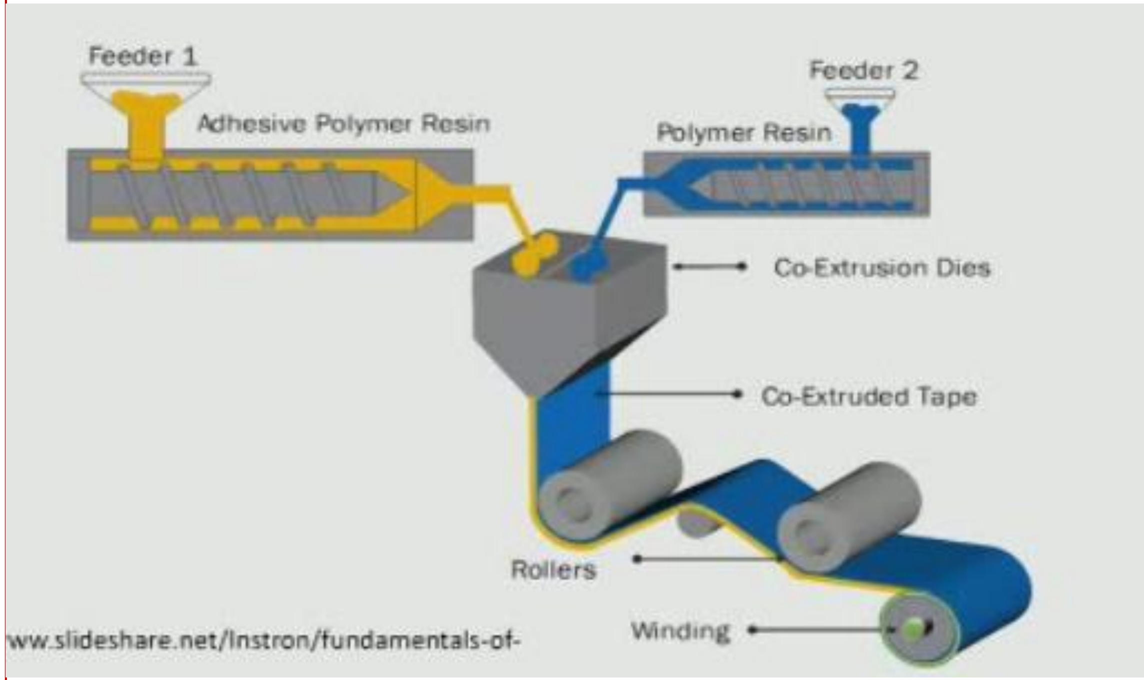


Pressure coating



Co-Extrusion

- Co-extrusion allows different materials to be extruded together
- Multiple colors can be co-extruded together.
- Multiple types of materials can be combined to maximize properties It is used to apply one or more layers on top of the base material to obtain specific properties such as UV absorption, grip and reflection, while base material is more suitable for other applications ex, impact resistance and structural performance.
- It may be used on any of the processes such as blown film, over jacketing, sheet/film extrusion.
- Two or more extruders are used to deliver materials which are combined into a single die that extrudes the materials in the desired shape.



The layer thickness is controlled by the speed and size of the individual extruders delivering the materials.

Materials Used

Different types of plastic material can be used in extrusion process are

Polyethylene, Polypropylene, Acrylic, Nylon, Polystyrene, Polyvinyl Chloride, ABS and Polycarbonate.

Advantages

- High production volume
- Relatively low cost
- Design flexibility
- Coating of wires can be done
- Continuous part can be produced.

Dis adv:

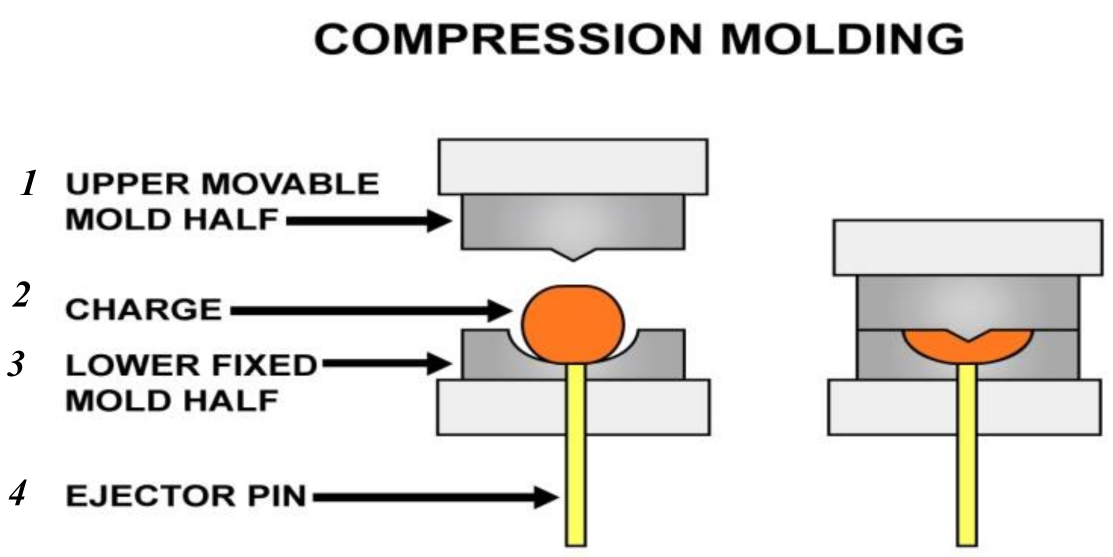
- Limited complexity of parts
- Uniform cross section can only be produced.

Applications

Rods, Plates and tubes, wire and cable coating, hose liners, sheets, multilayer film, food packaging etc.

Compression molding

- It is of the low cost molding methods as compared to injection molding and transfer molding
- It is a high pressure forming process in which the molten plastic material is squeezed directly into a mold cavity, by the application of heat and pressure to conform to the shape of the mold.



- In compressing molding of thermosets the mold remains hot throughout the entire process, as soon as a molded part is ejected, a new charge can be introduced.
- On the other hand, unlike thermosets, thermoplastics must be cooled to harden. So before a molded part is ejected, the entire mold must be cooled and as a result, the process of compression molding is quite low with the thermoplastics.
- Compression molding is thus commonly used for thermosetting plastics such as phenols, urea, melamine
- However in special ceases, such as when extreme accuracy is needed thermoplastics are also compression molded.

Mold design : Open flash, Fully positive, Semi-Positive

Open flash

- In an open flash mold a slight excess of molding powder is loaded into the cavity.
- On closing the top and bottom planes, the excess material is forced out and flash is formed.
- The flash blocks the plastic remaining in the cavity and causes mold plunger to exert pressure on it.

Fully positive

- In the fully positive mols no allowance is made for placing excess charge I the cavity.
- If excess charge is loaded, the mold will not close, an insufficient charge will result in reduced thickness of the molded article.
- A correctly measured charge must therefore be used with this mold, it is a advantage of the mold.
- Another disadvantage is t hat gases liberated during the chemical curing reaction are trapped inside and may show as blisters on the molded surface.

Semi Positive

- The semipositive mold combine certain features of the open flash and fully positive molds and it makes allowance for excess charge and flash.
- They are more expensive to manufacture and maintain than the other types, but they are much better from an applications point of view.

Process parameter

- Amount of plastic material
- Heating time
- Melting temperature of plastic material
- Process required to squeeze the material into the mold cavity
- Cooling time

Materials used:

- Epoxy, Polyester – TS
- PE, PP - TP

ADV:

- Low initial setup cost and fast setup time
- Heavy plastic parts can be molded
- Good surface finish of the molded parts

Limitation

Low production rate

Limiteed largely to flat or moderately curved parts with no undercuts.

Application

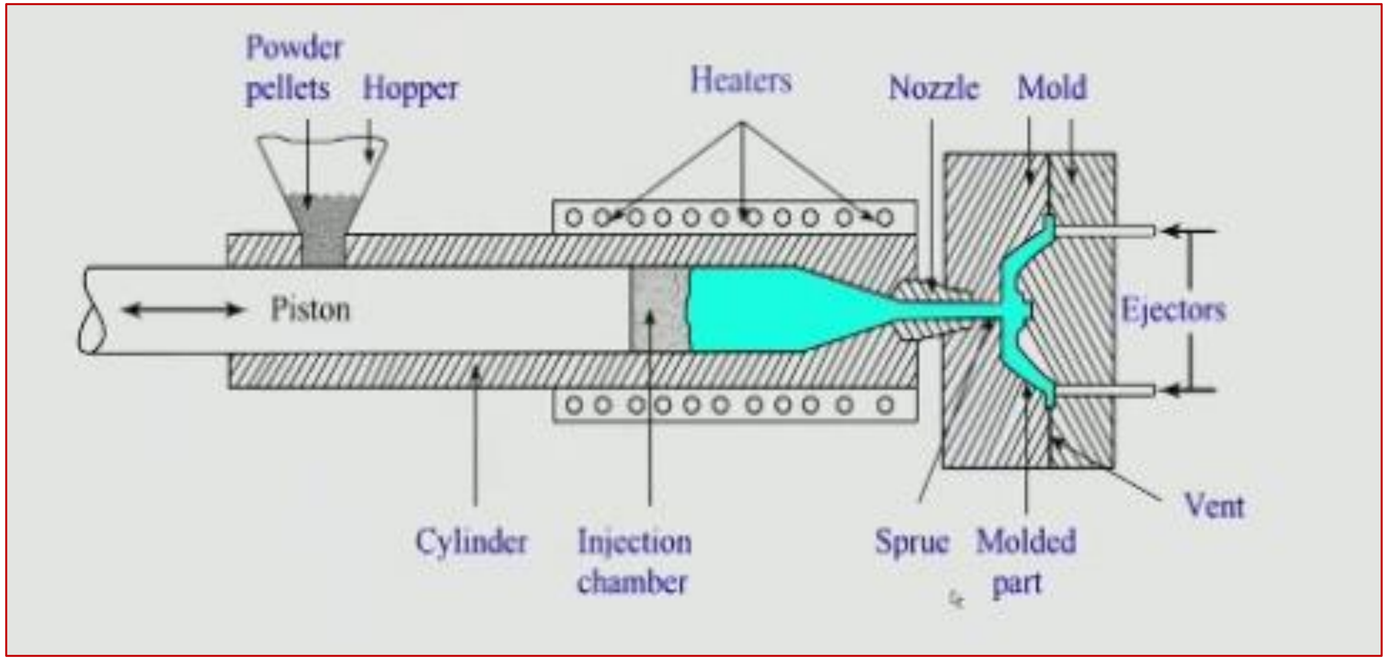
- Electrical and electronic equipments
- Brush and mirror handle
- Trays
- dinnerware
- Pot handles

Injection Molding

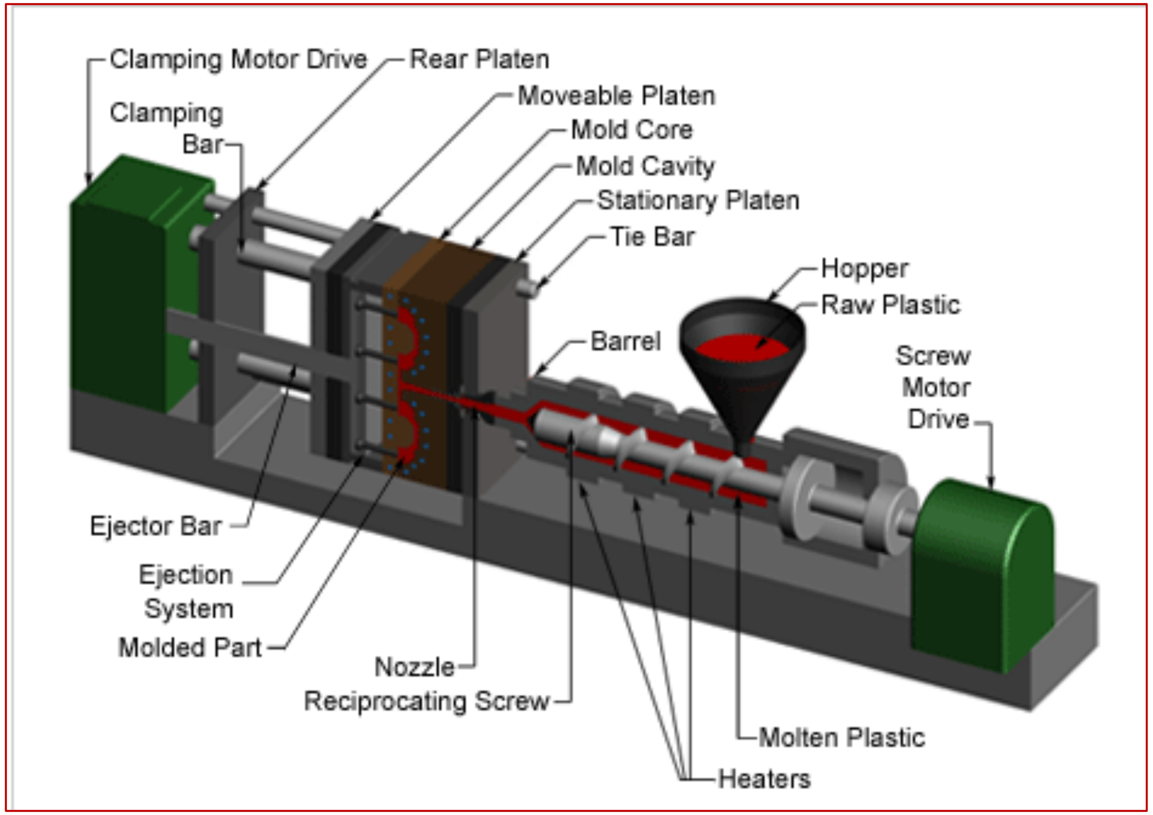
- It is one of the most commonly used processing techniques for the plastic components.
- It is used to manufacture thin walled plastic parts for a wide variety of shapes and sizes.
- Plastic material is melted in the heating chamber and then injected into the mold, where it cools and finally the finished plastic part is ejected.
- Three types: Hand injection, plunger type and reciprocating screw

Hand injection Molding Machine

Plunger type injection molding setup



Rotating Screw injection molding setup



Process

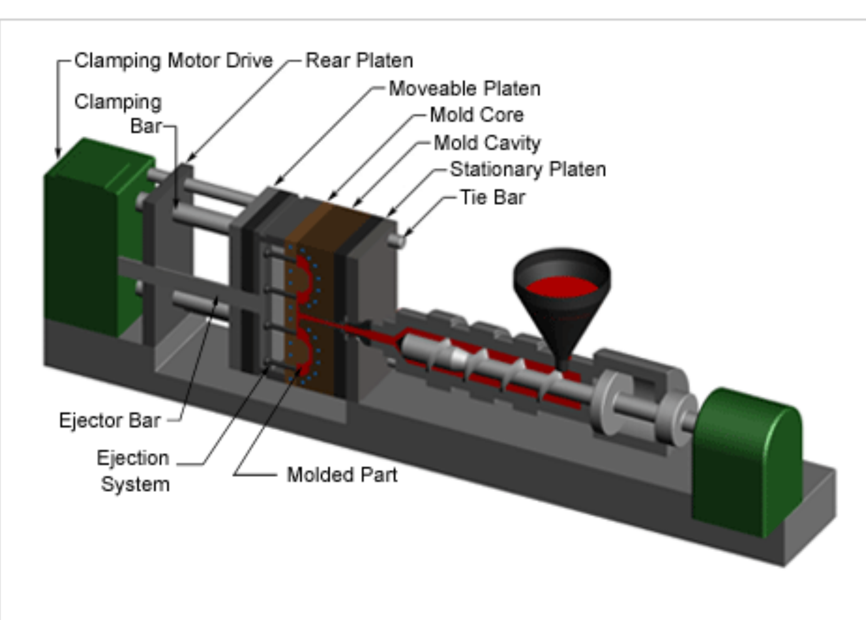
- Plastic materials usually in the form of powder or pellets are fed from hopper into the injection chamber.
- The piston and reciprocating screw arrangements is used to forward t eh material inserted from the hopper into the injection chamber.
- The material is heated in the injection chamber with application of heating elements.
- The molten plastic material is then injected into the mold through the nozzle
- The molded part is cooled quickly in the mold.
- Final plastic part is removed from the mold.
- The process cycle for injection molding is very short, typically between 2 to 60 seconds.

Four stages

- Clamping
- Injection
- Cooling
- Ejection

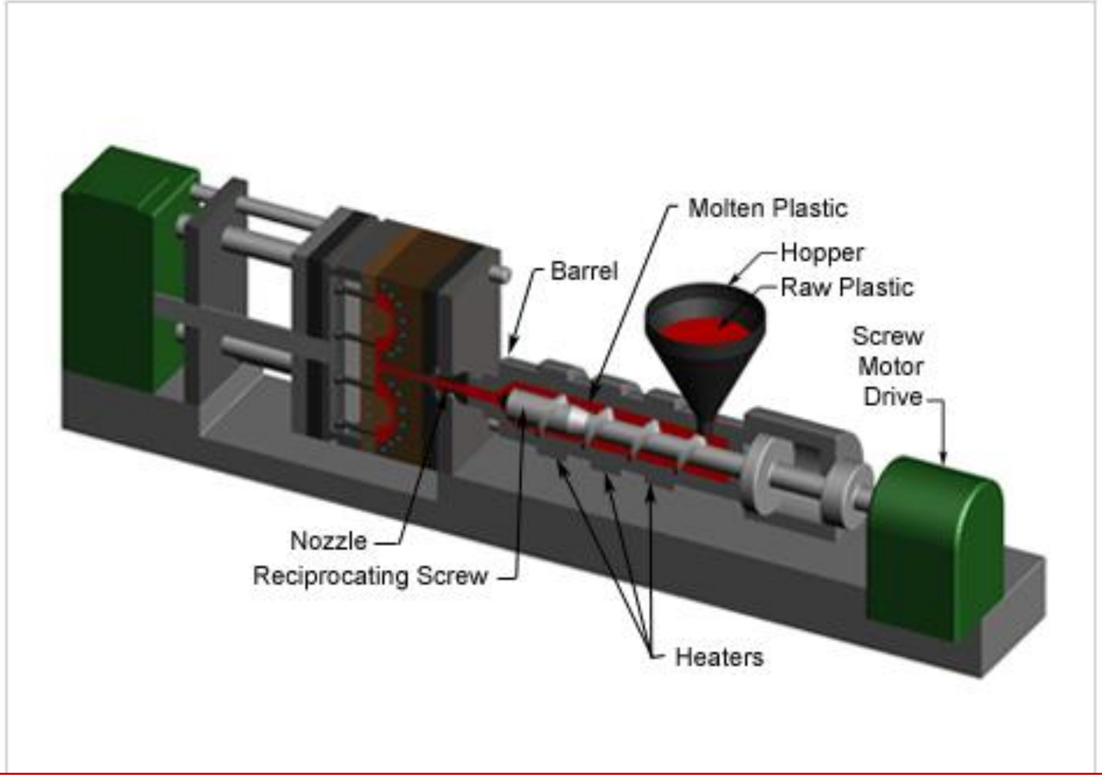
Clamping

- Two halves of the mold must be tightly closed before the plastic material is injected into the mold.
- One half of the mold is attached to the injection unit (nozzle) and other half is allowed to slide on the tie bars.
- The clamping of mold is operated hydraulically which pushes the moving half part of the mold towards the fixed part to make an air tight chamber.
- The pressure and the time required to close and open the mold depends upon the machine capability



Injection

- The plastic material is melted by the application of heat and forwarded through the piston towards the nozzle and finally into the mold.
- The molten plastic is then injected into the mold quickly.
- The amount of material that is injected into the mold is referred to as the shot volume
- The injection time can be estimated by the shot volume, injection power and pressure.



Cooling - The molten plastic that is inside the mold begins to cool as soon as it makes contact with the interior mold surfaces.

As the plastic cools, it will solidify into the shape of the desired part

Materials used:

Acetal Acrylic, Polycarbonate, PEEK, Polyethylene, PP, PVC

Applications

- The injection molding process can be used to manufacture thin walled plastic housing products which require many ribs and bosses on the interior surfaces.
- These housings are used in a variety of products including household appliance, electronics, power tools and as automotive dashboards.
- Thin walled products include different types of open containers such as buckets.
- It is also used to produce several daily use items such as toothbrushes or small plastic toys, many medical devices including valves and syringes.

Automotive console board mold

car door trim panel mold

Car bumper molds

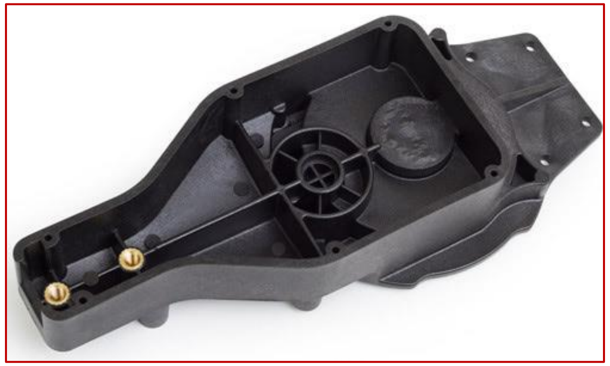
Plastic Material: ABS, PP, PS, PVC, HDPE

ADV:

- Higher production rate
- Close tolerances on small intricate parts
- Minimum wastage of material
- Complex geometry can be easily produced

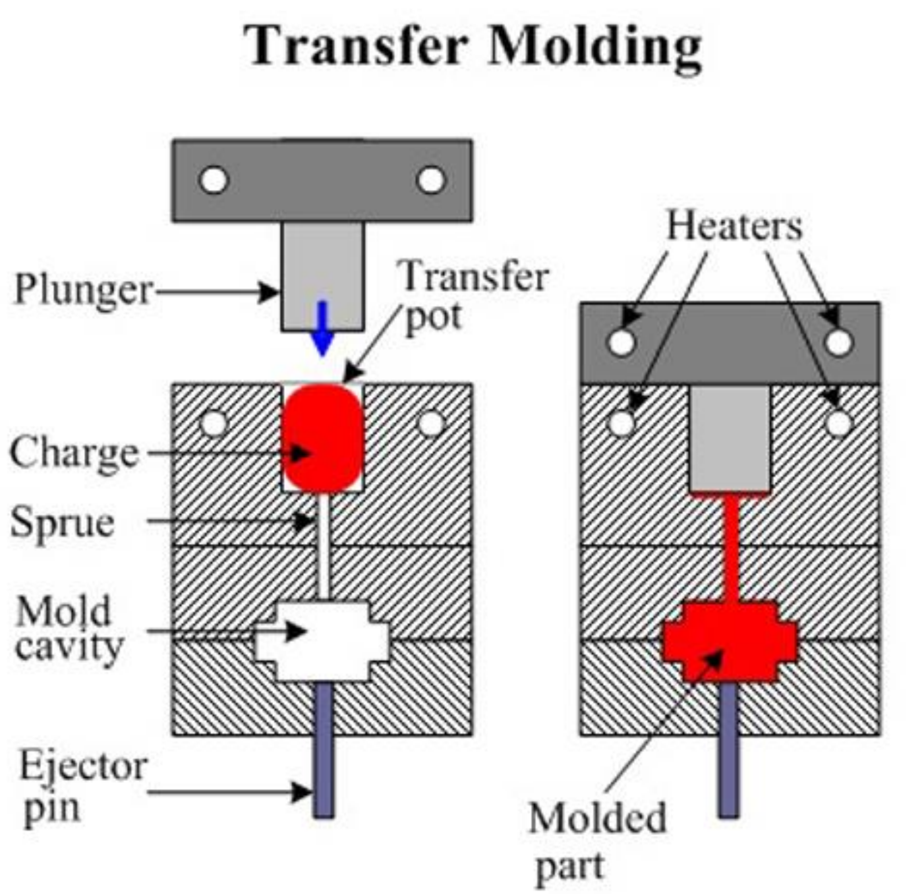
LIMITATIONS

- Tooling cost higher
- High setup cost



Transfer molding

- Transfer molding process combines the principle of compression and transfer of the polymer charge
- Resin is transferred from the transfer pot to the mold.
- No extra pressure is required
- The required amount of resin is weighed and inserted into the transfer pot before the molding process.
- The transfer pot is hated by the heating elements above the melting point of the resin.
- This allows a faster flow of material through the sprue into the mold cavity
- A plunger is used to push the material from the transfer pot through sprue into the mold cavity.
- The mold is held closed until the resin gets cured.



- The mold cavity is opened and the molded part can be removed with the help of ejector pin.

Process Parameters

- Heating time
- Melting temperature of the charge
- Applied pressure
- Cooling time

Materials used

Thermosets

- Epoxy
- Polyester
- Phenol-formaldehyde,
- Vinyl ester
- Silicone

Applications

- coils, integrated circuits, plugs, connectors, pins and studs

Adv:

- High production rate
- Fast setup time and lower cost
- Lower maintenance cost than injection molding
- Better flexibility
- Durable and dimensionally stable parts
- Uniform thickness of parts

Limitation

- Wastage of material
- Prodn. Rate lower than injection molding
- Air can be trapped in the mold.

Environmental degradation pathways

- Biodegradable polymer refers to a substance that can be broken down by microorganism within a fixed period of time.
- Effective bio degradation requires a specific environmental conditions including temperature, level of variation, allowing microorganisms to convert natural material into other natural substance such as compost, water and carbon dioxide.

Biodegradation
Photodegradation
Oxidants
Hydrolysis

→ Fragments

Aerobic biodegradation ----- Polymer + O_2 → $CO_2 + H_2O$ + biomass + residue

Anerobic Fermentation ------ Polymer → $CO_2 + CH_4 + H_2O$ + biomass + residue

Modes of biodegradation

- ***Microorganisms: Fungi, Bacteria***
- ***Enzymes:*** ***Physical factors affecting the activity of enzymes;***
  degradation by biological oxidation;
  degradation by biological hydrolysis.

Low energy degradation

- Ester
- Ether
- Ketone
- Hydroxyl

The overall strength and stability of polymers are due to

- Large molecular size
- Extensive molecular entanglement
- Huge cohesive force

Polymer materials have been designed in the past to resist degradation.

The challenge is to design polymers that have the necessary functionality during use but destruct under the stimulus of an environment trigger after use.

Trigger

- Microbial
- Hydrolysis
- Oxidation
- Catalyst

Caution

- Should not be toxic
- Should not persist in the environment
- Should be completely utilized by soil microorganisms.

Now biodegradable polymers are satisfying such niche markets

- Food wrappers
- Agricultural films
- Personal products
- Marine and fresh water applications
- Compost bags

Natural biodegradable polymers

- Polysaccharides: Starch, Cellulose, Chitin and Chitosan, Alginic acid
- Polypeptides of natural origin
- Bacterial Polyesters
- Animal origin: Collagen/gelatin
- Agricultural feedstock

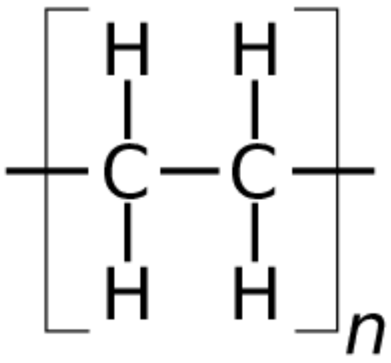
  Lipids/Fats

  - Bees wax
  - Carnauta wax
  - free fatty acids
- Microbial sources

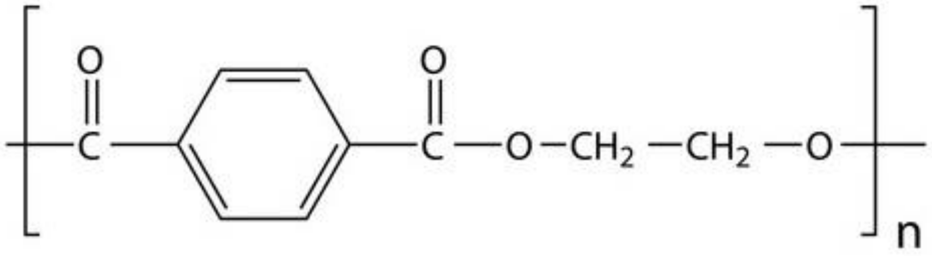
Factors that affect biodegradability

- Chemical bonding
- Polymer structure
- Microstructure/morphology
- Side chain
- Molecular weight
- Melting point
- Crystallinity

polyethylene



polyethylene tetra phthalate



Structural requirement for biodegradability

- Ester, ether and peptide bond
- Presence of branched hydrocarbons shows biodegradation.
- Hydroxyl and carboxyl groups accelerate biodegradation
- Unsaturated aliphatic compound is more biodegradation than the corresponding saturated compound.

Order of biodegradation

Aliphatic.>alicyclic>aromatic

Hydrophilic>hydrophobic

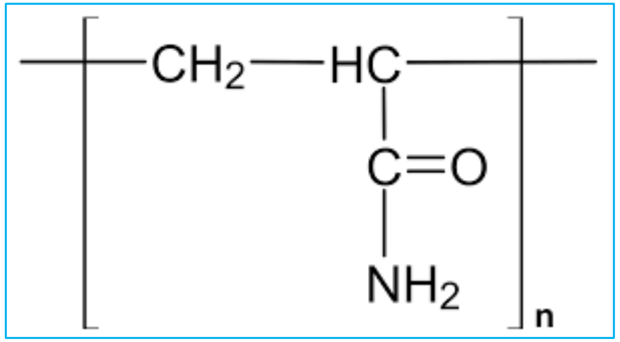
Low mol. Wt. > High mol. Wt.

Low mel. > High mel.

Amorphous>Crystalline

Polymers with hydrolysable backbones

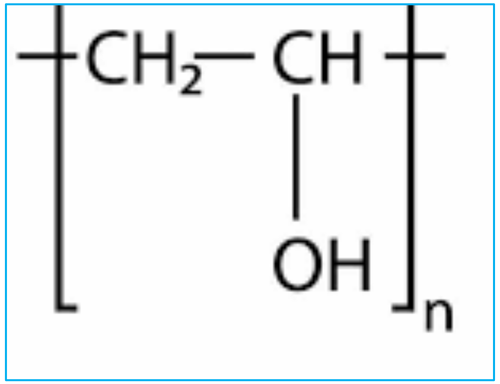
- ***Polyesters -PGA***
- ***Polycaprolactone***
- ***Polyamides***
- ***Polyurethanes and Polyurea***
- ***Polyanhydrides***
- ***Poly(amide-enamine)s***

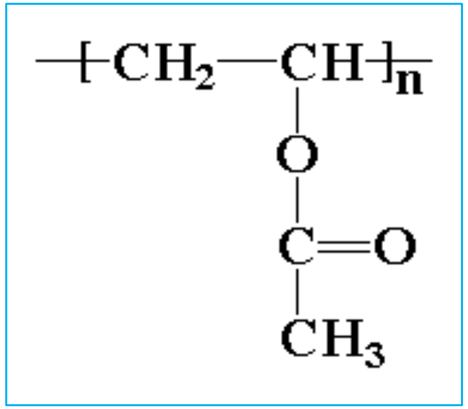


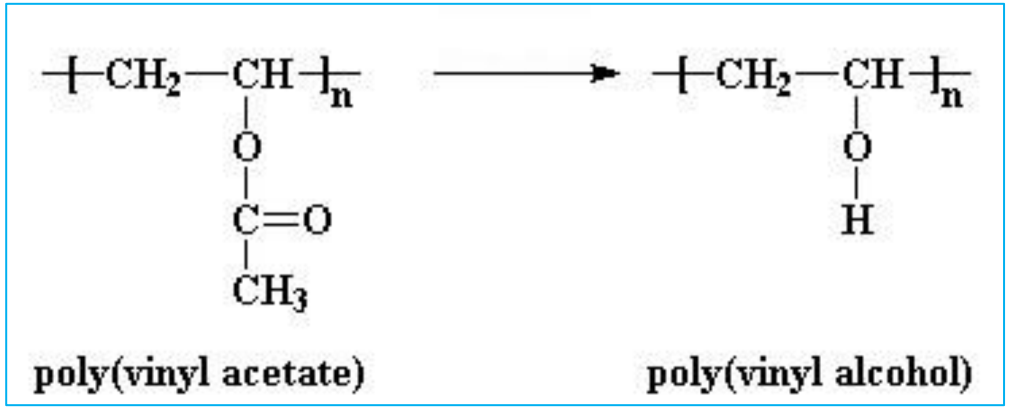
Types	*Application*	*Degradation*
PHA	Blow and injection moulded bottles and plastic films	10 weeks in compost
PHBH	Mono/multi layer films manufactured via casting or blowing methods	Under aerobic & anerobic conditions
PLA	Drink cups, take away food trays	2 weeks via hydrolysis
PCL	Food contact foam trays and film bags	6 weeks in compost

Polymers with carbon backbones

- ***Polyvinylalcohol:*** *Mostly readily biodegradable of vinyl polymers.Initial biodegradation of secondary alcohol group in PVA*
- ***Polyvinylacetatte****:* *Degrades more slowly. Degradation occurs through hydrolysis of PVAC to PVA followed by enzymatic oxidation and degradation.*
- ***Polyacrylates:*** *They generally resist biodegradation. Copolymers of ethylene and propylene with acrylic acid, acrylonitrile and acrylamide degrades under soil burial.*







PVA

Poly Vinyl Acetate

Polymer modification to facilitate Biodegradation

1.*Polymers with carbon chain backbone except for those with large number of polar groups on the main chain such as PVA are not susceptible to enzyme-catalyze degradation reactions*

Two approaches for the modification of such polymers

- *Insertion of functional groups in the main chain especially ester groups which can be cleaved by chemical hydrolysis.*
- *Insertion of functional groups in or on the main chain that can undergo photochemical chai-cleavage reactions*

2. *Blending*

- *Starch-glycerol-PVA to produce water soluble bags*
- *Starch-LDPE*
- *50% Starch-Polyethylene Films for packaging and agricultural mulch*
- *40% Starch-20% EAA-25%LDPE and 15% urea for agricultural mulch*

Starch based polymers

Thermoplastic starch products	Food packaging, disposable eatable utensils, protective packaging, compostable films and bags, retail and agriculture
Starch synthetic aliphatic polyester blends	High quality sheets and packaging films
Starch and PBS/PBSA polyester blends	Thermoformed biscuit trays
Starch-PVOH blends	Water soluble laundry bags, drug control release carrier.

Application of biodegradable polymers

- ***Medical applications***

  Surgical sutures: Most popular absorbable sutures such as Polyglycolide (PGA), poly-L-lactide (PLA) and their copolymers.

  Bone fixation devices: Biodegradable implants of PGA, PLA

  Vascular grafts: cross linked gelatin

  Adhesion prevention: Photocurable mucopolysaccharides

  Artificial skin: Collagen, chitin and poly-L-leucine

  Drug delivery systems: Poly(lactic acid) and poly(ortho ester)s

- ***Agricultural applications***

  Agricultural mulches: starch with poly(vinyl alcohol); poly(ethylene-co-acrylic acid) and PVC; polylactone and poly(vinyl alcohol) films.

- ***Packaging:***

  Poly((hydroxybutyrate-co-valerate) (PHBV), starch and chitosan films for food packaging.

  LDPE blends with 10% corn starch made into bags for groceries.

Essential guidelines to reach the goal of biodegradability

- ***Selection of a suitable polymer class.***
- ***Engineering essential structural units in the polymer.***
- ***Easy access of the polymer to the biodegradation agencies.***
- ***Creating accelerated biodegradation environment biodegradation.***
- ***Chemical modification of commodity polymers for biodegradation.***